

Efficient Pseudorandom Correlation Generators over $\mathbb{Z}/p^k\mathbb{Z}$

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Abstract. Modern efficient secure multi-party computation (MPC) protocols typically follow an offline-online design, where offline protocols produce a sufficient amount of correlated randomness that would be consumed during the online phases. The past decades have witnessed maturing of efficient online protocols, for computing circuits over either arbitrary finite fields or rings $\mathbb{Z}_{p^k}$. In particular, protocols tailored for $\mathbb{Z}_{2^k}$ arithmetic have achieved better concrete efficiency in most real-life applications, as it naturally captures modern CPU architectures. On the other hand, a recent paradigm of *pseudorandom correlation generator* (PCG) initiated by Boyle et al. (CCS'18, Crypto'19) opens a door to efficient preprocessing with sublinear communication. Since then, PCGs have been extensively studied and developed to produce various types of correlations required from online protocols. Although Li et al. (Euro-Crypt'25) recently put a significant step forward and propose efficient PCGs for arbitrary finite fields, the current state of PCGs for rings is not satisfying at all. Towards the great demand for efficiently generating correlations over rings, we investigate PCGs for general Galois rings, which simultaneously unify finite fields and integer rings modulo p^k . In summary, we establish the following results:

- (i) We generalize the state-of-the-art PCG constructions for oblivious linear evaluations (OLE) over Galois fields to *arbitrary Galois rings*, basing on Galois theory and the Hensel lift. Moreover, our PCGs for Galois rings are as efficient as PCGs for fields. Concretely, for mN OLE correlations over $\mathbb{Z}_{2^k}$, we require $O(m \log N)$ communication and $O(m^2 N \log N)$ computation, where m is an arbitrary integer ≥ 2 . In comparison, to our best knowledge, previous approaches incur communication at least linear in N .
- (ii) We extend the above OLE construction to provide various types of correlations over any Galois ring. One of the fascinating applications is an efficient PCG for two-party SPD $\mathbb{Z}_{2^k}$ authenticated multiplication triples (Crypto'18). For mN SPD $\mathbb{Z}_{2^k}$ triples, our approach requires only $O(m \log N)$ communication and $O(m^2 N \log N)$ computation. Concrete evaluations show that our method significantly outperforms existing schemes based on homomorphic encryption.

(iii) In addition, our PCGs for Galois rings also enable multi-party multiplication triple generation, yielding the first efficient MPC protocol for arithmetic circuits over $\mathbb{Z}_{2^k}$ with *silent* and *sublinear* preprocessing. Additional applications include circuit-dependent preprocessing and matrix multiplication triples, etc, which are of independent interest.

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1 Introduction

Secure multi-party computation (MPC) protocols allow n parties $P_1, \dots, P_n$ to jointly and securely compute a function $f : \mathcal{R}^n \rightarrow \mathcal{R}$ on an input $(x_1, \dots, x_n)$, where each party P_i holds x_i , without revealing the inputs to each other. Over

the past decades, significant advancements have improved the concrete efficiency of MPC, making it practical for various real-world applications. A key technique driving this progress is Beaver’s circuit randomization method [Bea91], commonly known as Beaver’s triples. This approach allows modern MPC protocols to adopt a modular design, where the computation is divided into two phases: (1) an information-theoretic *online* phase, which evaluates f assuming access to precomputed correlated randomness, and (2) a (probably cryptographic) *offline* phase protocol that produces the desired correlated randomness.

A central goal in the MPC literature lies in minimizing both communication complexity and round complexity. Recently, Boyle et al. [BCG18, BCG+19b] put forward the notion of pseudorandom correlation generators (PCGs), which offers an appealing solution to the offline phase. Loosely speaking, after a seed generation procedure, each party holds a *short* PCG seed, which can then be *efficiently* and *locally* expanded into *long* target correlations. Moreover, many PCGs enjoy a *programmable* property, reusing seeds to construct PCGs for multiparty correlations (e.g., Beaver triples) from simpler 2-party correlations (e.g., oblivious linear evaluation (OLE)). Previously, programmable PCGs for OLEs with quasi-linear computation and sublinear communication were only known for finite fields of size greater than 2 [BCG+20b, BCCD23]. Very recently, Li et al. [LXY25] filled in the final puzzle piece of $\mathbb{F}_2$ by a novel use of trace functions, yielding efficient PCGs for Beaver triples over any finite field and thereby semi-honestly secure MPC over any finite field with a silent and sublinear preprocessing.

On the other hand, numerous efforts have been made to extend the types of computation operations supported with high efficiency. Typically, most MPC protocols focus on field operations and assume that f can be efficiently represented by an arithmetic circuit over a finite field. However, it is not always the case in practice. In many real-world applications, f is more naturally expressed as a computer program, and converting it into an arithmetic circuit over fields incurs substantial overhead, as field operations do not align well with standard CPUs. Instead, computations in $\mathbb{Z}_{2^k}$ are often more practical, as they naturally support operations such as comparisons and bitwise manipulations, which are fundamental in many applications and efficiently implemented using modulo 2^k arithmetic. Consequently, designing MPC protocols that operate over $\mathbb{Z}_{2^k}$ is of particular interest.

Existing works [CDE+18, OSV20, CKL21, EXY22] on maliciously secure MPC over $\mathbb{Z}_{2^k}$ primarily employ two different flavors of “ring extension”: (1) extending $\mathbb{Z}_{2^k}$ to a larger ring $\mathbb{Z}_{2^\ell}$ [CDE+18, OSV20, CKL21], and (2) using a Galois extension of $\mathbb{Z}_{2^k}$ [EXY22]. Regardless of the extension flavor, all these works need to consume random Beaver triples defined over the corresponding ring in the online phase. However, in the current state, approaches to generating Beaver triples over these extensions are quite limited. In a bird’s eye view, for field cases, one can generate Beaver triples upon oblivious transfer (OT) [KOS16], or homomorphic encryption (HE) [BDOZ11, DPSZ12, BEP+22], or PCGs [BCG+20b, BCCD23, BBC+24, LXY25]. In contrast, for ring cases, only the first two approaches have

been explored [CDE⁺18, OSV20, CKL21, EXY22], and to our best knowledge, no efficient PCG-based approach exists. Moreover, existing methods suffer from significant overheads. The OT-based approach [CDE⁺18, EXY22] generally incurs a quadratic blow-up in the ring size due to bit-decomposition of ring elements while the HE-based approach [OSV20, CKL21] has high computational overhead and linear communication cost in the number of triples. Consequently, compared to the field setting, the current state of preprocessing MPC is less efficient, leaving substantial room for improvement. Since programmable PCGs for OLEs are the key ingredient of building Beaver triple correlations, we ask the following question:

Can we construct efficient programmable PCGs for OLEs over any Galois ring, and in particular, over the integer ring modulo 2^k , i.e., $\mathbb{Z}_{2^k}$?

1.1 Our Contributions

In this work, we revisit and generalize the state-of-the-art PCG paradigms [BCG⁺20b, BCCD23, LXY25] from Galois fields to *arbitrary Galois rings*, leveraging Galois theory and Hensel’s lifting techniques to achieve efficiency comparable to the field setting. A Galois ring denoted by $\mathbb{GR}(p^k, m)$, is a Galois extension of $\mathbb{Z}_{p^k}$, namely $\mathbb{Z}_{p^k}[x]/(g(x))$, where $\bar{g}(x) = g(x) \bmod p$ is a degree- m monic irreducible polynomial over $\mathbb{F}_p$. Furthermore, Galois rings also generalize Galois fields, because $\mathbb{GR}(p^k, m)/(p) \simeq \mathbb{F}_{p^m}$ and $\mathbb{GR}(p^1, m) = \mathbb{F}_{p^m}$. This means that our PCGs for arbitrary Galois rings are *universal*, not only subsuming the state-of-the-art results for Galois fields, but also enabling new application scenarios beyond the field setting.

PCG for OLE over any Galois ring. We exhibit PCG constructions for random OLE correlations over arbitrary Galois rings for the first time. In an OLE correlation, for $\theta \in \{0, 1\}$, party P_θ holds $b_\theta, z_\theta \in \mathbb{GR}(p^k, m)$, such that

$$b_0 \cdot b_1 = z_0 + z_1.$$

In other words, each party holds an additive share of $b_0 \cdot b_1$, which we denote as $[b_0 \cdot b_1]$. As in the literature, our constructions build upon Galois ring analogues of existing, well-studied Ring-LPN/QA-SD assumptions [BCG⁺20b, BCCD23, LXY25] and function secret sharing (FSS) [GH14, BGI15, BGI16, GYW⁺23] for sums of point functions (SPFSS).

Theorem 1.1 (PCG for OLE, informal). *Assuming Ring-LPN (Definition 3.2) or QA-SD over Galois ring (Definition 3.4) and the SPFSS scheme in [BGI16], there exists a secure programmable PCG construction for generating $O(N)$ random OLE correlations over any Galois ring $\mathbb{GR}(p^k, m)$, with seed size $O(\lambda^3 \log N)$ and computational cost $O(N \log N)$, where λ is the security parameter.*

Note that Theorem 1.1 includes the PCG for OLE over both $\mathbb{Z}_{p^k}$ and $\mathbb{GR}(p^k, m)$ employing distinct techniques as in Theorem 5.1 and Theorem 5.4. We then

compare our approach with existing methods for generating OLE correlations over Galois rings. Details are provided in [Table 1](#).

Methods	N OLEs	Comm.	Comp.	Assumption
Classical OT	Any $\mathbb{GR}(p^k, m)$	$k^2 m^2 \cdot \Omega(\lambda N)$	$\Omega(N)$	OT
[EXY22]	Any $\mathbb{GR}(p^k, m)$	$k^2 m \cdot \Omega(\lambda N)$	$\Omega(N)$	OT
MHz2k [CKL21]	$\mathbb{GR}(2^k, m), k > \lambda$	$\Omega(\lambda N)$	$\Omega(N)$	Ring-LWE
[LXY25]	Any $\mathbb{F}_{p^m}$	$O(\lambda^3 \log N + \lambda^2 m)$	$O(N \log N)$	QA-SD/Ring-LPN
This work	Any $\mathbb{GR}(p^k, m)$	$O(\lambda^3 \log N + \lambda^2 km)$	$O(N \log N)$	QA-SD/Ring-LPN

Table 1. Comparisons to previous approaches for OLE. Let λ denote the computational security parameter. “Comm.” stands for the communication complexity, and “Comp.” refers to the number of underlying arithmetic operations.

Applications to MPC for ring arithmetic. We then extend our PCG constructions to generate additional correlations over Galois rings. One of the fascinating applications is an efficient PCG for SPD $\mathbb{Z}_{2^k}$ -type [\[CDE+18\]](#) authenticated multiplication triples. In SPD $\mathbb{Z}_{2^k}$, to authenticate a multiplication triple $([x], [y], [z])$ over $\mathbb{Z}_{2^k}$ with $x, y, z \in \mathbb{Z}_{2^k}, x \cdot y = z \pmod{2^k}$, parties additionally holds $([x \cdot \Delta], [y \cdot \Delta], [z \cdot \Delta])$ over a larger ring $\mathbb{Z}_{2^\ell}$, where $\ell = k + r$ and $\Delta \in \mathbb{Z}_{2^r}$ is the global authentication key and keeps identical among triples. We denote such a triple by $(\llbracket x \rrbracket, \llbracket y \rrbracket, \llbracket z \rrbracket)$. Our PCG efficiently generates SPD $\mathbb{Z}_{2^k}$ triples in the two-party setting.

Theorem 1.2 (PCG for SPD $\mathbb{Z}_{2^k}$ triples, informal). *Assuming Ring-LPN ([Definition 3.2](#)) or QA-SD over Galois ring ([Definition 3.4](#)) and the SPFSS in [\[BGI16\]](#), there exists a secure PCG construction for generating $O(N)$ two-party SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples, with seed size $O(\lambda^3 \log N)$ and computational complexity $O(N \log N)$, where λ is the security parameter.*

The SPD $\mathbb{Z}_{2^k}$ authenticated multiplication triple requires maliciously secure distributed setup protocols for PCG seed generation, which rely on malicious DPF constructions for Galois rings. In particular, such DPFs can be naturally adapted from [\[BCG+20b, BCCD23\]](#), and the resulting distributed setup protocols have the same asymptotic efficiency as [Theorem 1.2](#). We implement and evaluate the concrete efficiency of our PCG for SPD $\mathbb{Z}_{2^k}$ triples based on QA-SD assumptions, which significantly outperforms previous approaches. Details are presented in [Table 2](#).

PCG for multi-party correlations. Since our PCG for OLE over arbitrary Galois rings is programmable, in general it enables generating typical multi-party degree-2 correlations *efficiently* and *silently*, as shown in [\[BCG+20b\]](#). For instance, we obtain multi-party PCGs for Beaver triples over arbitrary Galois rings, which admit efficient MPC for circuits over $\mathbb{Z}_{2^k}$ with sublinear preprocessing. To our best

Parameters	Protocol	Assumption	Triple Generation	Comm.
$(k, s) = (32, 26)$	SPD $\mathbb{Z}_{2^k}$ [CDE ⁺ 18]	VOLE/OT	$\approx 12500/s$	4.56 TB
	This work	QA-SD	$\approx 109,000/s$	≈ 115 MB
$(k, s) = (64, 57)$	Overdrive2k [OSV20]	BGV SHE	Unclear	17.85 TB
	MHz2k [CKL21]	BGV SHE	Unclear	3.71 TB
	Mon $\mathbb{Z}_{2^k}$ a [CRFG20]	Joye-Libert HE	19/s	13.72 TB
	SPD $\mathbb{Z}_{2^k}$ [CDE ⁺ 18]	VOLE/OT	$< 5000/s$	18.20 TB
	This work	QA-SD	$\approx 52,000/s$	≈ 175 MB

Table 2. The cost refers to the generation of SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples for two parties. The communication complexity is estimated for 10^9 triples as per [BBC⁺24, LXY25]. The runtime of SPD $\mathbb{Z}_{2^k}$ is from [DEF⁺19, Figure 9] and the communication cost is computed from the formula $(k + 2r)(4k + 9r)$ [CDE⁺18, Section 7]. The Mon $\mathbb{Z}_{2^k}$ a running time is taken from [CRFG20]. The running time and the communication cost of this work are obtained from our evaluations in the way similar to [BBC⁺24, LXY25]. Experimental details are referred to Section 7.2.

knowledge, prior approaches either relies on homomorphic encryption or builds upon oblivious transfers, and have at least linear communication complexity. In addition, following the routine of [BCG⁺20b, BCCD23, LXY25], our approach also supports circuit-dependent preprocessing. For computing N copies of a circuit $\mathcal{C}$ over GR with different inputs, we achieve a silent PCG with seed size $O(|\mathcal{C}| \log N)$. Finally, we can extend PCGs for matrix multiplication triples from Galois fields [LXY25] to Galois rings. Concretely, for N/m triples of $m \times m$ matrices over $\mathbb{Z}_{2^k}$, we have a similar seed size of $O(m\lambda^3 \log N)$.

1.2 Technical Overview

Recap: PCG constructions for OLE over Galois fields. Before describing our PCG for OLE over arbitrary Galois rings, it is instructive to briefly review existing frameworks of PCGs for OLE over Galois fields.

Framework of [BCG⁺19b]. This framework relies on a dual variant of the learning parity with noise (LPN) assumptions [BFKL93]. Given a public matrix $H \xleftarrow{\$} \mathbb{F}_q^{M \times N}$ and random sparse vectors $\mathbf{e}_0, \mathbf{e}_1 \in \mathbb{F}_q^N$ with Hamming weight t , the dual-LPN over $\mathbb{F}_q$ assumption states that $\mathbf{b}_0 := H \cdot \mathbf{e}_0, \mathbf{b}_1 := H \cdot \mathbf{e}_1 \in \mathbb{F}_q^M$ are computationally indistinguishable from uniform distribution. The goal is to obtain $[\mathbf{b}_0 * \mathbf{b}_1]$, where $*$ denotes the entry-wise multiplication. The main observation is that $\mathbf{b}_0 * \mathbf{b}_1$ lies on the main diagonal of $\mathbf{b}_0 \cdot \mathbf{b}_1^\top \in \mathbb{F}_q^{M \times M}$, and the following holds:

$$\mathbf{b}_0 \cdot \mathbf{b}_1^\top = (H \cdot \mathbf{e}_0) \cdot (H \cdot \mathbf{e}_1)^\top = H \cdot (\mathbf{e}_0 \cdot \mathbf{e}_1^\top) \cdot H^\top.$$

Hence, to compute additive shares of $\mathbf{b}_0 * \mathbf{b}_1$, it suffices to compute the additive shares of $\mathbf{e}_0 \cdot \mathbf{e}_1^\top \in \mathbb{F}_q^{N \times N}$, which has at most t^2 non-zero entries. Note that

$\mathbf{e}_0 \cdot \mathbf{e}_1^\top$ can be viewed as the sum of t^2 point functions with domain $[0, N^2 - 1]$ and range $\mathbb{F}_q$, so that efficient SPFSS [GI14, BGI15, BGI16, GYW⁺23] can be applied. However, as noted in [BCG⁺20b] this approach takes at least $O(N^2 \log N)$ computations, even if H is highly structured.

Framework of [BCG⁺20b, BCCD23]. At a high-level, this framework crucially relies on a ring homomorphism between the polynomial ring $\mathcal{R}$ and the vector space $\mathbb{F}_q^N$, reducing multiple OLE correlations over $\mathbb{F}_q$ to one OLE correlation over the polynomial ring $\mathcal{R}$. Specifically, let $\mathcal{R} := \mathbb{F}_q[\mathbf{x}_1, \dots, \mathbf{x}_n]/(g(\mathbf{x}_1), \dots, g(\mathbf{x}_n))$ with $n \geq 1$, $g(\mathbf{x}) = \mathbf{x}^d - 1$, and $d \mid q - 1$.

Since $\mathbf{x}^d - 1$ is fully splittable over $\mathbb{F}_q$, the Chinese remainder theorem gives an isomorphism $\mathcal{R} \simeq \mathbb{F}_q^N$ for $N = d^n$. To obtain OLEs over $\mathcal{R}$, the framework relies on Ring-LPN/QA-SD assumptions and function secret sharing schemes (SPFSS). In more detail, let $a \xleftarrow{\$} \mathcal{R}$ be the public input, and s_θ, e_θ be t -sparse elements of $\mathcal{R}$, where $\theta \in \{0, 1\}$. The Ring-LPN/QA-SD assumption³ implies that $b_\theta := a \cdot s_\theta + e_\theta$ is pseudorandom. The key observation is the following:

$$b_0 \cdot b_1 = (a \cdot s_0 + e_0) \cdot (a \cdot s_1 + e_1) = a^2 \cdot (s_0 s_1) + a \cdot (s_0 e_1 + s_1 e_0) + e_0 e_1,$$

which means that additive shares of $b_0 \cdot b_1$ can be derived from the additive shares of the four cross-terms $s_0 s_1, s_0 e_1, s_1 e_0, e_0 e_1$ as a is public. In addition, the four cross-terms are t^2 -sparse in $\mathcal{R}$ and thus can be succinctly shared by SPFSS. Putting all pieces together, the resulting PCG constructions have communication complexity $O(\lambda^3 \log N)$ and computation complexity $O(N \log N)$. However, as note in [BCCD23], this framework inherently requires $d \mid q - 1$ hence the field size q needs to be at least ≥ 3 .

Framework of [LXY25]. Li et al. [LXY25] overcome the field size limitations of [BCG⁺20b, BCCD23] by introducing a novel idea based on *trace* functions. At a high-level, they employ trace functions to allow efficient and local conversion of correlations over polynomial rings over $\mathbb{F}_q$ into multiple OLEs over $\mathbb{F}_p$, where $q = p^m$. Similarly, the polynomial rings are defined as $\mathcal{R} := \mathbb{F}_q[\mathbf{x}_1, \dots, \mathbf{x}_n]/(g(\mathbf{x}_1), \dots, g(\mathbf{x}_n))$, with $g(\mathbf{x}) = \mathbf{x}^d - 1$ and $d \mid q - 1$, so that there exists a ring isomorphism $\phi : \mathcal{R} \rightarrow \mathbb{F}_q^N$, where $N = d^n$. At the heart of their techniques is a properly defined trace functions over $\mathcal{R}$, denoted by $\text{Tr}_{\mathcal{R}}(\cdot)$, which enjoys a “commutative” property with ϕ . Namely, the following commutative diagram holds:

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\phi} & \mathbb{F}_q^N \\ & \searrow \text{Tr}_{\mathcal{R}} & \searrow \text{Tr}_{\mathbb{F}_q} \\ & \text{Im}(\text{Tr}_{\mathcal{R}}) & \xrightarrow{\phi} \mathbb{F}_p^N \end{array}$$

This implies that the correlation $\text{Tr}_{\mathcal{R}}(b_0) \cdot \text{Tr}_{\mathcal{R}}(b_1) = \text{Tr}_{\mathcal{R}}(z_0) + \text{Tr}_{\mathcal{R}}(z_1)$ over $\mathcal{R}$ is equivalent to N OLE correlations over $\mathbb{F}_p$, with respect to the isomorphism $\mathcal{R} \simeq$

³ Actually, Ring-LPN can be viewed as a special case of QA-SD with $n = 1$.

$\mathbb{F}_q^N$. Hence, it suffices to determine how party P_θ can efficiently obtain such $b_\theta, z_\theta \in \mathcal{R}$, for $\theta \in \{0, 1\}$. To this end, [LXY25] further exploits algebraic properties of the trace functions over $\mathcal{R}$, and eventually achieves $O(m\lambda^3 \log N)$ communication complexity and $O(m^2 N \log N)$ computation complexity for generating mN OLEs over arbitrary $\mathbb{F}_p$.

PCG for OLE over arbitrary Galois rings. We first extend the underlying assumptions from Galois fields to Galois rings. As shown in [BCG+19b, BCG+20b], LPN/Ring-LPN assumptions can be naturally defined over rings, and no significant weakness or structural difference has been discovered compared to the prime field cases. For detailed security analyses of LPN assumptions over Galois rings, we refer to [BBMS22, LWYY24, LXY24]. We systematically study the hardness of Ring-LPN/QA-SD assumptions over Galois rings following the analyses of [BCCD23, LWYY24, LXY25]. Details are presented in Appendix B.

Assuming the hardness of underlying Galois ring analogues, we observe that the first two frameworks [BCG+19b, BCG+20b, BCCD23] generalize straightforwardly. Similarly, the LPN-based approach [BCG+19b] admits programmable PCGs for OLE over any Galois ring, with $O(N^2 \log N)$ computation and $O(\lambda^3 \log N)$ communication. As for the Ring-LPN/QA-SD based approach [BCG+20b, BCCD23], the key observation is that a similar ring isomorphism can be established by using Hensel’s lift techniques [Wan03]. However, similar limitations on the Galois rings still persist. Consequently, we obtain programmable PCGs for OLEs with $O(N \log N)$ computation and $O(\lambda^3 \log N)$ communication, which generally works for arbitrary Galois rings except for $\mathbb{Z}_{2^k}$, i.e., Galois rings of characteristic 2^k , and extension degree 1.

Challenges of extending the trace-based paradigm. Although Galois theory apply to ring extensions and trace functions over Galois rings are known, extending the approach of [LXY25] is significantly more challenging. Intuitively, the key challenge is to properly define trace functions over polynomial rings over Galois rings, while preserving as many algebraic properties used in [LXY25] as possible.

Let $\mathcal{R} := \mathbb{F}_q[X_1, \dots, X_n]/(g(X_1), \dots, g(X_n))$, where $q = p^m$ and $g(X) = X^d - 1$, the trace function over $\mathcal{R}$ in [LXY25] is defined as

$$\text{Tr}_{\mathcal{R}} : \mathcal{R} \rightarrow \mathcal{R}, \quad f \mapsto \sum_{i=0}^{m-1} f^{p^i}.$$

We explicitly summarize the necessary algebraic properties of the trace function as follows:

1. **Commutativity.** The “commutative” property is critical for the trace idea to work. It ensures that applying trace on $\mathcal{R}$ is essentially equivalent to applying trace on $\mathbb{F}_q^N$.
2. **Additive Homomorphism.** The trace function is additive, enabling compatibility with function secret sharing schemes (SPFSS) for efficient sharing of cross-terms.

3. **Sparsity Property** For any t -sparse element $f \in \mathcal{R}$, f^{p^i} is still t -sparse, for any integer i , so that the cross-terms can be succinctly shared by SPFSS.
4. **$\mathbb{F}_p$ -linearity.** For any $b_0, b_1 \in \mathcal{R}$, it holds that $\text{Tr}_{\mathcal{R}}(b_0) \cdot \text{Tr}_{\mathcal{R}}(b_1) = \text{Tr}_{\mathcal{R}}(b_0 \cdot \text{Tr}_{\mathcal{R}}(b_1))$, which would significantly reduce the number of cross-terms to be shared by a multiplicative factor m .
5. **Extraction property.** Given a basis $(b_1, \dots, b_m)$ of $\mathbb{F}_{p^m}$ over $\mathbb{F}_p$, $(\text{Tr}_{\mathcal{R}}(b_1 \cdot x), \dots, \text{Tr}_{\mathcal{R}}(b_m \cdot x))$ is uniformly random in $\mathbb{F}_p^{mN}$ with respect to the isomorphism $\phi : \mathcal{R} \rightarrow \mathbb{F}_p^N$, for $x \xleftarrow{\$} \mathcal{R}$. This enables to extract mN OLE correlations over $\mathbb{F}_p$ rather than only N from a single PCG seed.

Let $a \xleftarrow{\$} \mathcal{R}$ be the public input, and s_θ, e_θ be t -sparse elements of $\mathcal{R}$, where $\theta \in \{0, 1\}$. Assuming QA-SD($\mathcal{R}, t$) (or Ring-LPN($\mathcal{R}, t$), if $n = 1$ and p^m is sufficiently large) and by the above five properties, the trace-based paradigm exploits the following observation: for each $j \in [0, m-1]$,

$$\begin{aligned}
\text{Tr}_{\mathcal{R}}(\xi^j b_0) \cdot \text{Tr}_{\mathcal{R}}(\xi^j b_1) &= \text{Tr}_{\mathcal{R}}(\xi^j b_0 \cdot \text{Tr}_{\mathcal{R}}(\xi^j b_1)) \\
&= \text{Tr}_{\mathcal{R}}(\xi^j \cdot (as_0 + e_0) \cdot \sum_{i=0}^{m-1} \xi^{j \cdot p^i} \cdot (as_1 + e_1)^{p^i}) \\
&= \text{Tr}_{\mathcal{R}}(\xi^{j \cdot (p^i+1)} \cdot \sum_{i=0}^{m-1} (as_0 + e_0)(a^{p^i} s_1^{p^i} + e_1^{p^i})) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}(\xi^{j \cdot (p^i+1)} \cdot (a^{p^i+1} \cdot s_0 s_1^{p^i} + a^{p^i} \cdot e_0 s_1^{p^i} + a \cdot s_0 e_1^{p^i} + e_0 e_1^{p^i})),
\end{aligned}$$

where $\xi \in \mathbb{F}_q$ such that $\mathbb{F}_q = \mathbb{F}_p(\xi)$. Hence it suffices to succinctly share the $4m$ cross-terms $s_0 s_1^{p^i}, e_0 s_1^{p^i}, s_0 e_1^{p^i}, e_0 e_1^{p^i}$ by SPFSS, as these are t^2 -sparse elements of $\mathcal{R}$.

Unfortunately, none of the above five properties can be easily satisfied for Galois ring cases. Basically, the most effective method is to find a natural generalization of the trace functions for Galois field cases introduced in [LXY25]. Let $\text{GR} := \mathbb{GR}(p^k, m) := \mathbb{Z}_{p^k}[x]/(g(x))$ be a Galois ring, where $\bar{g}(x) = g(x) \bmod p$ is a degree- m irreducible polynomial over $\mathbb{F}_p$. Here, for simplicity, we redefine $\mathcal{R}$ as a univariate polynomial ring $\mathcal{R} := \text{GR}[\mathbf{X}]/(f(\mathbf{X}))$ and we will assume Ring-LPN over $\mathcal{R}$ (Definition 3.2). A quick observation is that the sparsity property no longer holds if one naively define trace function over $\mathcal{R}$ in the same way as for fields. Because the sparsity property for fields $\mathbb{F}_q$ is implied by the fact that $\mathbb{F}_q$ has characteristic p . Namely, for any $f_0, \dots, f_{d-1} \in \mathbb{F}_q$, $(\sum_{i=0}^{d-1} f_i \cdot \mathbf{X}^i)^p = \sum_{i=0}^{d-1} f_i^p \cdot \mathbf{X}^{ip}$. However, this does not hold for $f_0, \dots, f_{d-1} \in \mathbb{GR}(p^k, m)$, where the characteristic is p^k rather than p .

Beyond that, we discuss the above trace functions for fields from the perspective of Galois theory. Let $\sigma_{\mathbb{F}_q}(\cdot)$ denote the Frobenius map over $\mathbb{F}_q$. We can

restate the trace function in [LXY25] in a more formal way:

$$\mathrm{Tr}_{\mathcal{R}}(f) := \sum_{i=0}^{m-1} f^{p^i} = \sum_{i=0}^{m-1} \left(\sum_{j=0}^{d-1} f_j \cdot \mathbf{X}^j \right)^{p^i} = \sum_{i=0}^{m-1} \left(\sum_{j=0}^{d-1} f_j^{p^i} \cdot \mathbf{X}^{j \cdot p^i} \right) = \sum_{i=0}^{m-1} \sigma_{\mathbb{F}_q}^i \left(\sum_{j=0}^{d-1} f_j \cdot \mathbf{X}^j \right).$$

Note that $\mathbf{X}$ is an indeterminate with respect to $\mathbb{F}_q$ and $\sigma_{\mathbb{F}_q}(\mathbf{X}) := \mathbf{X}^p$. Let $\sigma_{\mathrm{GR}}(\cdot)$ denote the generalized Frobenius map over GR (Definition 2.11). By viewing $\mathbb{F}_q$ as a special case of $\mathbb{G}\mathbb{R}(p^k, m)$ where $k = 1$, a natural extension of the trace functions for $\mathcal{R} := \mathrm{GR}[\mathbf{X}]/(f(\mathbf{X}))$ would be

$$\mathrm{Tr}_{\mathcal{R}}(f) := \sum_{i=0}^{m-1} \sigma_{\mathrm{GR}}^i \left(\sum_{j=0}^{d-1} f_j \cdot \mathbf{X}^j \right),$$

which collapses to the definition of [LXY25] when $k = 1$. However, this natural definition rules out the sparsity property in the sense that the number of indeterminates would blow up. At a high-level, this is because the generalized Frobenius map over GR is defined as the “composition” of m Frobenius maps over $\mathbb{F}_q$ in some sense. Specifically, assume $\zeta \in \mathrm{GR}$ with order $p^m - 1$, so that $(1, \zeta, \dots, \zeta^{m-1})$ is a basis of $\mathrm{GR} = \mathbb{Z}_{p^k}[\zeta]$ over $\mathbb{Z}_{p^k}$. Then by Definition 2.11, reasonably it should be the case that $\sigma_{\mathrm{GR}}(a) = \sum_{\kappa=0}^{m-1} a_{\kappa} \cdot \zeta^{\kappa p}$, where $a := \sum_{j=0}^{m-1} a_{\kappa} \cdot \zeta^{\kappa} \in \mathrm{GR}$ and each $a_{\kappa} \in \mathbb{Z}_{p^k}$. Anyway, $\mathrm{Tr}_{\mathcal{R}}(\cdot)$ no longer admits a concise form.

We remark that Galois ring elements have p -adic representations, so do the generalized Frobenius maps. However, the p -adic representation seems ineffective in this context, as it would similarly incur a blow-up of a multiplicative factor k . In particular, an intriguing trace function definition over GR is given in [Jan66, Theorem 2.2]⁴. However, its applicability to our setting remains unclear.

Our solution. Our key observation is the trace functions over $\mathcal{R} := \mathrm{GR}[\mathbf{X}]/(f(\mathbf{X}))$ have not to be as perfect as $\mathrm{Tr}_{\mathrm{GR}}(\cdot)$. Actually, in our case, trace functions make a difference only with respect to the isomorphism $\phi : \mathcal{R} \rightarrow \mathrm{GR}^d$. Hence, we take a step back, and propose a more pragmatic, albeit “artificial” definition as follows:

$$\mathrm{Tr}_{\mathcal{R}}(f) := \sum_{i=0}^{m-1} \sum_{j=0}^{d-1} \sigma_{\mathrm{GR}}^i(f_j) \cdot \mathbf{X}^{j \cdot p^i}.$$

At a first glance, the additive homomorphism and sparsity property of $\mathrm{Tr}_{\mathcal{R}}(\cdot)$ are straightforward. However, the remaining three required properties are nontrivial to prove. In particular, the most important “commutative” property seems no longer to hold: taking $f := \mathbf{X} \in \mathcal{R}$ as an example, typically

$$\phi(\mathrm{Tr}_{\mathcal{R}}(\mathbf{X})) = \phi\left(\sum_{i=0}^{m-1} \mathbf{X}^{p^i}\right) \neq \sum_{i=0}^{m-1} \sigma_{\mathrm{GR}}^i(\phi(\mathbf{X})) = \mathrm{Tr}_{\mathrm{GR}}(\phi(\mathbf{X})).$$

⁴ For $a = a_0\zeta^0 + a_1\zeta^1 + \dots + a_{m-1}\zeta^{m-1} \in \mathbb{G}\mathbb{R}(p^k, m)$, define the projection function $\pi_j(a) := a_j$. The trace map for a is defined as $\mathrm{Tr}_{\mathrm{GR}}(a) := \sum_j \pi_j(a \cdot \zeta^j)$.

To resolve this issue, we find a particular type of isomorphisms $\phi : \mathcal{R} \rightarrow \text{GR}^d$, such that the second inequality will not be the case. Specifically, we first define $\bar{g}(x)$ as a monic primitive polynomial of degree m over $\mathbb{F}_p$ and let $g(x) \in \mathbb{Z}_{p^k}[x]$ be the Hensel lift of $\bar{g}(x)$. This implies a root ζ of $g(x) \in \mathbb{Z}_{p^k}[x]$ has order $p^m - 1$, and moreover $x^{p^m-1} = \prod_{i=1}^{p^m-1} (x - \zeta^i)$ (see [Theorem 4.1](#)). Recall that $\mathcal{R} = \text{GR}[\mathbf{X}]/(\mathbf{X}^d - 1)$, where $d \mid p^m - 1$. Hensel's lemma implies $\mathbf{X}^d - 1 \mid \mathbf{X}^{p^m-1} - 1$ over GR as it holds over $\mathbb{F}_{p^m}$. Now, suppose $\alpha_1, \dots, \alpha_d$ are the roots of $\mathbf{X}^d - 1 = 0$ over GR , then it is readily to see that $\alpha_1, \dots, \alpha_d$ all are some powers of ζ . Formally, the isomorphism ϕ is explicitly given as follows:

$$\phi : \mathcal{R} \rightarrow \text{GR}^d, \quad f \mapsto (f(\alpha_1), \dots, f(\alpha_d)).$$

Intuitively, such ϕ offers the desired commutativity in the sense that now $\mathbf{X}$ is an indeterminate with respect to the set $\{\alpha_1, \dots, \alpha_d\}$, which are powers of ζ . In fact, the commutative property of $\text{Tr}_{\mathcal{R}}(\cdot)$ is implied by that of the generalized Frobenius map $\sigma_{\text{GR}}(\cdot)$. Basically, for any $f \in \mathcal{R}$, we have the following:

$$\sigma_{\text{GR}}(f(\zeta^j)) = \sigma_{\text{GR}}\left(\sum_{i=0}^{m-1} f_i \cdot \zeta^{ji}\right) = \sum_{i=0}^{m-1} \sigma_{\text{GR}}(f_i) \sigma_{\text{GR}}(\zeta^{ji}) = \sum_{i=0}^{m-1} \sigma_{\text{GR}}(f_i) \zeta^{ji \cdot p}$$

and thus

$$\phi(\sigma_{\mathcal{R}}(f)) = \sigma_{\text{GR}}(\phi(f)).$$

We formally prove the desired algebraic properties of $\text{Tr}_{\mathcal{R}}(\cdot)$ in [Section 4](#), which fits the framework of [\[LXY25\]](#) well. Ultimately, we obtain efficient PCGs for arbitrary Galois rings, which have $O(m^2 N \log N)$ computation and $O(m \lambda^3 \log N)$ communication for generating mN OLE correlations, under either Ring-LPN or QA-SD assumptions.

2 Preliminaries

Notations. Let $\mathbf{A}[i]$ denote the i -th entry of $\mathbf{A}$. Given $\ell, r \in \mathbb{Z}$, denote $[\ell, r] = [\ell, r+1) \subset \mathbb{Z}$ as the set of integers starts from ℓ and ends with r . If a value x is additively shared, we denote it as $[x]$. If a value x is shared via a $\text{SPD}\mathbb{Z}_{2^k}$ -style authenticated sharing [\[CDE+18\]](#), for instance let $\ell = k + r$, $\llbracket x \rrbracket := (\llbracket x \rrbracket_0, \llbracket x \rrbracket_1)$ and $\llbracket x \rrbracket_\theta := (x_\theta, M_\theta[x])$ where $x = x_0 + x_1 \pmod{2^k}$, $M_0[x] + M_1[x] = x \cdot (\Delta_0 + \Delta_1) \pmod{2^\ell}$ and $(\Delta_0, \Delta_1) \in \mathbb{Z}_{2^r}^2$ is a sharing of the global key $\Delta \in \mathbb{Z}_{2^r}$.

2.1 Function Secret Sharing

Function secret sharing (FSS) [\[BGH15, BGH16\]](#) allows to share functions succinctly among two parties. More concisely, for some secret function $f : I \rightarrow \mathbb{G}$ where $\mathbb{G}$ is an Abelian group and I is an input domain, an FSS scheme splits it into two functions f_0, f_1 , such that: (1) $f_0(x) + f_1(x) = f(x)$ for every input $x \in I$, and (2) f_0, f_1 can be represented by short keys K_0, K_1 respectively, with each of K_0, K_1 individually hiding f . Distributed point function (DPF) [\[GI14, BGH15\]](#) is a useful and fundamental kind of FSS schemes, specialized for sharing point functions. We give a formal definition of DPF as follows.

Definition 2.1 (Distributed point function). Given an input domain $I = [0, N)$, and an Abelian group $(\mathbb{G}, +)$, the point function $f_{\alpha, \beta} : I \rightarrow \mathbb{G}$ is defined by $f_{\alpha, \beta}(x) = \beta$ if $x = \alpha$, and $f_{\alpha, \beta}(x) = 0$ if $x \neq \alpha$, where $\alpha \in I, \beta \in \mathbb{G}$. A DPF scheme for the class of point functions $\{f_{\alpha, \beta} : \alpha \in I, \beta \in \mathbb{G}\}$ consists of the following two algorithms.

- $\text{DPF.Gen}(1^\lambda, \alpha, \beta)$ is a PPT algorithm that, given the security parameter λ , a position index $\alpha \in I$ and an element $\beta \in \mathbb{G}$, outputs a pair of keys $(\mathbf{k}_0^{\text{dpf}}, \mathbf{k}_1^{\text{dpf}})$.
- $\text{DPF.Eval}(\sigma, \mathbf{k}_\sigma^{\text{dpf}}, x)$ is a polynomial-time algorithm that, given a key $\mathbf{k}_\sigma^{\text{dpf}}$ for party $\sigma \in \{0, 1\}$, and an input $x \in I$, outputs $z_\sigma \in \mathbb{G}$.

The above scheme should satisfy the following requirements.

- **Correctness:** For any $\alpha \in I, \beta, x \in \mathbb{G}$, it holds that $z_0 + z_1 = f(x)$, where $z_\sigma \leftarrow \text{DPF.Eval}(\sigma, \mathbf{k}_\sigma^{\text{dpf}}, x)$ for $\sigma \in \{0, 1\}$ and $(\mathbf{k}_0^{\text{dpf}}, \mathbf{k}_1^{\text{dpf}}) \xleftarrow{\$} \text{DPF.Gen}(1^\lambda, \alpha, \beta)$.
- **Security:** For any $\sigma \in \{0, 1\}$, there exists a PPT simulator $\mathcal{S}$ such that for any point function $f_{\alpha, \beta}$, the distributions $\{(\mathbf{k}_\sigma^{\text{dpf}} : (\mathbf{k}_0^{\text{dpf}}, \mathbf{k}_1^{\text{dpf}}) \xleftarrow{\$} \text{DPF.Gen}(1^\lambda, \alpha, \beta))\}$ and $\{\mathbf{k}_\sigma^{\text{dpf}} \xleftarrow{\$} \mathcal{S}(1^\lambda, I, \mathbb{G})\}$ are computationally indistinguishable.

In this paper, we will use a simple and generic extension of DPF to sums of point functions. Let $\mathbf{A} = (\alpha_1, \dots, \alpha_t) \in [0, N)^t$, $\mathbf{\beta} = (\beta_1, \dots, \beta_t) \in \mathbb{G}^t$, the sum of point functions $f_{\mathbf{A}, \mathbf{\beta}} : [0, N) \rightarrow \mathbb{G}$ is defined by $\sum_{i=1}^t f_{\alpha_i, \beta_i}$. And it is direct to see that an FSS for $f_{\mathbf{A}, \mathbf{\beta}}$ can be realized by invoking t parallel DPFs, with each for f_{α_i, β_i} . Similarly, we define $(\text{SPFSS.Gen}, \text{SPFSS.Eval})$ as FSS for sums of point functions, with security following that of DPF. For convenience, we define SPFSS.FullEval as evaluations of SPFSS.Eval on all possible $x \in [0, N)$. Notably, a recent work [BGH⁺24] presents improved FSS constructions for multi-point functions.

The typical DPF construction [BGI16] uses an arbitrary pseudorandom generator (PRG) $G : \{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda+2}$ and the parameter is given as follows. The key size is at most $\lceil \log N \rceil (\lambda+2) + \lambda + \lceil \log |\mathbb{G}| \rceil$ bits. Taking $m = \left\lceil \frac{\log |\mathbb{G}|}{\lambda+2} \right\rceil$, the key generation SPFSS.Gen algorithm makes at most $2(\lceil \log N \rceil + m)$ PRG calls and the full domain evaluation algorithm SPFSS.FullEval makes at most $N(1 + m)$ PRG calls. For regular multi-point distributions, the number of PRG calls sheaves by a factor t . Employing batching codes, the SPFSS.FullEval algorithm can be asymptotically improved at the cost of more complicated seed generation algorithm. Recently, the SPFSS.Gen and SPFSS.FullEval algorithms are improved via a half-tree technique [GYW⁺23].

2.2 Pseudorandom Correlation Generators

The notion pseudorandom correlation generators PCG was first introduced in [BCG⁺19a, BCG⁺19b]. Informally, a PCG for some correlation, allows to distribute short, correlated seeds to the parties, such that each party can locally compute long, correlated pseudorandomness (satisfying the desired correlation)

from expanding their own seed. To give a formal definition, we start with correlation generators, and reverse-sampleable correlation generators.

Definition 2.2 (Correlation generator). A correlation generator $\mathcal{C}$ is a PPT algorithm that on input 1^λ , outputs a pair of strings in $\{0, 1\}^\ell \times \{0, 1\}^\ell$, where $\ell \in \text{poly}(\lambda)$.

Definition 2.3 (Reverse-sampleable correlation generator). Let $\mathcal{C}$ be a correlation generator. We say $\mathcal{C}$ is reverse sampleable if there exists a PPT algorithm RSample such that for $\sigma \in \{0, 1\}$, the following distribution

$$\{(R'_0, R'_1) \mid (R_0, R_1) \xleftarrow{\$} \mathcal{C}(1^\lambda), R'_\sigma := R_\sigma, R'_{1-\sigma} \xleftarrow{\$} \text{RSample}(\sigma, R_\sigma)\}$$

is computationally indistinguishable from the distribution given by $\mathcal{C}(1^\lambda)$.

Definition 2.4 (PCG). Let $\mathcal{C}$ be a reverse-sampleable correlation generator. A PCG for $\mathcal{C}$ consists of two algorithms $(\text{PCG.Gen}, \text{PCG.Expand})$ with the following syntax:

- $\text{PCG.Gen}(1^\lambda)$ is a PPT algorithm that given the security parameter λ , outputs a pair of seeds $(\mathbf{k}_0, \mathbf{k}_1)$.
- $\text{PCG.Expand}(\sigma, \mathbf{k}_\sigma)$ is a polynomial-time algorithm that given a party index $i \in \{0, 1\}$ and a seed $\mathbf{k}_\sigma$, outputs a bit string $R_\sigma \in \{0, 1\}^\ell$.

The two algorithms $(\text{PCG.Gen}, \text{PCG.Expand})$ should satisfy the following:

- **Correctness.** The following distribution

$$\{(R_0, R_1) \mid (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda), R_\sigma \leftarrow \text{PCG.Expand}(\sigma, \mathbf{k}_\sigma)\}$$

is computationally indistinguishable from the distribution given by $\mathcal{C}(1^\lambda)$.

- **Security.** For any $\sigma \in \{0, 1\}$, the following two distributions are computationally indistinguishable:

$$\begin{aligned} &\{(k_{1-\sigma}, R_\sigma) \mid (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda), R_\sigma \leftarrow \text{PCG.Expand}(\sigma, \mathbf{k}_\sigma)\} \text{ and} \\ &\{(k_{1-\sigma}, R_\sigma) \mid (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda), R_{1-\sigma} \leftarrow \text{PCG.Expand}(\sigma, \mathbf{k}_{1-\sigma}), \\ &\quad R_\sigma \xleftarrow{\$} \text{RSample}(\sigma, R_{1-\sigma})\}, \end{aligned}$$

where RSample is the reverse sampling algorithm for correlation $\mathcal{C}$.

2.3 Galois Rings

We briefly introduce Galois rings, and refer more details to the lecture notes [Wan03]. Let p be a prime and $\mathbb{Z}_{p^k}$ be the integer residue classes modulo p^k .

Definition 2.5 (Galois Ring). *The Galois ring is a ring extension of $\mathbb{Z}_{p^k}$ and has characteristic p^k . Specifically, let $m \geq 1$ be an integer and $g(x) \in \mathbb{Z}_{p^k}[x]$ be a monic polynomial of degree m such that $\bar{g}(x) := g(x) \bmod p$ is an irreducible polynomial over $\mathbb{F}_p$. The Galois ring of extension degree m is defined as*

$$\text{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(g(x)).$$

Let $\zeta = x + (g(x)) \in \text{GR}(p^k, m)$ be a root of $g(x)$. Then any $a \in \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$ can be expressed uniquely in the form of $a = a_0 + a_1\zeta + \cdots + a_{m-1}\zeta^{m-1}$, $a_i \in \mathbb{Z}_{p^k}$.

Definition 2.6 (Monic Basic Primitive Polynomials). *Let $g(x)$ be a monic polynomial over $\mathbb{Z}_{p^k}$. We say $g(x) \in \mathbb{Z}_{p^k}[x]$ is monic basic primitive, if $\bar{g}(x) := g(x) \bmod p$ is a monic primitive polynomial over $\mathbb{F}_p$.*

Lemma 2.7 (Hensel's Lemma). *[Wan03, Lemma 13.6] Let $f(x)$ be a monic polynomial in $\mathbb{Z}_{p^k}[x]$ and assume that $\bar{f} = \bar{f}_1\bar{f}_2$ in $\mathbb{F}_p[x]$, where $\bar{f} = f \bmod p$ and $\bar{f}_1, \bar{f}_2$ are coprime monic polynomials in $\mathbb{F}_p[x]$. Then there exist coprime monic polynomials f_1, f_2 in $\mathbb{Z}_{p^k}[x]$ such that*

$$f = f_1f_2 \text{ in } \mathbb{Z}_{p^k}[x], \text{ with } \bar{f}_1 = f_1 \bmod p, \text{ and } \bar{f}_2 = f_2 \bmod p.$$

Definition 2.8. *[Wan03, Definition 13.1] Let $\bar{f}(x)$ be a monic polynomial over $\mathbb{F}_p$. A monic polynomial $f(x)$ over $\mathbb{Z}_{p^k}[x]$ is called a Hensel lift of $\bar{f}(x)$ if $\bar{f} = f \bmod p$ and there is a positive integer n not dividing p such that*

$$f(x) \mid (x^n - 1) \text{ in } \mathbb{Z}_{p^k}[x].$$

Theorem 2.9. *[Wan03, Theorem 13.9] For any integer $m \geq 1$, there exist monic basic primitive polynomials of degree m over $\mathbb{Z}_{p^k}$ and dividing $x^{p^m-1} - 1$ in $\mathbb{Z}_{p^k}[x]$.*

Theorem 2.10. *[Wan03, Theorem 13.11] Let $\bar{f}(x)$ be a monic polynomial over $\mathbb{F}_p$ without multiple roots and $x \nmid \bar{f}$ in $\mathbb{F}_p[x]$, then $f(x)$ has a unique Hensel lift in $\mathbb{Z}_{p^k}[x]$.*

2.4 Galois Theories and Trace Functions

According to Galois theories, the generalized Frobenius automorphisms for Galois rings are defined in the following way.

Definition 2.11 (Generalized Frobenius Automorphisms for Galois Rings). *[Wan03, Theorem 14.30] Let $\text{GR} := \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$, where ζ has order $p^m - 1$. The generalized Frobenius automorphism $\sigma_{\text{GR}} : \text{GR} \rightarrow \text{GR}$ is defined as follows:*

$$\sigma(a) := a_0 + a_1\zeta^p + \cdots + a_{m-1}\zeta^{p(m-1)} = \sum_{j \in [0, m-1]} a_j \zeta^{jp},$$

where $a = a_0 + a_1\zeta + \cdots + a_{m-1}\zeta^{m-1}$, with each $a_j \in \mathbb{Z}_{p^k}$.

It is readily to verify that σ_{GR} defines an automorphism over $\mathbb{GR}(p^k, m)$ fixing $\mathbb{Z}_{p^k}$. In particular, $\sigma_{\text{GR}}(a) = a$ if and only if $a \in \mathbb{Z}_{p^k}$. For simplicity, we recursively define $\sigma_{\text{GR}}^j(\cdot) := \sigma_{\text{GR}} \circ \sigma_{\text{GR}}^{j-1}(\cdot)$, for $j \geq 1$, where $\sigma_{\text{GR}}^m(\cdot) = \sigma_{\text{GR}}^0(\cdot)$ is an identity map.

Definition 2.12 (Trace Functions over Galois Rings). [*Wan03*, Chapter 14.7] For any $a \in \text{GR} := \mathbb{GR}(p^k, m)$, the trace function is defined as

$$\text{Tr}_{\text{GR}}(a) := a + \sigma_{\text{GR}}^1(a) + \cdots + \sigma_{\text{GR}}^{m-1}(a) = \sum_{j \in [0, m-1]} \sigma_{\text{GR}}^j(a).$$

Lemma 2.13 (Properties of Trace Functions over Galois Rings). For $a, b \in \text{GR} := \mathbb{GR}(p^k, m)$, it holds that

1. $\text{Tr}_{\text{GR}}(a) \in \mathbb{Z}_{p^k}$.
2. $\text{Tr}_{\text{GR}}(a + b) = \text{Tr}_{\text{GR}}(a) + \text{Tr}_{\text{GR}}(b)$.
3. $\forall \alpha \in \mathbb{Z}_{p^k}$, $\text{Tr}_{\text{GR}}(\alpha) = m \cdot \alpha$ and $\text{Tr}_{\text{GR}}(\alpha \cdot a) = \alpha \cdot \text{Tr}_{\text{GR}}(a)$.
4. $\text{Tr}_{\text{GR}}(a) \cdot \text{Tr}_{\text{GR}}(b) = \text{Tr}_{\text{GR}}(a \cdot \text{Tr}_{\text{GR}}(b)) = \text{Tr}_{\text{GR}}(\text{Tr}_{\text{GR}}(a) \cdot b)$
5. $\text{Tr}_{\text{GR}}(\sigma_{\text{GR}}(a)) = \text{Tr}_{\text{GR}}(a)$.

These properties are similar to the trace functions for field extensions. We will use $\text{Tr}_{\mathbb{F}}(\cdot)$ to specify the algebraic structure $\mathbb{F}$ of the trace function if necessary. For $\mathbf{a} \in \text{GR}^N = \mathbb{GR}(p^k, m)^N$, we naturally define the vector trace $\text{Tr}_{\text{GR}^N}(\mathbf{a})$ as

$$\text{Tr}_{\text{GR}^N}(\mathbf{a}) := (\text{Tr}_{\text{GR}}(\mathbf{a}_1), \dots, \text{Tr}_{\text{GR}}(\mathbf{a}_N)).$$

For simplicity, we sometimes abuse Tr_{GR} as Tr_{GR^N} , and σ_{GR} as σ_{GR^N} .

3 Ring-LPN and QA-SD over Galois Rings

Learning parity with noise (LPN) [*BFKL93*] has been studied for decades, which is highly related to the problem of syndrome decoding of random linear codes. LPN and its various variants have been widely used to build cryptographic primitives, especially for pseudorandom correlation generators (PCGs) [*BCGI18*, *BCG+19a*, *BCG+19b*, *YWL+20*, *BCG+20b*, *BCG+22*, *BCG+23*, *RRT23*, *BCCD23*, *ABG+24*, *LXY24*]. Specifically, most of these works rely on LPN assumptions over finite fields, while [*BCGI18*, *SGRR19*, *BBMS22*, *LXY24*] consider LPN over rings, e.g., integer rings $\mathbb{Z}_{2^k}$ and Galois rings $\mathbb{GR}(p^k, m)$, for constructing ring correlations.

In this section, we naturally generalize Ring-LPN assumptions [*HKL+12*] and syndrome decoding of quasi-abelian codes (QA-SD) assumptions [*BCCD23*] to be defined over Galois rings $\mathbb{GR}(p^k, m)$. As noted in [*BCG+20b*, Remark 3.3], Ring-LPN can similarly be defined over other rings, which are not known to introduce any significant weakness or structural difference compared to the version over prime-order fields. In particular, the security over Galois rings can be based on Galois fields, since $\mathbb{GR}(p^k, m)/(p) \simeq \mathbb{F}_{p^m}$. For example, a Ring-LPN/QA-SD instance over $\mathbb{GR}(p^k, m)$ with noise rate t induces a sub-instance over $\mathbb{F}_{p^m}$ via modulo p , where the effective noise rate is slightly reduced to $\approx t \cdot \frac{p^m-1}{p^m}$.

Moreover, as a unified template to analyze the security of variants of the LPN or syndrome decoding assumption against the most common attacks, the linear test framework [BCG⁺20a, BCCD23] also works compatibly with Galois rings. Generally speaking, the similarity between Galois rings and Galois fields allows for plausible security analyses and parameter choices. We defer detailed security analyses to Appendix B.

Definition 3.1 (Search Ring-LPN). Assume $\mathbb{GR} := \mathbb{GR}(p^k, m)$. Let $\mathcal{R} = \mathbb{GR}(p^k, m)[X]/(f(X))$, where $f(X)$ over $\mathbb{GR}(p^k, m)$ has degree N . Let $a \xleftarrow{\$} \mathcal{R}$ and e_0, e_1 be random t -sparse elements of $\mathcal{R}$. Given access to a pair of the form $(a, a \cdot e_0 + e_1)$, the goal is to recover e_0, e_1 .

Definition 3.2 (Decisional Ring-LPN). Let $\mathcal{R} = \mathbb{GR}(p^k, m)[X]/(f(X))$, where $f(X)$ over $\mathbb{GR}(p^k, m)$ has degree N . The goal is to distinguish the following two distributions, with a non-negligible advantage:

$$\mathcal{D}_0 : \{(a, u)\}, \text{ where } a, u \xleftarrow{\$} \mathcal{R};$$

$$\mathcal{D}_1 : \{(a, a \cdot e_0 + e_1)\}, \text{ where } a \xleftarrow{\$} \mathcal{R}, e_0, e_1 \text{ are random } t\text{-sparse elements of } \mathcal{R}.$$

The above problems are referred as search(decisional) Ring-LPN($\mathcal{R}, t$), respectively. And the search (decisional) Ring-LPN($\mathcal{R}, t$) assumption conjectures that there exists no PPT algorithm that solves the problem with a non-negligible advantage. In particular, we consider $f(X)$ to be a fully splittable polynomial over $\mathbb{GR}(p^k, m)$, similar to the field case in [BCG⁺20b].

QA-SD over Galois rings. We introduce a concrete instantiation of QA-SD assumptions over Galois rings, serving as a multivariate analogue of Ring-LPN. A more general definition via group rings, following [BCCD23, BBC⁺24], is given in Appendix A.

Definition 3.3 (Search QA-SD). Let $\mathcal{R} = \mathbb{GR}(p^k, m)[X_1, \dots, X_n]/(X_1^d - 1, \dots, X_n^d - 1)$, where $d \mid p^m - 1$. Let $c \geq 2$ be some constant integer called the compression factor. Let $\mathbf{a} = (a_1, \dots, a_{c-1})$, where $a_i \xleftarrow{\$} \mathcal{R}$, $i \in [1, c-1]$. Let $e_0, \mathbf{e} := (e_1, \dots, e_{c-1})$ be random t -sparse elements of $\mathcal{R}$. Given access to a pair of the form $(\mathbf{a}, \langle \mathbf{a}, \mathbf{e} \rangle + e_0)$, the goal is to recover $e_0, \mathbf{e}$.

Definition 3.4 (Decisional QA-SD). Let $\mathcal{R} = \mathbb{GR}(p^k, m)[X_1, \dots, X_n]/(X_1^d - 1, \dots, X_n^d - 1)$, where $d \mid p^m - 1$. Let $c \geq 2$ be some constant integer called the compression factor. The goal is to distinguish the following two distributions:

$$\mathcal{D}_0 : \{(a_1, \dots, a_{c-1}, u)\}, \text{ where } a_i, u \xleftarrow{\$} \mathcal{R}, i \in [1, c-1]$$

$$\mathcal{D}_1 : \{(a_1, \dots, a_{c-1}, \langle \mathbf{a}, \mathbf{e} \rangle + e_0)\}, \text{ where } a_i \xleftarrow{\$} \mathcal{R}, e_0, \mathbf{e} \text{ are random } t\text{-sparse elements of } \mathcal{R}, i \in [1, c-1].$$

In this work, for simplicity we set the compression factor $c = 2$, referring to the above problems as search QA-SD($\mathcal{R}, t$) and decisional QA-SD($\mathcal{R}, t$), respectively. Our constructions can be naturally extended to $c \geq 2$. We say that the search (decisional) QA-SD($\mathcal{R}, t$) assumption holds if no PPT algorithm solves the problem with non-negligible advantage.

4 Generalized Frobenius Maps and Trace Functions

In this section, we show how to define trace functions over polynomial rings $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$ properly, so that it can fit the trace-based PCG paradigm of [LXY25]. We first establish isomorphisms of polynomial rings over Galois rings and modules over Galois rings in Section 4.1. Then in Section 4.2 we study Galois theories over polynomial rings $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$ with respect to the ring isomorphisms. Finally, in Section 4.3, we define traces functions over $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$, satisfying all desired properties. Assume $\text{GR} := \mathbb{GR}(p^k, m)$.

4.1 Isomorphisms of Polynomial Rings over Galois Rings

Similar to the field case, there exist isomorphisms between $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$ and GR^N for fully splittable polynomials $f(\mathbf{X})$ over GR of degree- N , by the well-known Chinese Remainder Theorem (CRT). In particular, we carefully select the ring GR and construct polynomials $f(\mathbf{X})$ with specific structural properties, ensuring that their roots in GR correspond to exponentiated values of a chosen primitive element. Jumping ahead, we address that this property is crucial for our trace functions defined over $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$.

Before introducing the isomorphism, we prove the following theorem.

Theorem 4.1. *Let $\bar{g}(x)$ be a monic primitive polynomial of degree m over $\mathbb{F}_p$ and $g(x) \in \mathbb{Z}_{p^k}[x]$ be the Hensel lift of $\bar{g}(x)$. Then $g(x) \mid (x^{p^m-1} - 1)$ in $\mathbb{Z}_{p^k}[x]$, and moreover*

$$x^{p^m-1} - 1 = \prod_{i=1}^{p^m-1} (x - \zeta^i),$$

where ζ is a root of $g(x)$ in $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(g(x))$ and has order $p^m - 1$.

Proof. Let $\bar{\zeta} := \zeta \bmod p$ be a root of the monic primitive polynomial $\bar{g}(x)$ over $\mathbb{F}_p$. By the proof of Theorem 2.9 [Wan03, Theorem 13.9], $g(x) \mid (x^{p^m-1} - 1)$ in $\mathbb{Z}_{p^k}[x]$. By Theorem 2.10, $x^{p^m-1} - 1$ over $\mathbb{Z}_{p^k}[x]$ is the unique Hensel lift of $x^{p^m-1} - 1$ over $\mathbb{F}_p[x]$. Note that

$$\prod_{i=1}^{p^m-1} (x - \zeta^i) = \prod_{i=1}^{p^m-1} (x - \bar{\zeta}^i) = x^{p^m-1} - 1 \pmod{p}.$$

Hence, $x^{p^m-1} - 1 = \prod_{i=1}^{p^m-1} (x - \zeta^i) \pmod{p^k}$ by the uniqueness of the Hensel lift. Finally, as ζ is a root of the monic basic primitive polynomial $g(x)$ over $\mathbb{Z}_{p^k}$, ζ has order $p^m - 1$. $\square$

Theorem 4.1 immediately gives the following corollary.

Corollary 4.2. *For any $N \mid p^m - 1$, we have $x^N - 1 \mid x^{p^m-1} - 1$ in $\mathbb{Z}_{p^k}[x]$, and moreover roots of $x^N - 1 = 0$ in GR are of the form ζ^α , where $\alpha \in [1, p^m - 1]$.*

Now we define the isomorphism between $\text{GR}[\mathbf{X}]/(f(\mathbf{X}))$ and GR^N .

Lemma 4.3 (The Univariate Ring). *Let $\text{GR} := \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}(x)/(g(x))$, where $g(x)$ is the Hensel lift of a degree- m monic primitive polynomial over $\mathbb{F}_p$. Let $\mathcal{R} := \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1)$ with $N \mid (p^m - 1)$. Then there exists $\phi : \mathcal{R} \rightarrow \text{GR}^N$ such that ϕ is a ring isomorphism. In particular, the addition and multiplication operations on GR^N are defined component-wisely.*

Proof (proof of Lemma 4.3). Let ζ be a root of $g(x)$, then ζ has order $p^m - 1$ and $\text{GR} \simeq \mathbb{Z}_{p^k}[\zeta]$. By Theorem 4.1, we have that $\mathbf{x}^{p^m-1} - 1 = \prod_{i=1}^{p^m-1} (\mathbf{x} - \zeta^i)$. Then for $N \mid p^m - 1$, we have $\mathbf{x}^N - 1 \mid \mathbf{x}^{p^m-1} - 1$. Hence, all roots of $\mathbf{x}^N - 1 = 0$ over GR are of the form ζ^{α_i} , where $\alpha_i \in [1, p^m - 1]$, and $i \in [1, N]$. This implies that

$$\phi : \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1) \rightarrow \text{GR}^N, f(\mathbf{x}) \mapsto (f(\zeta^{\alpha_1}), \dots, f(\zeta^{\alpha_N}))$$

is an isomorphism. $\square$

The above results for univariate polynomial rings can be naturally generalized to multivariate polynomial rings.

Lemma 4.4 (The Multivariate Ring). *Let $\text{GR} := \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}(x)/(g(x))$, where $g(x)$ is the Hensel lift of a degree- m monic primitive polynomial over $\mathbb{F}_p$. Let $\mathcal{R} := \text{GR}[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$ with $d \mid (p^m - 1)$. Then there exists $\phi : \mathcal{R} \rightarrow \text{GR}^N$, where $N = d^n$, such that ϕ is a ring isomorphism.*

Proof (proof of Lemma 4.4). First, by CRT, we have the following isomorphism:

$$\text{GR}[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1) \simeq \text{GR}[\mathbf{x}_1]/(\mathbf{x}_1^d - 1) \times \dots \times \text{GR}[\mathbf{x}_n]/(\mathbf{x}_n^d - 1).$$

Then, iteratively applying Lemma 4.3, we complete the proof. $\square$

For simplicity, for the rest of the section, we focus on the univariate polynomial ring case, i.e., $\mathcal{R} := \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1)$. Actually, all the results and proofs still hold for the multivariate ring case by the above isomorphism.

4.2 Generalized Frobenius Maps

With respect to the isomorphism ϕ defined in Lemma 4.3, we define the generalized Frobenius map $\sigma_{\mathcal{R}}$ over $\mathcal{R}$ as an extension of the generalized Frobenius map σ_{GR} previously defined over GR .

Definition 4.5 (Generalized Frobenius Maps). *Let $\text{GR} := \mathbb{GR}(p^k, m)$, and $\mathcal{R} := \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1)$ as defined above. For $f \in \mathcal{R}$, f can be uniquely expressed as*

$$f = f_0 + f_1 \cdot \mathbf{x} + f_2 \cdot \mathbf{x}^2 + \dots + f_{N-1} \cdot \mathbf{x}^{N-1} = \sum_{i=0}^{N-1} f_i \cdot \mathbf{x}^i.$$

The generalized Frobenius map $\sigma_{\mathcal{R}} : \mathcal{R} \rightarrow \mathcal{R}$ is defined by

$$\sigma_{\mathcal{R}}(f) := \sum_{i=0}^{N-1} \sigma_{\text{GR}}(f_i) \cdot \mathbf{x}^{i \cdot p} \pmod{\mathbf{x}^N - 1}$$

Similarly, we recursively define $\sigma_{\mathcal{R}}^j(\cdot)$ as $\sigma_{\mathcal{R}}^j(f) := \sigma_{\mathcal{R}} \circ \sigma_{\mathcal{R}}^{j-1}(f)$, for $j \geq 2$. Now we study the properties that the generalized Frobenius map over $\mathcal{R}$ inherits from the Galois extension $\mathbb{Z}_{p^k}[\zeta]$ over $\mathbb{Z}_{p^k}$, respect to the isomorphism ϕ .

Lemma 4.6 (Properties of $\sigma_{\mathcal{R}}$). *Let $\text{GR} := \mathbb{GR}(p^k, m)$, and $\mathcal{R} = \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1)$ as defined above, $\sigma_{\mathcal{R}}$ satisfies the following properties:*

1. For arbitrary $a, b \in \mathcal{R}$ and $j \in \mathbb{N}$, then $\sigma_{\mathcal{R}}^j(a + b) = \sigma_{\mathcal{R}}^j(a) + \sigma_{\mathcal{R}}^j(b)$.
2. If $a \in \mathcal{R}$ is t -sparse, $\sigma_{\mathcal{R}}^j(a)$ is t -sparse for any integer j .
3. For arbitrary $a \in \mathcal{R}$, $\sigma_{\mathcal{R}}^m(a) = a$.

Proof (Proof of Lemma 4.6).

1. By the additive homomorphism of σ_{GR} , it is easy to verify that $\sigma_{\mathcal{R}}$ is additively homomorphic, so does $\sigma_{\mathcal{R}}^j$.
2. Assume a is t -sparse. Then then by Definition 4.5, $\sigma_{\mathcal{R}}$ applies to each monomial of a and thus $\sigma_{\mathcal{R}}(a)$ is still t -sparse, so as $\sigma_{\mathcal{R}}^j$.
3. Specifically, the following equation holds:

$$\sigma_{\mathcal{R}}^m(a) = \sum_{i=0}^{N-1} \sigma_{\text{GR}}^m(a_i) \cdot \mathbf{x}^{i \cdot p^m} = \sum_{i=0}^{N-1} a_i \cdot \mathbf{x}^{i \cdot p^m} = \sum_{i=0}^{N-1} a_i \cdot \mathbf{x}^i \pmod{\mathbf{x}^N - 1}.$$

The first equality follows from the definition of $\sigma_{\mathcal{R}}^m(\cdot)$, while the second equality holds since σ_{GR}^m is the identity map. The final equality follows from $\mathbf{x}^N - 1 \mid \mathbf{x}^{p^m-1} - 1$, which implies $\mathbf{x}^{i \cdot p^m} = \mathbf{x}^i \pmod{\mathbf{x}^N - 1}$. $\square$

Recall that for the field case, the trace-based PCG paradigm crucially relies on the “commutative” property of the underlying trace functions. To this end, we first establish a similar commutative property for the generalized Frobenius map $\sigma_{\mathcal{R}}(\cdot)$.

Lemma 4.7 (Commutative). *Let $\text{GR} := \mathbb{GR}(p^k, m)$, and $\mathcal{R} = \text{GR}[\mathbf{x}]/(\mathbf{x}^N - 1)$ as defined above. Let ϕ be the isomorphism defined in Lemma 4.3. Then, $\phi(\sigma_{\mathcal{R}}(f)) = \sigma_{\text{GR}^N}(\phi(f))$. Moreover, $\phi(\sigma_{\mathcal{R}}^j(f)) = \sigma_{\text{GR}^N}^j(\phi(f))$ holds for any $j \in \mathbb{N}$. Basically, we have the following commutative diagram:*

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\phi} & \text{GR}^N \\ & \searrow \sigma_{\mathcal{R}}^j & \searrow \sigma_{\text{GR}^N}^j \\ & \mathcal{R} & \xrightarrow{\phi} \text{GR}^N \end{array}$$

Proof (Proof of Lemma 4.7). Denote by $\vec{\zeta} = (\zeta^{\alpha_1}, \dots, \zeta^{\alpha_N})$ the N roots of $\mathbf{x}^N - 1$ over GR . We have the following:

$$\begin{aligned} \phi(\sigma_{\mathcal{R}}(f)) &= \phi\left(\sum_{i=0}^{N-1} \sigma_{\text{GR}}(f_i) \cdot \mathbf{x}^{ip}\right) = \phi\left(\sum_{i=0}^{N-1} \left(\sum_{j=0}^{m-1} f_{i,j} \cdot \zeta^{jp}\right) \cdot \mathbf{x}^{ip}\right), \\ &= \sum_{i=0}^{N-1} \sum_{j=0}^{m-1} f_{i,j} \cdot \zeta^{jp} \cdot \vec{\zeta}^{ip} = \sum_{i=0}^{N-1} \sum_{j=0}^{m-1} f_{i,j} \cdot (\zeta^j \cdot \vec{\zeta}^i)^p, \end{aligned}$$

and

$$\begin{aligned}\sigma_{\text{GR}^N}(\phi(f)) &= \sigma_{\text{GR}^N}\left(\phi\left(\sum_{i=0}^{N-1} f_i \cdot \mathbf{X}^i\right)\right) = \sigma_{\text{GR}^N}\left(\phi\left(\sum_{i=0}^{N-1} \sum_{j=0}^{m-1} f_{i,j} \cdot \zeta^j \cdot \mathbf{X}^i\right)\right) \\ &= \sigma_{\text{GR}^N}\left(\sum_{i=0}^{N-1} \sum_{j=0}^{m-1} f_{i,j} \cdot (\zeta^j \cdot \vec{\zeta}^i)\right) = \sum_{i=0}^{N-1} \sum_{j=0}^{m-1} f_{i,j} \cdot (\zeta^j \cdot \vec{\zeta}^i)^p,\end{aligned}$$

where $f_{i,j} \in \mathbb{Z}_{p^k}$. Now we can inductively prove that

$$\begin{aligned}\phi(\sigma_{\mathcal{R}}^j(f)) &= (\phi \circ \sigma_{\mathcal{R}} \circ \cdots \circ \sigma_{\mathcal{R}})(f) = (\sigma_{\text{GR}^N} \circ \phi \circ \cdots \circ \sigma_{\mathcal{R}})(f) \\ &= (\sigma_{\text{GR}^N} \circ \cdots \circ \sigma_{\text{GR}^N} \circ \phi)(f) = \sigma_{\text{GR}^N}^j(\phi(f)),\end{aligned}$$

for arbitrary $j \in \mathbb{N}$, as desired. This completes the proof. $\square$

Finally, we prove that $\sigma_{\mathcal{R}}^j$, $j \in \mathbb{N}$ are automorphisms over $\mathcal{R}$.

Lemma 4.8. *The generalized Frobenius map $\sigma_{\mathcal{R}}$ is an automorphism of $\mathcal{R}$. Moreover, $\mathbb{Z}_{p^k} \subset \mathcal{R}$ is fixed by $\sigma_{\mathcal{R}}$.*

Proof. Bijection and additive homomorphism are straightforward to verify. Hence it suffices to show that $\sigma_{\mathcal{R}} : \mathcal{R} \rightarrow \mathcal{R}$ is multiplicative homomorphism, i.e., for any $a, b \in \mathcal{R}$, $\sigma_{\mathcal{R}}(ab) = \sigma_{\mathcal{R}}(a) \cdot \sigma_{\mathcal{R}}(b)$. By [Lemma 4.7](#), it is equivalent to show $\phi(\sigma_{\mathcal{R}}(ab)) = \phi(\sigma_{\mathcal{R}}(a)) \cdot \phi(\sigma_{\mathcal{R}}(b))$, since ϕ is an isomorphism.

$$\begin{aligned}\phi(\sigma_{\mathcal{R}}(a)) \cdot \phi(\sigma_{\mathcal{R}}(b)) &= \sigma_{\text{GR}^N}(\phi(a)) \cdot \sigma_{\text{GR}^N}(\phi(b)) \\ &= \sigma_{\text{GR}^N}(\phi(a) \cdot \phi(b)) = \sigma_{\text{GR}^N}(\phi(ab)) = \phi(\sigma_{\mathcal{R}}(ab)).\end{aligned}$$

Hence, $\sigma_{\mathcal{R}}$ is an automorphism. It is easy to verify that $\mathbb{Z}_{p^k}$ is fixed by $\sigma_{\mathcal{R}}$, since $\mathbb{Z}_{p^k}$ is fixed by σ_{GR} and for any $a \in \mathbb{Z}_{p^k} \subset \mathcal{R}$, $\phi(a) = (a, \dots, a) \in \mathbb{Z}_{p^k}^N$. $\square$

4.3 Trace Functions

Let $\text{GR} := \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$ and define $\mathcal{R} := \text{GR}[\mathbf{X}]/(\mathbf{X}^N - 1)$ as above. Given the definition of generalized Frobenius map in [Definition 4.5](#), we define trace functions over $\mathcal{R}$ in the following way.

Definition 4.9 (Trace Functions over $\mathcal{R}$). *The trace function $\text{Tr}_{\mathcal{R}} : \mathcal{R} \rightarrow \mathcal{R}$ is defined as follows, for an arbitrary $f \in \mathcal{R}$,*

$$\text{Tr}_{\mathcal{R}}(f) := \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(f).$$

We first prove the commutative property of trace functions over $\mathcal{R}$ with respect to the isomorphism ϕ .

Theorem 4.10 (Commutative). Let $\mathcal{R} := \text{GR}[\mathbf{X}]/(\mathbf{X}^N - 1)$ and $\text{Tr}_{\mathcal{R}}(\cdot)$ defined as above. For an arbitrary $f \in \mathcal{R}$,

$$\phi(\text{Tr}_{\mathcal{R}}(f)) = \text{Tr}_{\text{GR}^N}(\phi(f)),$$

Moreover, $\phi(\text{Tr}_{\mathcal{R}}(f)) \in \mathbb{Z}_{p^k}^N$. Hence, we have the following commutative diagram:

$$\begin{array}{ccc} \mathcal{R} & \xrightarrow{\phi} & \text{GR}^N \\ & \searrow \text{Tr}_{\mathcal{R}} & \searrow \text{Tr}_{\text{GR}} \\ & & \text{Im}(\text{Tr}_{\mathcal{R}}) \xrightarrow{\phi} \mathbb{Z}_{p^k}^N \end{array}$$

Proof (Proof of Theorem 4.10). The commutative property is directly implied by Lemma 4.7. Namely,

$$\phi(\text{Tr}_{\mathcal{R}}(f)) = \phi\left(\sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(f)\right) = \sum_{j=0}^{m-1} \phi(\sigma_{\mathcal{R}}^j(f)) = \sum_{j=0}^{m-1} \sigma_{\text{GR}^N}^j(\phi(f)) = \text{Tr}_{\text{GR}^N}(\phi(f)).$$

This completes the proof, as $\text{Tr}_{\text{GR}} : \text{GR} \rightarrow \mathbb{Z}_{p^k}$ and thus $\text{Tr}_{\text{GR}^N} : \text{GR}^N \rightarrow \mathbb{Z}_{p^k}^N$. $\square$

By Theorem 4.10, we prove the following properties for $\text{Tr}_{\mathcal{R}}(\cdot)$, analogues to those of trace functions over GR.

Theorem 4.11 (Properties of $\text{Tr}_{\mathcal{R}}$). Let $\mathcal{R} := \text{GR}[\mathbf{X}]/(\mathbf{X}^N - 1)$ defined as above, $\text{Tr}_{\mathcal{R}}(\cdot)$ satisfies the following properties:

1. $\forall f_1, f_2 \in \mathcal{R}, \text{Tr}_{\mathcal{R}}(f_1 + f_2) = \text{Tr}_{\mathcal{R}}(f_1) + \text{Tr}_{\mathcal{R}}(f_2).$
2. $\forall a \in \mathbb{Z}_{p^k}, \forall f \in \mathcal{R}, \text{Tr}_{\mathcal{R}}(a \cdot f) = a \cdot \text{Tr}_{\mathcal{R}}(f).$
3. $\forall f \in \mathcal{R}, \text{Tr}_{\mathcal{R}}(\sigma_{\mathcal{R}}(f)) = \text{Tr}_{\mathcal{R}}(f),$ and $\text{Tr}_{\mathcal{R}}(\text{Tr}_{\mathcal{R}}(f)) = m \cdot \text{Tr}_{\mathcal{R}}(f).$
4. $\forall f_1, f_2 \in \mathcal{R}, \text{Tr}_{\mathcal{R}}(f_1) \cdot \text{Tr}_{\mathcal{R}}(f_2) = \text{Tr}_{\mathcal{R}}(\text{Tr}_{\mathcal{R}}(f_1) \cdot f_2) = \text{Tr}_{\mathcal{R}}(f_1 \cdot \text{Tr}_{\mathcal{R}}(f_2)).$

Proof (Proof of Theorem 4.11).

1. Additive homomorphism of $\text{Tr}_{\mathcal{R}}$ is directly implied by that of $\sigma_{\mathcal{R}}$.
2. Intuitively, $\mathbb{Z}_{p^k}$ -linearity of $\text{Tr}_{\mathcal{R}}$ is due to that $\mathbb{Z}_{p^k}$ is fixed by $\sigma_{\mathcal{R}}$. In more detail, $\text{Tr}_{\mathcal{R}}(a \cdot f) = \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(a \cdot f) = \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(a) \sigma_{\mathcal{R}}^j(f) = a \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(f) = a \cdot \text{Tr}_{\mathcal{R}}(f).$
3. Since $\sigma_{\mathcal{R}}^m(f) = f = \sigma_{\mathcal{R}}^0(f)$, $\text{Tr}_{\mathcal{R}}(\sigma_{\mathcal{R}}(f)) = \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^{j+1}(f) = \sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(f) = \text{Tr}_{\mathcal{R}}(f)$. Similarly, $\text{Tr}_{\mathcal{R}}(\text{Tr}_{\mathcal{R}}(f)) = \text{Tr}_{\mathcal{R}}(\sum_{j=0}^{m-1} \sigma_{\mathcal{R}}^j(f)) = \sum_{j=0}^{m-1} \text{Tr}_{\mathcal{R}}(\sigma_{\mathcal{R}}^j(f)) = \sum_{j=0}^{m-1} \text{Tr}_{\mathcal{R}}(f) = m \cdot \text{Tr}_{\mathcal{R}}(f).$
4. This property follows from the $\mathbb{Z}_{p^k}$ -linearity and commutativity of $\text{Tr}_{\mathcal{R}}$. Specifically, since ϕ is an isomorphism, it suffices to show

$$\phi(\text{Tr}_{\mathcal{R}}(f_1) \cdot \text{Tr}_{\mathcal{R}}(f_2)) = \phi(\text{Tr}_{\mathcal{R}}(\text{Tr}_{\mathcal{R}}(f_1) \cdot f_2)).$$

By Theorem 4.10 and Lemma 2.13, we have

$$\begin{aligned} \phi(\text{Tr}_{\mathcal{R}}(f_1) \cdot \text{Tr}_{\mathcal{R}}(f_2)) &= \phi(\text{Tr}_{\mathcal{R}}(f_1)) * \phi(\text{Tr}_{\mathcal{R}}(f_2)) \\ &= \phi(\text{Tr}_{\mathcal{R}}(f_1)) * \phi(\text{Tr}_{\mathcal{R}}(f_2)) = \phi(\text{Tr}_{\mathcal{R}}(f_1)) * \text{Tr}_{\text{GR}^N}(\phi(f_2)), \end{aligned}$$

and

$$\begin{aligned}\phi(\text{Tr}_{\mathcal{R}}(\text{Tr}_{\mathcal{R}}(f_1) \cdot f_2)) &= \text{Tr}_{\text{GR}^N}(\phi(\text{Tr}_{\mathcal{R}}(f_1) \cdot f_2)) \\ &= \text{Tr}_{\text{GR}^N}(\phi(\text{Tr}_{\mathcal{R}}(f_1)) * \phi(f_2)) = \phi(\text{Tr}_{\mathcal{R}}(f_1)) * \text{Tr}_{\text{GR}^N}(\phi(f_2)),\end{aligned}$$

where $\phi(\text{Tr}_{\mathcal{R}}(f_1)) \in \mathbb{Z}_{p^k}^N$. This completes the proof. $\square$

For arbitrary Galois ring elements $a, b \in \text{GR} = \mathbb{GR}(p^k, m)$, the trace function satisfies $\text{Tr}_{\text{GR}}(a) \cdot \text{Tr}_{\text{GR}}(b) = \text{Tr}_{\text{GR}}(\text{Tr}_{\text{GR}}(a) \cdot b) = \text{Tr}_{\text{GR}}(a \cdot \text{Tr}_{\text{GR}}(b))$ since $\text{Tr}_{\text{GR}}(a) \in \mathbb{Z}_{p^k}$ ([Lemma 2.13](#)). Extending this to vectors, for any $\mathbf{u}, \mathbf{v} \in \text{GR}^N$, it holds that $\text{Tr}_{\text{GR}^N}(\mathbf{u}) * \text{Tr}_{\text{GR}^N}(\mathbf{v}) = \text{Tr}_{\text{GR}^N}(\text{Tr}_{\text{GR}^N}(\mathbf{u}) * \mathbf{v})$ as multiplication is component-wise. By property 4 of [Theorem 4.11](#), an analogous result holds for $\mathcal{R}$.

Theorem 4.12. *Let $\text{GR} = \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$ defined as above. For $f \xleftarrow{\$} \mathcal{R}$,*

$$\left(\phi(\text{Tr}_{\mathcal{R}}(f)), \phi(\text{Tr}_{\mathcal{R}}(\zeta \cdot f)) \dots \phi(\text{Tr}_{\mathcal{R}}(\zeta^{m-1} \cdot f)) \right)$$

admits the uniform distribution over $\mathbb{Z}_{p^k}^N$.

Proof (Proof of Theorem 4.12). By [Theorem 4.10](#), it suffices to prove that for a random $x \in \text{GR}$, $(\text{Tr}_{\text{GR}}(\zeta^0 x), \text{Tr}_{\text{GR}}(\zeta^1 x) \dots \text{Tr}_{\text{GR}}(\zeta^{m-1} x)) \in \mathbb{Z}_{p^k}^m$ is a linear transformation of $(x_1 \dots x_m) \in \mathbb{Z}_{p^k}^m$ where $(x_1 \dots x_m) \in \mathbb{Z}_{p^k}^m$ is the coefficient of x with respect to a given basis $(b_1 \dots b_m)$ of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$.

Lemma 4.13. *Given $(b_1 \dots b_m) \in \text{GR}^m$ a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$ and an arbitrary $x = \sum_{i \in [1, m]} x_i \cdot b_i \in \text{GR}$ with $x_i \in \mathbb{Z}_{p^k}$, $(\text{Tr}_{\text{GR}}(x \cdot b_1) \dots \text{Tr}_{\text{GR}}(x \cdot b_m))$ is a linear transformation of $(x_1 \dots x_m)$.*

Proof. Recall the discriminant matrix definition of a tuple $(b_1 \dots b_m)$ as in [Definition C.2](#). By [Theorem C.5](#), if $(b_1 \dots b_m)$ is a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$, then $\det(\text{Disc}(b_1 \dots b_m)) \in \mathbb{Z}_{p^k}^*$. Then $\text{Tr}_{\text{GR}}(x \cdot b_j) = \sum_{i \in [1, m]} x_i \cdot \text{Tr}_{\text{GR}}(b_i b_j) = (x_1 \dots x_m) \cdot (\text{Tr}_{\text{GR}}(b_1 b_j) \dots \text{Tr}_{\text{GR}}(b_m b_j))^T$, for all $j \in [1, m]$. Thus,

$$(\text{Tr}_{\text{GR}}(x \cdot b_1) \dots \text{Tr}_{\text{GR}}(x \cdot b_m)) = (x_1 \dots x_m) \begin{bmatrix} \text{Tr}_{\text{GR}}(b_1 b_1) & \dots & \text{Tr}_{\text{GR}}(b_1 b_m) \\ \vdots & & \vdots \\ \text{Tr}_{\text{GR}}(b_m b_1) & \dots & \text{Tr}_{\text{GR}}(b_m b_m) \end{bmatrix} = (x_1 \dots x_m) \text{Disc}(b_1 \dots b_m).$$

Therefore, the trace function $(\text{Tr}_{\text{GR}}(x \cdot b_1) \dots \text{Tr}_{\text{GR}}(x \cdot b_m))$ of $x = \sum_{i \in [1, m]} x_i \cdot b_i$ induces a linear transformation from $\mathbb{Z}_{p^k}^m$ to $\mathbb{Z}_{p^k}^m$. $\square$

Hence, a uniformly random x in GR induces a uniform tuple $(\text{Tr}_{\text{GR}}(x \cdot b_1) \dots \text{Tr}_{\text{GR}}(x \cdot b_m))$ over $\mathbb{Z}_{p^k}^m$. In particular, $(1, \zeta^1 \dots \zeta^{m-1})$ and $(\zeta, \zeta^{p^1} \dots \zeta^{p^{m-1}})$ are two bases of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$. Consequently, for a uniformly random $x \in \text{GR}$, $(\text{Tr}_{\text{GR}}(x \cdot \zeta^0) \dots \text{Tr}_{\text{GR}}(x \cdot \zeta^{m-1}))$ is uniformly distributed over $\mathbb{Z}_{p^k}^m$. This completes the proof. $\square$

[Theorem 4.12](#) essentially holds for any basis $(b_1, \dots, b_m)$ of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$. In particular, the tuple $(\text{Tr}(\zeta^{p^0} \cdot f), \text{Tr}(\zeta^{p^1} \cdot f), \dots, \text{Tr}(\zeta^{p^{m-1}} \cdot f))$ are distributed uniformly at random in $\mathbb{Z}_{p^k}^{mN}$ as well, as $(\zeta^{p^0}, \zeta^{p^1}, \dots, \zeta^{p^{m-1}})$ is a normal basis.

Remark 4.14. It is readily to verify that taking $k = 1$, the above results essentially boil down to the results for field extensions [[LXY25](#)].

5 PCG for OLE over Any Galois Ring

In this section, we show how to construct efficient programmable PCGs for OLE over any Galois ring $\mathbb{GR}(p^k, m)$. For the case of extension degree $m \geq 2$, we follow the paradigm of [BCG⁺20b, BCCD23], while for OLEs over $\mathbb{Z}_{p^k}$, we instead generalize the framework of [LXY25] by introducing our generalized trace functions. W.l.o.g., we build our constructions on QA-SD assumptions, which can be easily extended to basing on Ring-LPN.

5.1 OLE over Galois Rings with Extension Degree ≥ 2

Notations. Let $\mathbb{GR} := \mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(f(x))$, where $f(x) \in \mathbb{Z}_{p^k}[x]$ of degree $m \geq 2$ is the Hensel lift of a primitive polynomial over $\mathbb{F}_p$. Then there exists a $\zeta \in \mathbb{GR}(p^k, m)$ with order $p^m - 1$ such that $f(\zeta) = 0$. Essentially, $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$. Let $\mathcal{R} := \mathbb{GR}(p^k, m)[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$ with $d \mid p^m - 1$. Let $N = d^n$. Fix an isomorphism ϕ of $\mathcal{R} \simeq \mathbb{GR}^N$, given by Lemma 4.4. The QA-SD($\mathcal{R}, t$) assumption conjectures that for a random public $a \xleftarrow{\$} \mathcal{R}$ and t -sparse elements $s, e \xleftarrow{\$} \mathcal{R}$, $b := as + e$ is computationally indistinguishable from uniform.

The goal is to obtain N random OLE correlations over $\mathbb{GR}$. We follow the PCG paradigm of [BCG⁺20b, BCCD23] and utilize the following observation:

$$b_0 \cdot b_1 = (as_0 + e_0)(as_1 + e_1) = a^2 s_0 s_1 + ae_0 s_1 + as_0 e_1 + e_0 e_1, \quad (1)$$

where s_0, s_1, e_0, e_1 are random t -sparse elements of $\mathcal{R}$. Note that by assuming QA-SD($\mathcal{R}, t$), b_0, b_1 are indistinguishable from uniform, and moreover $\mathcal{R} \simeq \mathbb{GR}^N$. Hence, it suffices to distribute additive shares of the four cross-terms $s_0 s_1, s_0 e_1, s_1 e_0, e_0 e_1$ to the two parties, which then can be locally converted into additive shares of $b_0 \cdot b_1$, and equivalent N OLE correlations over $\mathbb{GR}$, as desired. Since $s_0 s_1, s_0 e_1, s_1 e_0, e_0 e_1$ are essentially t^2 -sparse elements of $\mathcal{R}$, sharing of them can be done efficiently from using SPFSS. Moreover, the resulting construction has programmability, admitting PCGs for multi-party Beaver triples over $\mathbb{GR}$.

A self-contained construction $\text{Cons}_{\text{OLE}}^{\text{GR}}$ is explicitly given as follows. The construction relies on the hardness of QA-SD($\mathcal{R}, t$) and an SPFSS scheme for sums of point functions, with domain $[0, N)$ and range $\mathbb{GR}$. We address that here we define a bijection $\varphi : j \mapsto \prod_{i=1}^n \mathbf{x}_i^{j_i}$, where $j \in [0, N)$ and $(j_1, \dots, j_n)$ go through $[0, d)^n$. Thus, any $s \in \mathcal{R}$ corresponds to a unique pair $(\mathbf{A}, \mathbf{s}) \in [0, N)^t \times \mathbb{GR}^t$ under φ . Similarly, for $\mathbf{A}_0, \mathbf{A}_1 \in [0, N)^t$, we define $\mathbf{A}_0 \boxplus \mathbf{A}_1 := \varphi^{-1}(\varphi(\mathbf{A}_0) \otimes \varphi(\mathbf{A}_1))$. Note that the above addition operations on the positions of $[0, N)$ are essentially nothing more than standard additions on a group $\mathbb{G} = \mathbb{Z}_d^n$, and ‘ $\otimes$ ’ refers to tensor product.

Construction 1: Cons_{OLE}^{GR}

PARAMETER: Security parameter λ , noise weight $t = t(\lambda)$, $N = d^n$. Let $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(f(x))$, where $f(x)$ is the Hensel lift of a primitive polynomial over $\mathbb{F}_p$. Let $\zeta \in \mathbb{GR}$ be the root of $f(x)$, which has order $p^m - 1$. Then $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$, and let $\mathcal{R} = \mathbb{GR}[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$, where $d \mid p^m - 1$. An FSS scheme (SPFSS.Gen, SPFSS.FullEval) for sums of t^2 point functions, with domain $[0, N]^a$ and range $\mathbb{GR}(p^k, m)$.

PUBLIC INPUT: A uniformly random $a \in \mathcal{R}$.

CORRELATION: After expansion, outputs $(\mathbf{x}_0, \mathbf{z}_0) \in \mathbb{GR}^{2N}$ and $(\mathbf{x}_1, \mathbf{z}_1) \in \mathbb{GR}^{2N}$ such that $\mathbf{x}_0 * \mathbf{x}_1 = \mathbf{z}_0 + \mathbf{z}_1$.

Gen: On input 1^λ :

1. For $\theta \in \{0, 1\}$, sample random vectors $\mathbf{A}_\theta^0, \mathbf{A}_\theta^1 \leftarrow [0, N]^t$ and $\mathbf{s}_\theta, \mathbf{e}_\theta \leftarrow (\mathbb{GR}(p^k, m)^*)^t$.
2. Sample FSS keys according to Eq.(1), namely:

$$(K_0^0, K_1^0) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^0 \boxplus \mathbf{A}_1^0, \mathbf{s}_0 \otimes \mathbf{s}_1),$$

$$(K_0^1, K_1^1) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^1 \boxplus \mathbf{A}_1^0, \mathbf{e}_0 \otimes \mathbf{s}_1),$$

$$(K_0^2, K_1^2) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^0 \boxplus \mathbf{A}_1^1, \mathbf{s}_0 \otimes \mathbf{e}_1),$$

$$(K_0^3, K_1^3) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^1 \boxplus \mathbf{A}_1^1, \mathbf{e}_0 \otimes \mathbf{e}_1),$$
3. For $\theta \in \{0, 1\}$, let $\mathbf{k}_\theta = ((K_\theta^i)_{i \in [0, 3]}, (\mathbf{A}_\theta^0, \mathbf{s}_\theta), (\mathbf{A}_\theta^1, \mathbf{e}_\theta))$.
4. Output $(\mathbf{k}_0, \mathbf{k}_1)$.

Expand: On input $(\theta, \mathbf{k}_\theta)$:

1. Parse $\mathbf{k}_\theta$ as $((K_\theta^i)_{i \in [0, 3]}, (\mathbf{A}_\theta^0, \mathbf{s}_\theta), (\mathbf{A}_\theta^1, \mathbf{e}_\theta))$.
2. Define elements of $\mathcal{R}$

$$\mathbf{s}_\theta = \sum_{l=1}^t \mathbf{s}_\theta[l] \cdot \mathbf{A}_\theta^0[l], \quad \mathbf{e}_\theta = \sum_{l=1}^t \mathbf{e}_\theta[l] \cdot \mathbf{A}_\theta^1[l].$$

3. Compute $\mathbf{x}_\theta = \phi(a \cdot \mathbf{s}_\theta + \mathbf{e}_\theta) \in \mathbb{GR}^N$.
4. For $i \in [0, 3]$, compute $u_{\theta, i} \leftarrow \text{SPFSS.FullEval}(\theta, K_\theta^i)$, viewed as $\mathcal{R}$ elements.
5. According to Eq.(1), compute

$$\mathbf{z}_\theta = \phi(a^2 \cdot u_{\theta, 0} + a \cdot u_{\theta, 1} + a \cdot u_{\theta, 2} + u_{\theta, 3}) \in \mathbb{GR}^N.$$

6. Output $(\mathbf{x}_\theta, \mathbf{z}_\theta)$.

^a Each $j \in [0, N)$ corresponds to a basis $\prod_{i=1}^n \mathbf{x}_i^{j_i}$ of $\mathcal{R}$, where $j_i \in [0, d)$. The addition on $[0, N)$ is actually the addition on $\mathbb{G} = \mathbb{Z}_d^n$.

Theorem 5.1. Assume a secure FSS scheme SPFSS for sums of point functions and QA-SD($\mathcal{R}, t$) is hard. Then there exists a PCG construction Cons_{OLE}^{GR} that generates OLE correlations over $\mathbb{GR}(p^k, m)$. If the SPFSS is based on a PRG : $\{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda+2}$ via the PRG-based construction from [BGI16], for generating N OLE correlations over $\mathbb{GR}$, we have that:

- Each party's seed has maximum size around: $4t^2((\log N - \log t + 1)(\lambda + 2) + \lambda + km \log p) + 2t(\log N + km \log p)$ bits.
- The computation of **Expand** can be done with at most $(2 + \lceil (km \log p)/\lambda \rceil)4Nt$ PRG operations and $O(N \log N)$ operations in $\mathbb{GR}(p^k, m)$.

Remark 5.2. Regarding the seed size of different parameters, we follow optimizations of [BCG⁺20b, BCCD23], i.e., assuming the QA-SD assumption holds also for a regular noise distributions. In particular, one can directly employ the improved multi-point FSS construction in [BGH⁺24] to avoid assuming regular noise distributions. For fairness, the seed size and computations are obtained by following a similar argument as [BCG⁺20b, BCCD23]. Moreover, by the same analyses of [BCCD23], we immediately have the following claim.

Corollary 5.3. $\text{Cons}_{\text{OLE}}^{\text{GR}}$ is a secure programmable PCG for OLE over GR.

Note that to achieve security, choosing $t = O(\lambda)$ is sufficient.

5.2 PCG for OLE over Any $\mathbb{Z}_{p^k}$

The above PCG construction $\text{Cons}_{\text{OLE}}^{\text{GR}}$ has an inherent restriction on the extension degree of the underlying Galois ring. Namely, $d \mid p^m - 1$, hence $\text{Cons}_{\text{OLE}}^{\text{GR}}$ fails in the most significant case when $\text{GR} = \mathbb{GR}(2^k, 1) = \mathbb{Z}_{2^k}$. To solve this issue, we resort to the trace-based framework [LXY25]. We first recall notations.

Notations. Let $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(f(x))$, where $f(x) \in \mathbb{Z}_{p^k}[x]$ of degree m is the Hensel lift of a primitive polynomial over $\mathbb{F}_p$. Then there exists a $\zeta \in \mathbb{GR}(p^k, m)$ with order $p^m - 1$ such that $f(\zeta) = 0$. Essentially, $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$. Let $\mathcal{R} = \mathbb{GR}(p^k, m)[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$ with $d \mid p^m - 1$. We know that by Theorem 4.1 $x^{p^m-1} - 1 = \prod_{i=1}^{p^m-1} (x - \zeta^i)$, and $f(x) \mid x^{p^m-1} - 1$ since $d \mid p^m - 1$. Hence, $\mathbf{x}^d - 1 \mid \mathbf{x}^{p^m-1} - 1$ and the roots of $\mathbf{x}^d - 1 = 0$ over GR are of the form ζ^j , where $j \in [1, p^m - 1]$. Let $N = d^n$. Let $a \xleftarrow{\$} \mathcal{R}$ be the public input, and s_θ, e_θ be random t -sparse elements of $\mathcal{R}$, where $\theta \in \{0, 1\}$. Assuming the hardness of QA-SD($\mathcal{R}, t$), $b_\theta := a \cdot s_\theta + e_\theta$ is pseudorandom for $\theta \in \{0, 1\}$.

We show how to generate mN OLEs over $\mathbb{Z}_{p^k}$ in a batch, following the trace-based PCG framework of [LXY25]. The main innovation here is a properly defined trace function $\text{Tr}_{\mathcal{R}}(\cdot)$ in Def. 4.9, which fits the framework well. We briefly illustrate the idea as follows. At a high-level, the goal is to share the product of $\text{Tr}_{\mathcal{R}}(b_0), \text{Tr}_{\mathcal{R}}(b_1)$ over $\mathcal{R}$, and then locally convert the relation over $\mathcal{R}$ to multiple OLEs over $\mathbb{Z}_{p^k}$. In more detail, by the commutative property of $\text{Tr}_{\mathcal{R}}(\cdot)$ (Theorem 4.10), we can reduce the task to distributing $(b_0, z_0), (b_1, z_1) \in \mathcal{R}^2$ to parties P_0, P_1 respectively, satisfying

$$\text{Tr}_{\mathcal{R}}(b_0) \cdot \text{Tr}_{\mathcal{R}}(b_1) = \text{Tr}_{\mathcal{R}}(z_0) + \text{Tr}_{\mathcal{R}}(z_1),$$

which are equivalent to N OLEs over $\mathbb{Z}_{p^k}$, with respect to the isomorphism $\mathcal{R} \simeq \mathbb{Z}_{p^k}^N$. By exploiting the properties of $\text{Tr}_{\mathcal{R}}(\cdot)$, we have the following observation:

$$\begin{aligned}
\text{Tr}_{\mathcal{R}}(b_0) \cdot \text{Tr}_{\mathcal{R}}(b_1) &= \text{Tr}_{\mathcal{R}}(b_0 \cdot \text{Tr}_{\mathcal{R}}(b_1)) = \text{Tr}_{\mathcal{R}}\left(b_0 \cdot \left(\sum_{i=0}^{m-1} \sigma_{\mathcal{R}}^i(b_1)\right)\right) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}(b_0 \cdot \sigma_{\mathcal{R}}^i(b_1)) = \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}((as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(as_1 + e_1)) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}((as_0 + e_0) \cdot (\sigma_{\mathcal{R}}^i(a)\sigma_{\mathcal{R}}^i(s_1) + \sigma_{\mathcal{R}}^i(e_1))) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}(a\sigma_{\mathcal{R}}^i(a)s_0\sigma_{\mathcal{R}}^i(s_1) + \sigma_{\mathcal{R}}^i(a)e_0\sigma_{\mathcal{R}}^i(s_1) + as_0\sigma_{\mathcal{R}}^i(e_1) + e_0\sigma_{\mathcal{R}}^i(e_1))
\end{aligned} \tag{2}$$

We first briefly explain the correctness of the above [Eq.\(2\)](#). The first equality is due to Property 4 of [Theorem 4.11](#), while the second equality follows by the definition [Definition 4.9](#). The third equality holds due to the additive homomorphism of $\text{Tr}_{\mathcal{R}}$ (Property 2 of [Theorem 4.11](#)), and the remaining equalities follow by calculations and the fact that $\sigma_{\mathcal{R}}$ is an automorphism over $\mathcal{R}$ ([Lemma 4.8](#)).

Note that by [Lemma 4.6](#), $\sigma_{\mathcal{R}}^i(s_1), \sigma_{\mathcal{R}}^i(e_1)$ are t -sparse elements of $\mathcal{R}$. Therefore, for each $i \in [0, m-1]$, we can use SPFSS to succinctly share the t^2 -sparse elements

$$((s_0 \cdot \sigma_{\mathcal{R}}^i(s_1), e_0 \cdot \sigma_{\mathcal{R}}^i(s_1), s_0 \cdot \sigma_{\mathcal{R}}^i(e_1), e_0 \cdot \sigma_{\mathcal{R}}^i(e_1)))$$

of $\mathcal{R}$. Hence, the communication complexity scales linearly with $4m$. Moreover, by [Theorem 4.12](#), our trace function $\text{Tr}_{\mathcal{R}}(\cdot)$ essentially offers extractions of mN OLE correlations over $\mathbb{Z}_{p^k}$ from these $4m$ shares. Specifically, let $(\zeta^{p^0}, \zeta^{p^1}, \dots, \zeta^{p^{m-1}})$ be a basis of $\mathbb{G}\mathbb{R}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$ over $\mathbb{Z}_{p^k}$. Then, for each $j \in [0, m-1]$, we have that

$$\begin{aligned}
\text{Tr}_{\mathcal{R}}(\zeta^{p^j} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{p^j} b_1) &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{p^j} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{p^j} \cdot (as_1 + e_1))\right) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}(\zeta^{p^j} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{p^j}) \cdot \sigma_{\mathcal{R}}^i(as_1 + e_1)) \\
&= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{p^j+p^{j+i}} (a\sigma_{\mathcal{R}}^i(a)s_0\sigma_{\mathcal{R}}^i(s_1) + \sigma_{\mathcal{R}}^i(a)e_0\sigma_{\mathcal{R}}^i(s_1) + as_0\sigma_{\mathcal{R}}^i(e_1) + e_0\sigma_{\mathcal{R}}^i(e_1))\right).
\end{aligned} \tag{3}$$

Hence, the $4m$ cross-terms of

$$(s_0 \cdot \sigma_{\mathcal{R}}^i(s_1), e_0 \cdot \sigma_{\mathcal{R}}^i(s_1), s_0 \cdot \sigma_{\mathcal{R}}^i(e_1), e_0 \cdot \sigma_{\mathcal{R}}^i(e_1))$$

from [Eq.\(2\)](#) can be *reused*. Recall that $\sigma_{\mathcal{R}}^i(\zeta^{p^j}) := (\zeta^{p^j})^{p^i} = \zeta^{p^{j+i}}$. With [Eq.\(2\)](#) & [Eq.\(3\)](#) as above, we are able to explicitly present a construction [Cons_{OLE} ^{\$\mathbb{Z}_{p^k}\$}](#) that

works with any $p, m \geq 2$. The construction relies on the hardness of $\text{QA-SD}(\mathcal{R}, t)$ and an SPFSS scheme for sums of point functions, with domain $[0, N)$ and range $\mathbb{GR}(p^k, m)$. The operations ‘ $\boxplus$ ’ and ‘ $\boxtimes$ ’ are defined in the same way as Section 5.1.

Construction 2: $\text{Cons}_{\text{OLE}}^{\mathbb{Z}_{p^k}}$

PARAMETER: Security parameter λ , noise weight $t = t(\lambda)$, $N = d^n$. Let $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[x]/(f(x))$, where $f(x)$ is the Hensel lift of a primitive polynomial over $\mathbb{F}_p$. Let $\zeta \in \mathbb{GR}$ be the root of $f(x)$, which has order $p^m - 1$. Then $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$, and let $\mathcal{R} = \mathbb{GR}(p^k, m)[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$, where $d \mid p^m - 1$. An FSS scheme ($\text{SPFSS.Gen}, \text{SPFSS.FullEval}$) for sums of t^2 point functions, with domain $[0, N)$ ^a and range $\mathbb{GR}(p^k, m)$. Fix a basis $(\zeta, \zeta^{p^1}, \dots, \zeta^{p^{m-1}})$ of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$.

PUBLIC INPUT: A uniformly random $a \in \mathcal{R}$.

CORRELATION: After expansion, outputs for all $j \in [0, m)$, $(\mathbf{x}_0^{(j)}, \mathbf{z}_0^{(j)}) \in \mathbb{Z}_{p^k}^{2N}$ and $(\mathbf{x}_1^{(j)}, \mathbf{z}_1^{(j)}) \in \mathbb{Z}_{p^k}^{2N}$ such that $\mathbf{x}_0^{(j)} * \mathbf{x}_1^{(j)} = \mathbf{z}_0^{(j)} + \mathbf{z}_1^{(j)}$.

Gen: On input 1^λ :

1. For $\theta \in \{0, 1\}$, sample random vectors $\mathbf{A}_\theta^0, \mathbf{A}_\theta^1 \leftarrow [0, N)^t$ and $\mathbf{s}_\theta, \mathbf{e}_\theta \leftarrow (\mathbb{GR}(p^k, m)^*)^t$.
2. Sample FSS keys according to Eq.(2), namely for each $i \in [0, m-1]$:

$$\begin{aligned} (K_0^{4i}, K_1^{4i}) &\stackrel{\$}{\leftarrow} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^0 \boxplus (\mathbf{A}_1^0)^{p^i}, \mathbf{s}_0 \otimes \sigma_{\mathcal{R}}^i(\mathbf{s}_1)), \\ (K_0^{4i+1}, K_1^{4i+1}) &\stackrel{\$}{\leftarrow} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^1 \boxplus (\mathbf{A}_1^0)^{p^i}, \mathbf{e}_0 \otimes \sigma_{\mathcal{R}}^i(\mathbf{s}_1)), \\ (K_0^{4i+2}, K_1^{4i+2}) &\stackrel{\$}{\leftarrow} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^0 \boxplus (\mathbf{A}_1^1)^{p^i}, \mathbf{s}_0 \otimes \sigma_{\mathcal{R}}^i(\mathbf{e}_1)), \\ (K_0^{4i+3}, K_1^{4i+3}) &\stackrel{\$}{\leftarrow} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^1 \boxplus (\mathbf{A}_1^1)^{p^i}, \mathbf{e}_0 \otimes \sigma_{\mathcal{R}}^i(\mathbf{e}_1)), \end{aligned}$$
3. For $\theta \in \{0, 1\}$, let $\mathbf{k}_\theta = ((K_\theta^i)_{i \in [0, 4k-1]}, (\mathbf{A}_\theta^0, \mathbf{s}_\theta), (\mathbf{A}_\theta^1, \mathbf{e}_\theta))$.
4. Output $(\mathbf{k}_0, \mathbf{k}_1)$.

Expand: On input $(\theta, \mathbf{k}_\theta)$:

1. Parse $\mathbf{k}_\theta$ as $((K_\theta^i)_{i \in [0, 4k-1]}, (\mathbf{A}_\theta^0, \mathbf{s}_\theta), (\mathbf{A}_\theta^1, \mathbf{e}_\theta))$.
2. Define elements of $\mathcal{R}$

$$\mathbf{s}_\theta = \sum_{l=1}^t \mathbf{s}_\theta[l] \cdot \mathbf{A}_\theta^0[l], \quad \mathbf{e}_\theta = \sum_{l=1}^t \mathbf{e}_\theta[l] \cdot \mathbf{A}_\theta^1[l].$$

3. For each $j \in [0, m-1]$, compute $\mathbf{x}_\theta^{(j)} = (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{p^j} \cdot (a\mathbf{s}_\theta + \mathbf{e}_\theta))$.
4. For $i \in [0, 4k-1]$, compute $u_{\theta,i} \leftarrow \text{SPFSS.FullEval}(\theta, K_\theta^i)$, viewed as $\mathcal{R}$ elements.
5. For each $j \in [0, m-1]$, according to Eq.(3), compute

$$\mathbf{z}_\theta^{(j)} = \sum_{i=0}^{m-1} (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{p^j + p^{j+i}} (a\sigma_{\mathcal{R}}^i(a)u_{\theta,4i} + \sigma_{\mathcal{R}}^i(a)u_{\theta,4i+1} + au_{\theta,4i+2} + u_{\theta,4i+3}))$$

6. Output $\{(\mathbf{x}_\theta^{(j)}, \mathbf{z}_\theta^{(j)})\}_{j \in [0, m-1]}$.

^a Each $j \in [0, N)$ corresponds to a basis $\prod_{i=1}^n \mathbf{x}_i^{j_i}$ of $\mathcal{R}$, where $j_i \in [0, d)$. The addition on $[0, N)$ is actually the addition on $\mathbb{G} = \mathbb{Z}_d^n$.

Theorem 5.4. Assume a secure FSS scheme SPFSS for sums of point functions and QA-SD($\mathcal{R}, t$) is hard. Then there exists a PCG construction $\text{Cons}_{\text{OLE}}^{\mathbb{Z}_{p^k}}$ that generates OLE correlations over $\mathbb{Z}_{p^k}$. If the SPFSS is based on a PRG : $\{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda+2}$ via the PRG-based construction from [BGI16], for generating mN OLE correlations over $\mathbb{Z}_{p^k}$, we have that:

- Each party's seed has maximum size around: $4mt^2((\log N - \log t + 1)(\lambda + 2) + \lambda + km \log p) + 2t(\log N + km \log p)$ bits.
- The computation of **Expand** can be done with at most $(2 + \lceil (km \log p) / \lambda \rceil) 4mNt$ PRG operations and $O(m^2 N \log N)$ operations in $\mathbb{GR}(p^k, m)$.

Remark 5.5. Regarding the size of the different parameters, we also follow optimizations of [BCG+20b, BCCD23, LXY25], e.g., assuming the QA-SD assumption holds also for a regular noise distributions. Then the seed size and computations can be obtained by following a similar argument as theirs.

Proof (of Theorem 5.4). Correctness is straightforward given by the Construction $\text{Cons}_{\text{OLE}}^{\mathbb{Z}_{p^k}}$ and Eq.(3). Security is based on the QA-SD assumption and the underlying SPFSS scheme, and we prove similarly to [BCG+20b, BCCD23, LXY25]. For completeness, we give a sketched proof as follows. It suffices to consider Party P_0 's view (i.e., $\theta = 0$). Let $(\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda)$ with associated expanded outputs $(\mathbf{x}_0^{(j)}, \mathbf{z}_0^{(j)})$ and $(\mathbf{x}_1^{(j)}, \mathbf{z}_1^{(j)})$, for $j \in [0, m-1]$. We need to show that

$$\{(\mathbf{k}_0, (\mathbf{x}_1^{(j)}, \mathbf{z}_1^{(j)})_{j \in [0, m-1]})\} \equiv \left\{(\mathbf{k}_0, (\tilde{\mathbf{x}}_1^{(j)}, \tilde{\mathbf{z}}_1^{(j)})_{j \in [0, m-1]}) \mid \begin{array}{l} j \in [0, m-1], \tilde{\mathbf{x}}_1^{(j)} \xleftarrow{\$} \mathbb{Z}_{p^k}^N, \\ \tilde{\mathbf{z}}_1^{(j)} = \mathbf{x}_0^{(j)} * \tilde{\mathbf{x}}_1^{(j)} - \mathbf{z}_0^{(j)} \end{array} \right\}$$

To show this, we use a sequence of hybrid distributions.

- Replace $\mathbf{z}_1^{(j)}$ by $\mathbf{x}_0^{(j)} * \mathbf{x}_1^{(j)} - \mathbf{z}_0^{(j)}$, for all $j \in [0, m-1]$.
- Step by step replace each SPFSS key K_0^i in $\mathbf{k}_0$ by a simulated key generated only with the range and the domain of the function. Due to the correctness and the security properties of the SPFSS scheme, this distribution is indistinguishable from the original distribution.
- Replace $\mathbf{x}_1^{(j)}$ by a fresh $\tilde{\mathbf{x}}_1^{(j)}$ for all $j \in [0, m-1]$. It is also impossible to distinguish this distribution from the previous one, since the K_0^i are now completely independent of $(\mathbf{x}_1^{(0)}, \dots, \mathbf{x}_1^{(m-1)})$, and we can rely on the QA-SD assumption and Theorem 4.12.
- Reverse step 2 by using the SPFSS security property once again.

We remark that QA-SD assumption implies $b_1 \in \mathcal{R}$ can not be distinguished from uniform, and Theorem 4.12 implies that $(\mathbf{x}_1^{(0)}, \dots, \mathbf{x}_1^{(m-1)})$ computed from a random $b_1 \in \mathcal{R}$ is uniformly distributed in $\mathbb{Z}_{p^k}^{mN}$. $\square$

Theorem 5.6. The PCG construction $\text{Cons}_{\text{OLE}}^{\mathbb{Z}_{p^k}}$ has programmability.

Proof (Proof of Theorem 5.6). To show that $\text{Cons}_{\text{OLE}}^{\mathbb{Z}_{p^k}}$ is programmable, we can re-define **Gen** by letting it take additional inputs (ρ_0, ρ_1) where $\rho_\theta = (\mathbf{A}_\theta^0, \mathbf{A}_\theta^1, \mathbf{s}_\theta, \mathbf{e}_\theta)$, for $\theta \in \{0, 1\}$. Note that ρ_θ corresponds to two t -sparse elements $\mathbf{s}_\theta, \mathbf{e}_\theta$ of $\mathcal{R}$ as well as an element $b_\theta := a \cdot \mathbf{s}_\theta + \mathbf{e}_\theta \in \mathcal{R}$. According to Eq.(3), party P_θ can locally compute $(\text{Tr}(\zeta^{p^0} b_\theta), \dots, \text{Tr}(\zeta^{p^{m-1}} b_\theta)) \in \mathbb{Z}_{p^k}^{mN}$ from ρ_θ . This essentially proves the programmability. Correctness follows that of Theorem 5.4. The programmable security can be proved in a very similar way to that of Theorem 5.4. $\square$

6 PCG for SPD $\mathbb{Z}_{2^k}$ -Type Authenticated Triples

Cramer et al. [CDE⁺18] proposed SPD $\mathbb{Z}_{2^k}$, an efficient dishonest-majority MPC protocol over $\mathbb{Z}_{2^k}$, with numerous real-world applications. In this section, we extend our PCG for OLE correlations over $\mathbb{Z}_{p^k}$ to generate SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples over $\mathbb{Z}_{2^k}$ in the two-party setting. W.l.o.g., we build our constructions on QA-SD assumptions, which can be easily extended to basing on Ring-LPN.

Secret-sharing over $\mathbb{Z}_{2^k}$ with MACs. We consider authenticated secret-sharing over $\mathbb{Z}_{2^k}$ based on SPD $\mathbb{Z}_{2^k}$ MACs [CDE⁺18] among T parties. Let $\ell = k + r$, where r is related to the security parameter λ . Specifically, the authenticated secret-sharing of $x \in \mathbb{Z}_{2^k}$ is defined as:

$$[\![x]\!] = (\Delta_i, x_i, M_{x,i})_{i=1}^T \in (\mathbb{Z}_{2^r} \times \mathbb{Z}_{2^\ell} \times \mathbb{Z}_{2^\ell})^T, \sum_i M_{x,i} = (\sum_i x_i) \cdot (\sum_i \Delta_i) \mod 2^\ell,$$

where $x = \sum_{i=1}^T x_i \mod 2^k$. We call $\Delta := \sum_i \Delta_i \in \mathbb{Z}_{2^\ell}$ the global key and $M := \sum_i M_{x,i}$ the MAC key. Note that the global key shares Δ_i are fixed for every shared x . An *authenticated multiplication triple* for $\mathbb{Z}_{2^k}$ is a tuple of random sharings $([\![x]\!], [\![y]\!], [\![z]\!])$, satisfying $x, y \xleftarrow{\$} \mathbb{Z}_{2^k}$ and $z = x \cdot y \mod 2^k$. We show an efficient PCG that generates a batch of such type of random authenticated triples for $\mathbb{Z}_{2^k}$ in the two-party setting.

Our PCG construction. We first recall notations. Let $\text{GR} = \mathbb{GR}(2^\ell, m) = \mathbb{Z}_{2^\ell}[x]/(f(x))$, where $f(x) \in \mathbb{Z}_{2^\ell}[x]$ of degree m is the Hensel lift of a primitive polynomial over $\mathbb{F}_2$. Then there exists $\zeta \in \mathbb{GR}(2^\ell, m)$ of order $2^m - 1$ such that $f(\zeta) = 0$. Essentially, $\mathbb{GR}(2^\ell, m) = \mathbb{Z}_{2^\ell}[\zeta]$. Let $\mathcal{R} = \mathbb{GR}(2^\ell, m)[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$ with $d \mid 2^m - 1$. From Theorem 4.1, it holds that $x^{2^m-1} - 1 = \prod_{i=1}^{2^m-1} (x - \zeta^i)$ and $f(x) \mid x^{2^m-1} - 1$ since $d \mid 2^m - 1$. Hence, $\mathbf{x}^d - 1 \mid \mathbf{x}^{2^m-1} - 1$ and the roots of $\mathbf{x}^d - 1 = 0$ over GR are of the form ζ^j , where $j \in [1, 2^m - 1]$. Let $N = d^n$ and fix a basis $(\zeta^{2^0}, \dots, \zeta^{2^{m-1}})$ of $\mathbb{GR}(2^\ell, m)$ over $\mathbb{Z}_{2^\ell}$. Let $a \xleftarrow{\$} \mathcal{R}$ be the public input, and $\mathbf{s}_\theta, \mathbf{e}_\theta$ be random t -sparse elements of $\mathcal{R}$, where $\theta \in \{0, 1\}$. Assuming the hardness of QA-SD($\mathcal{R}, t$), $b_\theta := a \cdot \mathbf{s}_\theta + \mathbf{e}_\theta$ is pseudorandom for $\theta \in \{0, 1\}$.

We show how to generate mN SPD $\mathbb{Z}_{2^k}$ -type authenticated triples in a batch, where $N = d^n$. To start with, we consider distribute additive secret sharings without MACs. Firstly, the goal is to obtain $[\text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_1)]_{\mathcal{R}}$ for $s \in$

$[0, m-1]$. This can be done by the approach of **Cons_{OLE} ^{$\mathbb{Z}_{p^k}$}** . Namely, it suffices to use SPFSS to distribute $\zeta^{2^s} b_\theta, z_\theta^{(s)} \in \mathcal{R}$ to party P_θ , with $\text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_1) = \text{Tr}_{\mathcal{R}}(z_0^{(s)}) + \text{Tr}_{\mathcal{R}}(z_1^{(s)})$, for $\theta \in \{0, 1\}$, $s \in [0, m-1]$, induced by the following observation (restatement of **Eq.(3)**):

$$\begin{aligned} \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_1) &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{2^s} \cdot (as_1 + e_1))\right) \\ &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{2^s}) \cdot \sigma_{\mathcal{R}}^i(as_1 + e_1)\right) \\ &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s+2^{s+i}} (a\sigma_{\mathcal{R}}^i(a)s_0\sigma_{\mathcal{R}}^i(s_1) + \sigma_{\mathcal{R}}^i(a)e_0\sigma_{\mathcal{R}}^i(s_1) + as_0\sigma_{\mathcal{R}}^i(e_1) + e_0\sigma_{\mathcal{R}}^i(e_1))\right). \end{aligned}$$

Secondly, it remains to obtain $[\text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_\theta)]_{\mathcal{R}}$ for $\theta \in \{0, 1\}$, $s \in [0, m-1]$. This can be done by using SPFSS as well, since the following holds:

$$[\text{Tr}_{\mathcal{R}}(\zeta^{2^s} \cdot b_\theta)]_{\mathcal{R}} = \text{Tr}_{\mathcal{R}}\left([\zeta^{2^s} \cdot b_\theta]_{\mathcal{R}}\right) = \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s} \cdot [b_\theta]_{\mathcal{R}}\right).$$

Note that $[b_\theta]_{\mathcal{R}} = a \cdot [s_\theta]_{\mathcal{R}} + [e_\theta]_{\mathcal{R}}$ and s_θ, e_θ are random t -sparse elements of $\mathcal{R}$.

Regarding the authentication part, we first assume each party P_θ has been given a global key share $\Delta_\theta \in \mathbb{Z}_{2^r}$, with $\Delta = \Delta_0 + \Delta_1 \pmod{2^\ell}$. Now the first goal is to obtain $[\Delta \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_\theta)]_{\mathcal{R}}$ for $\theta \in \{0, 1\}$, $s \in [0, m-1]$. Since $\Delta \in \mathbb{Z}_{2^\ell}$ and $\text{Tr}_{\mathcal{R}}(\cdot)$ has $\mathbb{Z}_{2^\ell}$ -linearity (Property 2 of **Theorem 4.11**), we have the following:

$$[\Delta \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} \cdot b_\theta)]_{\mathcal{R}} = [\text{Tr}_{\mathcal{R}}(\Delta \cdot \zeta^{2^s} \cdot b_\theta)]_{\mathcal{R}} = \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s} \cdot [\Delta \cdot b_\theta]_{\mathcal{R}}\right).$$

Note that $[\Delta \cdot b_\theta]_{\mathcal{R}} = a \cdot [\Delta \cdot s_\theta]_{\mathcal{R}} + [\Delta \cdot e_\theta]_{\mathcal{R}}$ and $\Delta \cdot s_\theta, \Delta \cdot e_\theta$ are random t -sparse elements of $\mathcal{R}$. Hence, it suffices to use SPFSS to succinctly share the 4 terms.

Finally, we need to obtain $[\Delta \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_1)]_{\mathcal{R}}$ for $s \in [0, m-1]$. Similarly, because $\Delta \in \mathbb{Z}_{2^\ell}$ and $\text{Tr}_{\mathcal{R}}(\cdot)$ has $\mathbb{Z}_{2^\ell}$ -linearity, we can extend **Eq.(3)** to the following form:

$$\begin{aligned} \Delta \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_0) \cdot \text{Tr}_{\mathcal{R}}(\zeta^{2^s} b_1) &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\Delta \cdot \zeta^{2^s} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{2^s} \cdot (as_1 + e_1))\right) \\ &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\Delta \cdot \zeta^{2^s} \cdot (as_0 + e_0) \cdot \sigma_{\mathcal{R}}^i(\zeta^{2^s}) \cdot \sigma_{\mathcal{R}}^i(as_1 + e_1)\right) \\ &= \sum_{i=0}^{m-1} \text{Tr}_{\mathcal{R}}\left(\zeta^{2^s+2^{s+i}} \cdot (a\sigma_{\mathcal{R}}^i(a) \cdot \Delta s_0\sigma_{\mathcal{R}}^i(s_1) + \sigma_{\mathcal{R}}^i(a) \cdot \Delta e_0\sigma_{\mathcal{R}}^i(s_1) \right. \\ &\quad \left. + a \cdot \Delta s_0\sigma_{\mathcal{R}}^i(e_1) + \Delta e_0\sigma_{\mathcal{R}}^i(e_1))\right). \end{aligned} \tag{4}$$

Note that there are $4m$ cross-terms $\Delta s_0\sigma_{\mathcal{R}}^i(s_1), \Delta e_0\sigma_{\mathcal{R}}^i(s_1), \Delta s_0\sigma_{\mathcal{R}}^i(e_1), \Delta e_0\sigma_{\mathcal{R}}^i(e_1)$ to shared, which are of sparsity- t^2 . Therefore, we can employ SPFSS for sums

of point functions as well. Putting all pieces together, there are $8m$ cross-terms of t^2 -sparse $\mathcal{R}$ elements and 8 terms of t -sparse $\mathcal{R}$ elements to be shared by SPFSS. Note that $s_0\sigma_{\mathcal{R}}^i(s_1)$ and $\Delta s_0\sigma_{\mathcal{R}}^i(s_1)$ share the same position, so that we can further reduce the number of SPFSS calls by half via reusing the position and expanding the payload. We remark that the cross-terms here are significantly fewer than in [LXY25] for authenticated multiplication triples over $\mathbb{F}_2$, as its global key over a large enough extension field $\mathbb{F}_{2^n}$ lacks $\mathbb{F}_2$ -linearity.

We give a self-contained construction $\text{Cons}_{\text{Auth-Triple}}^{\mathbb{Z}_{2^k}}$ as follows. The construction relies on the hardness of QA-SD($\mathcal{R}, t$) and an SPFSS scheme for sums of point functions, with domain $[0, N)$ and range $\mathbb{GR}(2^\ell, m) \times \mathbb{GR}(2^\ell, m)$. The operations ‘ $\boxplus$ ’ and ‘ $\boxtimes$ ’ are defined similarly to Section 5.1. Finally, we remark that it is straightforward to adapt the malicious distributed seed generation protocol [BCG⁺20b, LXY25] for finite fields to our general Galois ring settings.

Construction 3: $\text{Cons}_{\text{Auth-Triple}}^{\mathbb{Z}_{2^k}}$

PARAMETER: Security parameter λ , noise weight $t = t(\lambda)$, $N = d^n$, $\ell = k + r$. Let $\mathbb{GR}(2^\ell, m) = \mathbb{Z}_{2^\ell}[x]/(f(x))$, where $f(x)$ is the Hensel lift of a primitive polynomial over $\mathbb{F}_2$. Let $\zeta \in \mathbb{GR}$ be the root of $f(x)$, which has order $2^m - 1$. Then $\mathbb{GR}(2^\ell, m) = \mathbb{Z}_{2^\ell}[\zeta]$, and let $\mathcal{R} = \mathbb{GR}(2^\ell, m)[x_1, \dots, x_n]/(x_1^d - 1, \dots, x_n^d - 1)$, where $d \mid 2^m - 1$. An FSS scheme (SPFSS.Gen, SPFSS.FullEval) for sums of t^2 point functions, with domain $[0, N)^a$ and range $\mathbb{GR}(2^\ell, m) \times \mathbb{GR}(2^\ell, m)$. Fix a basis $(\zeta, \zeta^{2^1}, \dots, \zeta^{2^{m-1}})$ of $\mathbb{GR}(2^\ell, m)$ over $\mathbb{Z}_{2^\ell}$.

PUBLIC INPUT: A uniformly random $a \in \mathcal{R}$.

CORRELATION: After expansion, for all $s \in [0, m)$, outputs SPD $\mathbb{Z}_{2^k}$ -type triples $(\llbracket \mathbf{X}^{(s)} \rrbracket, \llbracket \mathbf{Y}^{(s)} \rrbracket, \llbracket \mathbf{Z}^{(s)} \rrbracket)$, where $\mathbf{X}^{(s)}, \mathbf{Y}^{(s)}, \mathbf{Z}^{(s)} \in \mathbb{Z}_{2^k}^N$ satisfying $\mathbf{Z}^{(s)} = \mathbf{X}^{(s)} * \mathbf{Y}^{(s)} \pmod{2^k}$, and MAC key shares $\Delta_0, \Delta_1 \in \mathbb{Z}_{2^r}$.

Gen: On input 1^λ :

1. Sample $\Delta_0, \Delta_1 \xleftarrow{\$} \mathbb{Z}_{2^r}$ and compute $\Delta = \Delta_0 + \Delta_1 \pmod{2^\ell}$.
2. For $\theta \in \{0, 1\}$, sample random vectors $\mathbf{A}_\theta^0, \mathbf{A}_\theta^1 \leftarrow [0, N)^t$ and $\mathbf{s}_\theta, \mathbf{e}_\theta \leftarrow (\mathbb{GR}(2^\ell, m)^*)^t$.
3. Sample FSS keys (sharings of $\mathbf{Z}^{(s)}, \Delta \cdot \mathbf{Z}^{(s)}$) according to Eq.(2)&(4), namely for each $i \in [0, m - 1]$:

$$(K_0^{4i}, K_1^{4i}) \xleftarrow{\$} \text{SPFSS.Gen}\left(1^\lambda, \mathbf{A}_0^0 \boxplus (\mathbf{A}_1^0)^{2^i}, (\mathbf{s}_0 \boxtimes \sigma_{\mathcal{R}}^i(\mathbf{s}_1)) \cdot (1, \Delta)\right),$$

$$(K_0^{4i+1}, K_1^{4i+1}) \xleftarrow{\$} \text{SPFSS.Gen}\left(1^\lambda, \mathbf{A}_0^1 \boxplus (\mathbf{A}_1^0)^{2^i}, (\mathbf{e}_0 \boxtimes \sigma_{\mathcal{R}}^i(\mathbf{s}_1)) \cdot (1, \Delta)\right),$$

$$(K_0^{4i+2}, K_1^{4i+2}) \xleftarrow{\$} \text{SPFSS.Gen}\left(1^\lambda, \mathbf{A}_0^0 \boxplus (\mathbf{A}_1^1)^{2^i}, (\mathbf{s}_0 \boxtimes \sigma_{\mathcal{R}}^i(\mathbf{e}_1)) \cdot (1, \Delta)\right),$$

$$(K_0^{4i+3}, K_1^{4i+3}) \xleftarrow{\$} \text{SPFSS.Gen}\left(1^\lambda, \mathbf{A}_0^1 \boxplus (\mathbf{A}_1^1)^{2^i}, (\mathbf{e}_0 \boxtimes \sigma_{\mathcal{R}}^i(\mathbf{e}_1)) \cdot (1, \Delta)\right),$$
4. Sample FSS keys (sharings of $(\mathbf{X}^{(s)}, \Delta \cdot \mathbf{X}^{(s)}), (\mathbf{Y}^{(s)}, \Delta \cdot \mathbf{Y}^{(s)})$), for each $i \in [0, m - 1]$:

$$(K_0^{4m}, K_1^{4m}) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^0, (\mathbf{s}_0, \Delta \cdot \mathbf{s}_0)),$$

$$(K_0^{4m+1}, K_1^{4m+1}) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_0^1, (\mathbf{e}_0, \Delta \cdot \mathbf{e}_0)),$$

$$(K_0^{4m+2}, K_1^{4m+2}) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_1^0, (\mathbf{s}_1, \Delta \cdot \mathbf{s}_1)),$$

$$(K_0^{4m+3}, K_1^{4m+3}) \xleftarrow{\$} \text{SPFSS.Gen}(1^\lambda, \mathbf{A}_1^1, (\mathbf{e}_1, \Delta \cdot \mathbf{e}_1)),$$

5. For $\theta \in \{0, 1\}$, let $\mathbf{k}_\theta = (\Delta_\theta, (K_\theta^i)_{i \in [0, 4m+3]})$.
6. Output $(\mathbf{k}_0, \mathbf{k}_1)$.

Expand: On input $(\theta, \mathbf{k}_\theta)$:

1. Parse $\mathbf{k}_\theta$ as $(\Delta_\theta, (K_\theta^i)_{i \in [0, 4m-1]}, (K_\theta^{4m+i})_{i \in [0, 3]})$.
2. For each $i \in [0, 4m)$, compute $u_{\theta,i}, u'_{\theta,i} \leftarrow \text{SPFSS.FullEval}(\theta, K_\theta^i)$, viewed as $\mathcal{R}$ elements. And for each $i \in [0, 4)$, compute $v_{\theta,i}, v'_{\theta,i} \leftarrow \text{SPFSS.FullEval}(\theta, K_\theta^{4m+i})$, viewed as $\mathcal{R}$ elements.
3. For each $s \in [0, m-1]$, compute

$$\mathbf{X}_\theta^{(s)} = (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s} \cdot (av_{\theta,0} + v_{\theta,1})), \mathbf{M}_{\mathbf{X},\theta}^{(s)} = (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s} \cdot (av'_{\theta,0} + v'_{\theta,1})),$$

$$\mathbf{Y}_\theta^{(s)} = (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s} \cdot (av_{\theta,2} + v_{\theta,3})), \mathbf{M}_{\mathbf{Y},\theta}^{(s)} = (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s} \cdot (av'_{\theta,2} + v'_{\theta,3})).$$

4. For each $s \in [0, m-1]$, according to Eq.(3)&(4), compute

$$\mathbf{Z}_\theta^{(s)} = \sum_{i=0}^{m-1} (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s+2^{s+i}} (a\sigma_{\mathcal{R}}^i(a)u_{\theta,4i} + \sigma_{\mathcal{R}}^i(a)u_{\theta,4i+1} + au_{\theta,4i+2} + u_{\theta,4i+3})),$$

$$\mathbf{M}_{\mathbf{Z},\theta}^{(s)} = \sum_{i=0}^{m-1} (\phi \circ \text{Tr}_{\mathcal{R}})(\zeta^{2^s+2^{s+i}} (a\sigma_{\mathcal{R}}^i(a)u'_{\theta,4i} + \sigma_{\mathcal{R}}^i(a)u'_{\theta,4i+1} + au'_{\theta,4i+2} + u'_{\theta,4i+3})).$$

5. Output $\Delta_\theta \in \mathbb{Z}_{2^r}$, and $(\mathbf{X}_\theta^{(s)}, \mathbf{Y}_\theta^{(s)}, \mathbf{Z}_\theta^{(s)}, \mathbf{M}_{\mathbf{X},\theta}^{(s)}, \mathbf{M}_{\mathbf{Y},\theta}^{(s)}, \mathbf{M}_{\mathbf{Z},\theta}^{(s)}) \in \mathbb{Z}_{2^\ell}^{6N}$, for $s \in [0, m)$. Note that $\mathbf{X}^{(s)} = \mathbf{X}_0^{(s)} + \mathbf{X}_1^{(s)} \bmod 2^k$, $\mathbf{Y}^{(s)} = \mathbf{Y}_0^{(s)} + \mathbf{Y}_1^{(s)} \bmod 2^k$, $\mathbf{Z}^{(s)} = \mathbf{Z}_0^{(s)} + \mathbf{Z}_1^{(s)} \bmod 2^k$.

^a Each $j \in [0, N)$ corresponds to a basis $\prod_{i=1}^n \mathbf{x}_i^{j_i}$ of $\mathcal{R}$, where $j_i \in [0, d)$. The addition on $[0, N)$ is actually the addition on $\mathbb{G} = \mathbb{Z}_d^n$.

Theorem 6.1. Assume a secure FSS scheme SPFSS for sums of point functions and $QA\text{-}SD(\mathcal{R}, t)$ is hard. Then there exists a PCG construction $\text{Cons}_{\text{Auth-Triple}}^{\mathbb{Z}_{2^k}}$ that generates $SPD\mathbb{Z}_{2^k}$ -type authenticated triples for $\mathbb{Z}_{2^k}$. If the SPFSS is based on a PRG $:\{0, 1\}^\lambda \rightarrow \{0, 1\}^{2\lambda+2}$ via the PRG-based construction from [BGI16], for generating mN triples over $\mathbb{Z}_{2^k}$, we have that:

- Each party's seed has maximum size around: $(4mt^2 + 4t)((\log N - \log t + 1)(\lambda + 2) + \lambda + 2\ell m) + r$ bits.
- The computation of **Expand** can be done with at most $(2 + \lceil (\ell m)/\lambda \rceil)4mNt$ PRG operations and $O(m^2N \log N)$ operations in $\mathbb{GR}(2^\ell, m)$.

Remark 6.2. Note that $\text{Cons}_{\text{Auth-Triple}}^{\mathbb{Z}_{2^k}}$ in fact guarantees a stronger condition that $(\mathbf{X}_0^{(s)} + \mathbf{X}_1^{(s)}) \cdot (\mathbf{Y}_0^{(s)} + \mathbf{Y}_1^{(s)}) = (\mathbf{Z}_0^{(s)} + \mathbf{Z}_1^{(s)}) \bmod 2^\ell$. Directly using such correlations in the $SPD\mathbb{Z}_{2^k}$ online protocol would leak information when reconstructing outputs. This issue can be easily solved by masking the higher $\ell - k$ bits of the shares in the opening procedure as similar to [BBMS22].

7 Efficiency

7.1 Parameters

According to the analyses and attacks in [Section B](#), we present various parameters for the QA-SD assumptions in our PCG constructions, focusing on OLEs and authenticated multiplication triples over $\mathbb{Z}_{2^k}$. Note that it suffices to select parameters against attacks over the residue field $\mathbb{F}_{2^m}$ with respect to the QA-SD instance modulo 2, where the effective noise weight is roughly reduced by $\frac{1}{2^m}$. In QA-SD-based PCGs, we have $d \mid 2^m - 1$ and $N = d^n$, where n is the number of variables. For faster SPFSS evaluations (approximately $t \times$ computational savings), the error vector can be assumed to have additional regular structure, requiring $t \mid N$ and thus constraining parameter choices. To generate suitable parameters, we follow the assumption:

$$(a_0, a_1 \dots a_c, b = \sum_{i \in [0, c-1]} a_i \cdot e_i),$$

as in [Definition 3.4](#) [[BCG+20b](#), [BCCD23](#), [BBC+24](#), [LXY25](#)], where $a_0 = 1$, $a_i \xleftarrow{\$} \mathcal{R}$ and $e_i \xleftarrow{\$} \mathcal{R}$ of t -sparsity. The parameter c , termed *compression factor*, determines the actual error weight as ct . For simplicity, we set $c = 2$ in our constructions, i.e., $(1, a_1, b = e_0 + a_1 \cdot e_1)$, though our approach extends naturally to $c \geq 2$. Below, we summarize key sizes of our PCG constructions, incorporating $c \geq 2$ by replacing t with $ct/2$ in the relevant formulas (e.g., [Theorem 5.1](#)).

PCG seed size with compression factor c :

- OLEs over $\mathbb{Z}_{2^k}$: $mc^2t^2((\log N - \log t + 1)(\lambda + 2) + \lambda + km) + ct(\log N + km)$.
- SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples for $\mathbb{Z}_{2^k}$: $(mc^2t^2 + ct)((\log N - \log t + 1)(\lambda + 2) + \lambda + 2\ell m) + r$.

In [Table 3](#) and [Table 4](#), we present PCG seed sizes for OLE over $\mathbb{Z}_{2^k}$ and authenticated multiplication triples over $\mathbb{Z}_{2^k}$ based on QA-SD assumptions. In particular, we select parameters to be resilient against a recent attack [[BDHV25](#)]. As shown in the tables, our constructions achieve efficiency comparable to those in [[BCG+20b](#), [BCCD23](#), [BBC+24](#)], for the two common 80-bit and 128-bit security levels.

7.2 Performance Evaluation

We implement our PCGs⁵ based on the $\mathbb{F}_4$ OLEAGE [[BBC+24](#)] source code and run them on a PC with an AMD EPYC 9754 2.30GHz CPU and 512GB RAM.

For fairness, we mainly follow the evaluation method of [[BBC+24](#), [LXY25](#)] for generating authenticated multiplication triples. Our results are reported in [Table 5](#). In particular, we test with various QA-SD parameters to generate SPD $\mathbb{Z}_{2^k}$ triples of a wide range (from 3^{13} to 3^{16}). As we use a more powerful

⁵ The code is available at <https://github.com/zhli271828/Trace-F2-OLE-PCG/>.

Error Type	λ	c	t	$\text{GR}(2^k, m)$	n	N	Seed Size	mN OLEs	Stretch
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	15	3^{15}	$2^{24.58}$	$2^{24.77}$	1.14
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	16	3^{16}	$2^{24.68}$	$2^{26.36}$	3.21
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	17	3^{17}	$2^{24.77}$	$2^{27.94}$	9.05
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	18	3^{18}	$2^{24.85}$	$2^{29.53}$	25.61
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	6	15^6	$2^{25.87}$	$2^{25.44}$	0.74
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	7	15^7	$2^{26.10}$	$2^{29.35}$	9.53
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	8	15^8	$2^{26.29}$	$2^{33.26}$	129.41
Regular	128	5	27	$\text{GR}(2^{64}, 2)$	15	3^{15}	$2^{26.60}$	$2^{24.77}$	0.28
Regular	128	5	27	$\text{GR}(2^{64}, 2)$	16	3^{16}	$2^{26.70}$	$2^{26.36}$	0.79
Regular	128	6	27	$\text{GR}(2^{64}, 2)$	17	3^{17}	$2^{27.33}$	$2^{27.94}$	1.53
Regular	128	6	27	$\text{GR}(2^{64}, 2)$	18	3^{18}	$2^{27.42}$	$2^{29.53}$	4.33
Regular	128	3	49	$\text{GR}(2^{64}, 3)$	10	7^{10}	$2^{27.66}$	$2^{29.66}$	3.99
Regular	128	3	49	$\text{GR}(2^{64}, 3)$	11	7^{11}	$2^{27.81}$	$2^{32.47}$	25.15
Regular	128	9	15	$\text{GR}(2^{64}, 4)$	7	15^7	$2^{27.90}$	$2^{29.35}$	2.73
Regular	128	9	15	$\text{GR}(2^{64}, 4)$	8	15^8	$2^{28.10}$	$2^{33.26}$	35.68
Regular	128	5	31	$\text{GR}(2^{64}, 5)$	5	31^5	$2^{28.42}$	$2^{27.09}$	0.40
Regular	128	5	31	$\text{GR}(2^{64}, 5)$	6	31^6	$2^{28.70}$	$2^{32.05}$	10.15

Table 3. Seed size for OLEs over $\mathbb{Z}_{2^k}$ from QA-SD.

Error Type	λ	c	t	$\text{GR}(2^k, m)$	n	N	Seed Size	mN Triples	Stretch
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	15	3^{15}	$2^{24.73}$	$2^{24.77}$	1.03
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	16	3^{16}	$2^{24.82}$	$2^{26.36}$	2.91
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	17	3^{17}	$2^{24.90}$	$2^{27.94}$	8.26
Regular	80	9	9	$\text{GR}(2^{64}, 2)$	18	3^{18}	$2^{24.97}$	$2^{29.53}$	23.50
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	6	15^6	$2^{26.14}$	$2^{25.44}$	0.62
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	7	15^7	$2^{26.33}$	$2^{29.35}$	8.10
Regular	80	6	15	$\text{GR}(2^{64}, 4)$	8	15^8	$2^{26.50}$	$2^{33.26}$	108.19
Regular	128	5	27	$\text{GR}(2^{64}, 2)$	15	3^{15}	$2^{26.70}$	$2^{24.77}$	0.26
Regular	128	5	27	$\text{GR}(2^{64}, 2)$	16	3^{16}	$2^{26.80}$	$2^{26.36}$	0.74
Regular	128	6	27	$\text{GR}(2^{64}, 2)$	17	3^{17}	$2^{27.41}$	$2^{27.94}$	1.34
Regular	128	6	27	$\text{GR}(2^{64}, 2)$	18	3^{18}	$2^{27.50}$	$2^{29.53}$	4.08
Regular	128	3	49	$\text{GR}(2^{64}, 3)$	10	7^{10}	$2^{27.79}$	$2^{29.66}$	3.66
Regular	128	3	49	$\text{GR}(2^{64}, 3)$	11	7^{11}	$2^{27.93}$	$2^{32.47}$	23.25
Regular	128	9	15	$\text{GR}(2^{64}, 4)$	7	15^7	$2^{28.05}$	$2^{29.35}$	2.45
Regular	128	9	15	$\text{GR}(2^{64}, 4)$	8	15^8	$2^{28.23}$	$2^{33.26}$	32.45
Regular	128	5	31	$\text{GR}(2^{64}, 5)$	5	31^5	$2^{28.64}$	$2^{27.09}$	0.34
Regular	128	5	31	$\text{GR}(2^{64}, 5)$	6	31^6	$2^{28.88}$	$2^{32.05}$	8.96

Table 4. Seed size for SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples over $\mathbb{Z}_{2^k}$ from QA-SD.

platform than previous works, we also include the benchmark results for the recently broken QA-SD parameters $c = 3$, $t = 27$, $m = 2$, which were used in previous works.

For comparison in communication, we calculate the seed size for generating $2 \cdot 3^{15}$ or $2 \cdot 3^{16}$ triples and scale it to 10^9 . That is, suppose a seed for generating $2 \cdot 3^{16}$ triples has size S , the seed size for 10^9 triples is then $S \cdot \frac{10^9}{2 \cdot 3^{16}}$. A comparison to other methods is given in Table 2, in which the seed sizes for $(k, s) = (32, 26)$ and $(k, s) = (64, 57)$ are computed from parameters $n = 15, c = 9, t = 9, m = 2$, and $n = 16, c = 5, t = 27, m = 2$, respectively. If using a distributed setup protocol for the seed generation, our communication would only be slighter higher than the seed size as in [BCG⁺20b, BCCD23]. Notably, to generate 10^9 SPD $\mathbb{Z}_{2^k}$ triples, prior methods require terabytes of communication, whereas ours transfers only megabytes. In conclusion, our PCG for two-party SPD $\mathbb{Z}_{2^k}$ authenticated multiplication triples is not only significantly faster than the fastest existing OT/OLE-based method [CDE⁺18], but also reducing communication by at least 4 magnitudes for generating more than $2 \cdot 3^{15}$ multiplication triples.

Parameters		Number of triples			
		3^{13}	3^{14}	3^{15}	3^{16}
$(k, s) = (32, 26)$	$c = 3, t = 27$	290,000/s	284,000/s	276,000/s	270,000/s
	$c = 5, t = 27$	115,000/s	113,000/s	109,000/s	102,000/s
	$c = 9, t = 9$	39,000/s	38,000/s	34,000/s	NaN
$(k, s) = (64, 57)$	$c = 3, t = 27$	163,000/s	161,000/s	156,000/s	151,000/s
	$c = 5, t = 27$	65,000/s	64,000/s	62,000/s	52,000/s
	$c = 9, t = 9$	22,000/s	21,000/s	17,500/s	NaN

Table 5. The concrete efficiency of our PCG for the generation of two-party SPD $\mathbb{Z}_{2^k}$ -type authenticated multiplication triples. Note that results with respect to $c = 3, t = 27$ over $\mathbb{GR}(\mathbb{Z}_{2^k}, 2)$ are only for benchmark, and should never be used in practice since it suffers from a recent attack by [BDHV25]. “NaN” means that the program ran out of memory.

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Supplementary Material

A More Preliminaries

A.1 Programmable PCG

Programmability is a crucial property that allows for extending 2-party correlations from PCG to multi-party correlations. To define programmability of PCG, suppose $\mathcal{C}^N$ is a simple bilinear 2-party correlation (specified by a bilinear map $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$ for some groups $\mathbb{G}_1, \mathbb{G}_2, \mathbb{G}_T$).

Definition A.1 (Programmability). We say a PCG $\text{PCG} = (\text{PCG.Gen}, \text{PCG.Expand})$ for a simple bilinear 2-party correlation $\mathcal{C}$ satisfies programmability, if $\text{PCG.Gen}(1^\lambda)$ takes additional random inputs $\rho_0, \rho_1 \in \{0, 1\}^\ell$, for a $\text{poly}(\lambda)$ size ℓ , such that:

- **Programmability.** There exist public efficiently computable functions $\psi_0 : \{0, 1\}^\ell \rightarrow \mathbb{G}_1^N$, $\psi_1 : \{0, 1\}^\ell \rightarrow \mathbb{G}_2^N$ such that

$$\Pr \left[\begin{array}{l} \rho_0, \rho_1 \xleftarrow{\$} \{0, 1\}^\ell, (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda, \rho_0, \rho_1) \\ (R_0, S_0) \leftarrow \text{PCG.Expand}(0, \mathbf{k}_0) \\ (R_1, S_1) \leftarrow \text{PCG.Expand}(1, \mathbf{k}_1) \end{array} : \begin{array}{l} R_0 = \psi_0(\rho_0) \\ R_1 = \psi_1(\rho_1) \end{array} \right] \geq 1 - \text{negl}(\lambda),$$

- **Programmable security.** The following pair of distributions are computationally indistinguishable:

$$\left\{ (\mathbf{k}_1, (\rho_0, \rho_1)) \mid \rho_0, \rho_1 \xleftarrow{\$} \{0, 1\}^\ell, (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda, \rho_0, \rho_1) \right\} \text{ and } \\ \left\{ (\mathbf{k}_1, (\rho_0, \rho_1)) \mid \rho_0, \rho_1, \tilde{\rho}_0 \xleftarrow{\$} \{0, 1\}^\ell, (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda, \tilde{\rho}_0, \rho_1) \right\}$$

as well as

$$\left\{ (\mathbf{k}_0, (\rho_0, \rho_1)) \mid \rho_0, \rho_1 \xleftarrow{\$} \{0, 1\}^\ell, (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda, \rho_0, \rho_1) \right\} \text{ and } \\ \left\{ (\mathbf{k}_0, (\rho_0, \rho_1)) \mid \rho_0, \rho_1, \tilde{\rho}_1 \xleftarrow{\$} \{0, 1\}^\ell, (\mathbf{k}_0, \mathbf{k}_1) \xleftarrow{\$} \text{PCG.Gen}(1^\lambda, \rho_0, \tilde{\rho}_1) \right\}$$

A.2 Module LPN

We introduce a natural generalization of Ring-LPN, referred as module-LPN, which is commonly used in the literature [BCG⁺20b].

Definition A.2 (Decisional Module-LPN). Let $\mathcal{R} = \text{GR}[\mathbf{X}]/(f(\mathbf{X}))$, where $f(\mathbf{X})$ over $\mathbb{G}\mathbb{R}(p^k, m)$ has degree N . Let $c \geq 2$ be some constant integer called the compression factor. The goal is to distinguish the following two distributions:

$$\mathcal{D}_0 : \{(a_1, \dots, a_{c-1}, u)\}, \text{ where } a_i, u \xleftarrow{\$} \mathcal{R}, i \in [1, c-1]; \\ \mathcal{D}_1 : \{(a_1, \dots, a_{c-1}, \langle \mathbf{a}, \mathbf{e} \rangle + e_0)\}, \text{ where } a_i \xleftarrow{\$} \mathcal{R}, e_0, e_i \text{ are random } t\text{-sparse elements of } \mathcal{R}, i \in [1, c-1].$$

It is not hard to see that when $c = 2$, module-LPN boils down to Ring-LPN. Analogue to [BCG⁺20b], we use a fully-reducible variant of the Ring-LPN. Namely, we consider $f(\mathbf{X}) = \mathbf{X}^N - 1$, where $N \mid p^m - 1$. This implies there exist distinct $\alpha_i \in \text{GR}$ such that $f(\mathbf{X}) = \prod_{i=1}^N (\mathbf{X} - \alpha_i)$, which moreover, allows to establish an isomorphism $\phi : \mathcal{R} \rightarrow \text{GR}^N, h(\mathbf{X}) \mapsto (h(\alpha_1), \dots, h(\alpha_N))$.

A.3 Group Rings

Definition A.3 (Group Ring). Let R be a ring, and G be a group. Define the set $R[G] = \{\sum_{g \in G} a_g g \mid a_g \in R, \text{ and there are finitely many } a_g \neq 0\}$. The two elements $\sum_{g \in G} a_g g, \sum_{g \in G} b_g g \in R[G]$ are said to be equal if and only if $a_g = b_g$ for any $g \in G$. Define addition and multiplication on $R[G]$ as follows:

$$\begin{aligned} \left(\sum_{g \in G} a_g g\right) + \left(\sum_{g \in G} b_g g\right) &= \sum_{g \in G} (a_g + b_g)g, \\ \left(\sum_{g \in G} a_g g\right) \cdot \left(\sum_{g \in G} b_g g\right) &= \sum_{g \in G} \left(\sum_{h \in G} (a_h b_{h^{-1}g})\right)g. \end{aligned}$$

$R[G]$ with addition and multiplication defined as above is called a group ring.

Lemma A.4 (Properties of Group Rings). Let R be a ring, G be a multiplicative group, and $R[G]$ be a group ring in Def. A.3. The followings hold:

1. $R[G]$ is commutative if and only if R is commutative and G is Abelian.
2. If R has identity 1, and G has identity e , then $1e$ is the identity of $R[G]$.

In this work, we particularly instantiate $R[G]$ with R a Galois ring $\mathbb{GR}(p^k, m)$ and G an Abelian group. By fixing an ordering $g_1, \dots, g_n$ of the elements in G , $\mathbb{GR}[G]$ is isomorphic to $\mathbb{GR}^n$ with respect to $\varphi : \mathbb{GR}[G] \rightarrow \mathbb{GR}^n, \sum_{i=1}^n a_i g_i \mapsto (a_1, \dots, a_n)$, where the multiplication over $\mathbb{GR}^n$ is defined as follows:

$$(a_1, \dots, a_n) \cdot (b_1, \dots, b_n) = \left(\sum_{g_i g_j = g_1} a_i b_j, \dots, \sum_{g_i g_j = g_n} a_i b_j \right).$$

In fact, the multiplication over $\mathbb{GR}^n$ can be characterized by standard matrix multiplications. For $a := \sum_{i=1}^n a_i g_i \in \mathbb{GR}[G]$, define

$$\mathbf{A} = \begin{pmatrix} \varphi(a \cdot g_1) \\ \vdots \\ \varphi(a \cdot g_n) \end{pmatrix} \in \mathbb{GR}^{n \times n}.$$

Essentially, the matrix $\mathbf{A}$ defines the multiply-by- a map $m \mapsto ma$ in $\mathbb{GR}[G]$ with respect to the basis $(g_1, \dots, g_n)$.

$$\varphi(ma) = \varphi(m)\mathbf{A} = (m_1, \dots, m_n) \begin{pmatrix} \varphi(a \cdot g_1) \\ \vdots \\ \varphi(a \cdot g_n) \end{pmatrix}.$$

Lemma A.5. Let G_1, G_2 be two finite Abelian groups. Then

$$\mathbb{GR}[G_1 \times G_2] \simeq \mathbb{GR}[G_1] \otimes_{\mathbb{GR}} \mathbb{GR}[G_2],$$

where $\times$ denotes the direct product, and $\otimes_{\mathbb{GR}}$ denotes the tensor product over $\mathbb{GR}$.

Examples. We give concrete examples of group rings.

- Let $G = \{1\}$ be the trivial group consisting of only an identity. Then $\text{GR}[G]$ is isomorphic to GR .
- Let $G = \mathbb{Z}/n\mathbb{Z}$ be the cyclic group with n elements. Assuming that n is coprime to p , then one can see $\text{GR}[G]$ is essentially the polynomial ring $\text{GR}[\mathbf{x}]/(\mathbf{x}^n - 1)$, with respect to the ordering $\bar{0}, \bar{1}, \dots, \bar{n-1}$ and the isomorphism $\varphi : \text{GR}[G] \rightarrow \text{GR}[\mathbf{x}]/(\mathbf{x}^n - 1)$, $\sum_{i=0}^{n-1} a_i \bar{i} \mapsto \sum_{i=0}^{n-1} a_i \mathbf{x}^i$.
- Let $G = \mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}$ with n_1, n_2 coprime to p . Then by Lemma A.5,
$$\begin{aligned} \text{GR}[\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2}] &\simeq \text{GR}[\mathbb{Z}_{n_1}] \otimes_{\text{GR}} \text{GR}[\mathbb{Z}_{n_2}] \simeq \text{GR}[\mathbf{x}]/(\mathbf{x}^{n_1} - 1) \otimes_{\text{GR}} \text{GR}[\mathbf{y}]/(\mathbf{y}^{n_2} - 1) \\ &\simeq \text{GR}[\mathbf{x}, \mathbf{y}]/(\mathbf{x}^{n_1} - 1, \mathbf{y}^{n_2} - 1). \end{aligned}$$

Moreover, assuming n divides $p^m - 1$, and $\alpha_0, \dots, \alpha_{n-1}$ are n roots of $\mathbf{x}^n - 1$ over $\mathbb{GR}(p^k, m)$. Then, $\phi : f(\mathbf{x}) \mapsto (f(\alpha_0), \dots, f(\alpha_{n-1}))$ is a ring isomorphism of $\text{GR}[\mathbf{x}]/(\mathbf{x}^n - 1)$ and the module GR^n . Hence, $\phi \circ \varphi : \text{GR}[G] \rightarrow \text{GR}^n$ is a ring isomorphism. Finally let $n \mid p^m - 1$ and $G = \mathbb{Z}_n^\ell$, then by Lemma A.5, $\text{GR}[G] \simeq \text{GR}[\mathbf{x}_1, \dots, \mathbf{x}_\ell]/(\mathbf{x}_1^n - 1, \dots, \mathbf{x}_\ell^n - 1) \simeq \text{GR}^N$, where $N = n^\ell$.

A.4 Syndrome Decoding of Quasi-Abelian Codes over Galois Rings

Let $\text{GR} = \mathbb{GR}(p^k, m)$ be a Galois ring and G a finite abelian group. Let ℓ be a positive integer, and consider the free $\text{GR}[G]$ -module of rank ℓ :

$$(\text{GR}[G])^\ell := \text{GR}[G] \oplus \dots \oplus \text{GR}[G] = \{(a_1, \dots, a_\ell) \mid a_i \in \text{GR}[G]\}.$$

Any $\text{GR}[G]$ -submodule of $(\text{GR}[G])^\ell$ is called a quasi- G code of index ℓ . If G is abelian, then a quasi- G code is quasi-abelian. Let $\mathbf{I} \in (\text{GR}[G])^{n \times \ell}$ be a generating matrix of rank $n \leq \ell$. Then the quasi- G code given by $\mathbf{I}$ is defined as follows:

$$\mathfrak{C} := \{\mathbf{m}\mathbf{I} \mid \mathbf{m} = (m_1, \dots, m_n) \in (\text{GR}[G])^n\}$$

The quasi- G code $\mathfrak{C}$ of index ℓ can be seen as a linear code of length $|G| \times \ell$ over GR , where the generating matrix is formed out by $n \times \ell$ square blocks of size $|G|$ (induced by $\mathbf{I}$). Assume $|G| = N$, and fix an ordering $g_1, \dots, g_N$ of elements in G and define the isomorphism $\varphi : \text{GR}[G] \rightarrow \text{GR}^N$. Let $\text{HW}_t(\text{GR}[G])$ denote the set $\{a \in \text{GR}[G] \mid \text{wt}(\varphi(a)) = t\}$, from which the noise will be uniformly sampled. A random quasi- G code of index 2, in systematic form, has a parity-check matrix $\mathbf{H} \in (\text{GR}[G])^2$ of the form $\mathbf{H} = (\mathbf{1} \mid \mathbf{a})$, where $\mathbf{a}$ is uniformly distributed in $\text{GR}[G]$.

Definition A.6 (Search QA-SD problem). Given $\mathbf{H} = (\mathbf{1} \mid \mathbf{a})$ a parity-check matrix of a random systematic quasi-abelian code, a target weight $t \in \mathbb{N}$ and a syndrome $\mathbf{s} \in \text{GR}[G]$, the goal is to recover an error vector $\mathbf{e} = (\mathbf{e}_0, \mathbf{e}_1)$, where $\mathbf{e}_i \xleftarrow{\$} \text{HW}_t(\text{GR}[G])$ such that $\mathbf{H}\mathbf{e}^T = \mathbf{s}$, i.e. $\mathbf{e}_0 + \mathbf{a} \cdot \mathbf{e}_1 = \mathbf{s}$.

Definition A.7 (Decisional QA-SD problem). Given a target weight t , the goal is to distinguish, with a non-negligible advantage, between the following two distributions:

$$\begin{aligned} \mathcal{D}_0 &: \{(\mathbf{a}, \mathbf{s})\}, \text{ where } \mathbf{a}, \mathbf{s} \xleftarrow{\$} \text{GR}[G], \\ \mathcal{D}_1 &: \{(\mathbf{a}, \mathbf{a} \cdot \mathbf{e}_0 + \mathbf{e}_1)\}, \text{ where } \mathbf{a} \xleftarrow{\$} \text{GR}[G], \text{ and } \mathbf{e}_0, \mathbf{e}_1 \xleftarrow{\$} \text{HW}_t(\text{GR}(G)). \end{aligned}$$

The above two problems can be generalized to random quasi- G codes of index $c \geq 2$. Then, the parity-check matrix would have the form $\mathbf{H} = (\mathbf{1} \mid \mathbf{a}_1 \mid \cdots \mid \mathbf{a}_{c-1})$. In this work, for simplicity we mostly consider $c = 2$, and the above problems are referred as search QA-SD($\text{GR}[G], t$), decisional QA-SD($\text{GR}[G], t$), respectively.

Note that the above definitions are naturally generalizations of Def.3.3 & 3.4, where we simply instantiate G with $\mathbb{Z}_d^n$, $d \mid p^m - 1$, so that $\text{GR}[G]$ is nothing else than the polynomial ring $\mathbb{GR}(p^k, m)[\mathbf{x}_1, \dots, \mathbf{x}_n]/(\mathbf{x}_1^d - 1, \dots, \mathbf{x}_n^d - 1)$.

A.5 The Linear Test Framework

The linear test framework provides a unified template to analyze the security of variants of the LPN or syndrome decoding assumptions against the most common attacks. It was first explicitly formalized for linear codes over finite fields in [BCG⁺20a]. We naturally generalize the framework to Galois rings. In sketch, an attack against syndrome decoding in the linear test framework performs in two stages:

1. First, a matrix $\mathbf{H}$ is uniformly sampled from a family of linear codes, and fed to the (unbounded) adversary $\mathcal{A}$. The adversary then returns a (nonzero) test vector $\mathbf{v} = \mathcal{A}(\mathbf{H})$.
2. Second, a noise vector $\mathbf{e}$ is sampled. The advantage of the adversary $\mathcal{A}$ in the linear test game is the bias of the induced distribution $\mathbf{v} \cdot \mathbf{H} \cdot \mathbf{e}^\top$.

The bias of a distribution over $\text{GR} = \mathbb{GR}(p^k, m)$ is defined as follows:

Definition A.8. *Given a distribution over $\mathcal{D}$ over GR^n and a vector $\mathbf{u} \in \text{GR}^n$, the bias of $\mathcal{D}$ with respect to $\mathbf{u}$, denoted by $\text{bias}_{\mathbf{u}}(\mathcal{D})$, is given by*

$$\text{bias}_{\mathbf{u}}(\mathcal{D}) = |\Pr_{\mathbf{x} \leftarrow \mathcal{D}}[\mathbf{u} \cdot \mathbf{x}^\top = 0] - \Pr_{\mathbf{x} \leftarrow \mathcal{U}_n}[\mathbf{u} \cdot \mathbf{x}^\top = 0]| = |\Pr_{\mathbf{x} \leftarrow \mathcal{D}}[\mathbf{u} \cdot \mathbf{x}^\top = 0] - \frac{1}{p^m}|,$$

where $\mathcal{U}_n$ denotes the uniform distribution over GR^n . The bias of $\mathcal{D}$, denoted by $\text{bias}(\mathcal{D})$, is the maximum bias of $\mathcal{D}$ with respect to any nonzero vector $\mathbf{u}$.

As in the literature, an instance of the syndrome decoding problem is secure against linear test if, with an overwhelm probability over the sampling of $\mathbf{H}$ in step 1, for any possible adversarial choice of $\mathbf{v} = \mathcal{A}(\mathbf{H})$, the bias of $\mathbf{v} \cdot \mathbf{H} \cdot \mathbf{e}^\top$ induced by the random sampling of $\mathbf{e}$ is negligible. Intuitively, the linear test framework captures any attack where the adversary is restricted to computing a linear function of the syndrome $\mathbf{b}^\top = \mathbf{H} \cdot \mathbf{e}^\top$, but the choice of the linear function itself can depend arbitrarily on the code. Hence, the adversary is restricted in one dimension (it has to be linear in $\mathbf{b}^\top$), but can run in unbounded time given $\mathbf{H}$. We refer concrete attacks to Appendix B.2.

Informally, resistance against linear test depends on both the code distribution and the noise distribution. At a high level, it suffices that

- (i) The code generated by $\mathbf{H}$ has large minimum distance modulo p , and
- (ii) for any large enough subset S of coordinates, with high probability over the choice of $\mathbf{e}$, one of the coordinates of $\mathbf{e}$ indexed by S will be non-zero.

Such characterization actually works for any noise distribution whose non-zero entries are uniformly random over $\mathbb{GR} \setminus \{0\}$, which is the case for all standard choices of noise distributions. We briefly explain why these conditions are sufficient. Recall that the adversarial advantage is the bias of $\mathbf{v} \cdot \mathbf{H} \cdot \mathbf{e}^\top$. By condition (i), $\mathbf{v} \cdot \mathbf{H} \bmod p$ has large Hamming weight for any possible (nonzero) vector $\mathbf{v}$, which is equivalent to saying that $\mathbf{v} \cdot \mathbf{H}$ has many invertible entries. As long as the subset S of invertible entries of $\mathbf{v} \cdot \mathbf{H}$ is sufficiently large, $\mathbf{e}$ will “overlap” S with high probabilities, and the output will be uniformly random.

Note that a random quasi- G code $\mathfrak{C}$ over Galois rings $\mathbb{GR}(p^k, m)$ implies a random quasi- G code $\bar{\mathfrak{C}}$ over Galois fields $\mathbb{F}_{p^m}$ through modulo p . Moreover, the minimal distance of $\mathfrak{C}$ would not be smaller than $\bar{\mathfrak{C}}$. Hence, the security in our Galois ring cases can be directly based on the analyses of the minimal distances of random quasi-abelian codes over finite fields [BCCD23, BBC⁺24].

B Security Analysis

B.1 Reductions

Following security analyses of [LWYY24, LXY24], there exists a reduction from Ring-LPN over Galois Rings to Ring-LPN over Galois fields. Let $\mathcal{R}_{p^k} = \mathbb{GR}(p^k, m)[\mathbf{X}]/(f(\mathbf{X}))$ and $\mathcal{R}_p = \mathbb{F}_{p^m}[\mathbf{X}]/(\bar{f}(\mathbf{X})) = \mathbb{GR}(p, m)[\mathbf{X}]/(\bar{f}(\mathbf{X}))$. Consider the fixed Hamming weight distribution:

Theorem B.1. *If decisional Ring-LPN($\mathcal{R}_{p^k}, t$) is (T, ϵ) -hard, then decisional Ring-LPN($\mathcal{R}_p, t'$) is $(T - \text{poly}(N), O(\epsilon))$ -hard, where $t' = \frac{p^{m(k-1)}(p^m-1)}{p^{mk}-1} \cdot t$.*

Proof. Let $(a, s := a \cdot e_0 + e_1)$ be a Ring-LPN instance over $\mathcal{R}_{p^k}$. As $\mathcal{R}_{p^k}/(p) \cong \mathcal{R}_p$, we observe that $(a' := a \bmod p, e'_0 := e_0 \bmod p, e'_1 := e_1 \bmod p)$ constitute exactly a Ring-LPN sample over $\mathcal{R}_p$. Since e_0, e_1 have fixed Hamming weight t , it can be observed that e'_0, e'_1 have expected weight $t' = \frac{p^{m(k-1)}(p^m-1)}{p^{mk}-1} \cdot t$. One can naturally generalize [LWYY24, Lemma 2] and obtain that e'_0, e'_1 are uniformly random t' -sparse elements of $\mathcal{R}_p$ with probability $\Omega(1/\sqrt{t})$. On the other hand, (a', u') are uniformly random as well. Therefore, one can use a distinguisher for Ring-LPN($\mathcal{R}, t'$) to distinguish (a', s') from uniform. $\square$

In the following, we consider Bernoulli distribution.

Theorem B.2. *If decisional Ring-LPN($\mathcal{R}_p, \frac{\rho(p^m-1)}{(1-\rho)p^{mk}+\rho p^m}$) is (T, ϵ) -hard, then decisional Ring-LPN($\mathcal{R}_{p^k}, \rho$) is $(T - \text{poly}(N), k\epsilon)$ -hard.*

Proof. Let $(a, s := a \cdot e_0 + e_1)$ be a Ring-LPN instance over $\mathcal{R}_{p^k}$. As $\mathcal{R}_{p^k}/(p) \cong \mathcal{R}_p$, for any $a \in \mathcal{R}_{p^k}$, it can be uniquely written in the following p -adic form

$$a = a^{(0)} + a^{(1)} \cdot p + \dots + a^{(k-1)} \cdot p^{k-1},$$

where $a^{(i)} \in \mathcal{R}_p, i \in [0, k)$. Thus, we can define a decomposition function Decom such that $(a^{(0)}, a^{(1)}, \dots, a^{(k-1)}) := \text{Decom}(a) \in \mathcal{R}_p^k$. We use Decom to decompose

elements of $\mathcal{R}_{p^k}$, and we obtain $(a^{(0)}, \dots, a^{(k-1)}) := \text{Decom}(a)$, $(e_0^{(0)}, \dots, e_0^{(k-1)}) := \text{Decom}(e_0)$, $(e_1^{(0)}, \dots, e_1^{(k-1)}) := \text{Decom}(e_1)$, and $(s^{(0)}, \dots, s^{(k-1)}) := \text{Decom}(s)$. Therefore, we have $s^{(0)} = a^{(0)} \cdot e_0^{(0)} + e_1^{(0)}$, and for $i \in [0, k)$, we have that

$$s^{(i)} = (a^{(i)} \cdot e_0^{(0)} + e_1^{(i)}) + f_i(a, s^{(0)}, \dots, s^{(i-1)}, e_0^{(0)}, \dots, e_0^{(i-1)}, e_1^{(0)}, \dots, e_1^{(i-1)}) \pmod{p},$$

where f_i is the sum of all other terms involving the individual $s^{(j)}$ and $e_0^{(j)}, e_1^{(j)}$ with $j \leq i-1$. Define the hybrid distributions $H_0, \dots, H_k$ as follows:

$$\begin{aligned} H_0 &= (a, u^{(0)}, \dots, u^{(i-1)}, u^{(i)}, \dots, u^{(k-1)}) \\ H_i &= (a, s^{(0)}, \dots, s^{(i-1)}, u^{(i)}, \dots, u^{(k-1)}) \\ H_k &= (a, s^{(0)}, \dots, s^{(i-1)}, s^{(i)}, \dots, s^{(k-1)}) \end{aligned}$$

where $u^{(i)} \xleftarrow{\$} \mathcal{R}_p$ for $i = 0, 1, \dots, k-1$. To distinguish the adjacent hybrids H_i and H_{i+1} , it suffices to distinguish $u^{(i)}$ and $s^{(i)}$ with the knowledge of $(s^{(0)}, \dots, s^{(i-1)})$. The f_i term can be neglected if the effective noise rate of $e_0^{(i)}, e_1^{(i)}$ conditioned on $e_0^{(0)}, \dots, e_0^{(i-1)}, e_1^{(0)}, \dots, e_1^{(i-1)}$ is sufficient to make $s^{(i)}$ pseudorandom. Consider a single noise sample $(e_{1,j}^{(0)}, \dots, e_{1,j}^{(k-1)}) \leftarrow \text{Decom}(\text{Ber}_\rho(\mathcal{R}_{p^k}))$, where $e_{1,j}^{(i)}$ is the j -th entry of $e_1^{(i)}$. If there exists a non-zero component in $(e_{1,j}^{(0)}, \dots, e_{1,j}^{(i-1)})$, $e_{1,j}^{(i)}$ must be uniformly random, which makes $s_j^{(i)}$ uniformly random. Thus, we have that

$$\Pr[e_{1,j}^{(i)} \neq 0 \mid (e_{1,j}^{(0)}, \dots, e_{1,j}^{(i-1)}) = \mathbf{0}] = \frac{\rho(p^m - 1)(p^{-m(i+1)})}{1 - \rho + \rho p^{-mi}}$$

is the noise rate needed to keep the computational indistinguishability between H_i and H_{i+1} , which reaches its minimum when $i = k-1$. $\square$

The above two reductions also naturally apply to the multi-variate case, i.e., the QA-SD assumption for GR.

B.2 Concrete Attacks

As indicated in [Section B.1](#), the Ring-LPN and QA-SD assumptions over Galois rings rely on the Ring-LPN and QA-SD assumptions over finite fields. In this section, we analyze concrete attacks on Ring-LPN/QA-SD over Galois rings. The concrete parameters are obtained from a SageMath script [\[Bom24\]](#). The essence of the attack was previously presented in [\[BCCD23, Section 6\]](#) and [\[BBC⁺24, Section 6\]](#) for the QA-SD assumption. For the Ring-LPN assumption, we rely on the same SageMath script. Note that the SageMath script [\[Bom24\]](#) and the resulting parameter selection are quite conservative.

The attacks in this section are generally presented for QA-SD/Ring-LPN over $\text{GR} := \mathbb{GR}(p^k, m)$. For concrete parameters, we focus on $p = 2, m = 2$. Let G be a finite abelian group. The goal of the attack is to distinguish $((a_1 \dots a_{c-1}), b := e_0 + a_1 e_1 + \dots + a_c e_c)$ from the uniform distribution over $\text{GR}[G]^{c-1} \times \text{GR}[G]$, where $a_i \xleftarrow{\$} \text{GR}[G]$ and e_i is a sparse element of $\text{GR}[G]$ with Hamming weight

t . The search Ring-LPN and search QA-SD problems are exactly the decoding problems of the following form as indicated in [Section A.4](#):

$$(I, A_1, \dots, A_{c-1}) \cdot \begin{pmatrix} e_0 \\ \vdots \\ e_{c-1} \end{pmatrix} = b,$$

where A_i is the matrix induced by a_i with respect to the group G and e_i is a sparse vector of Hamming weight t . Specifically, the error vector $\mathbf{e}$ is split into c blocks of weight t . The error vector with c -split structure has recently gained attention due to its use in SDitH [\[MFG⁺23\]](#). The hardness of c -split structure error vector can be bounded by the following result:

Lemma B.3. [\[FJR22, Theorem 1\]](#) *If there exists an algorithm solving a random instance of the c -split decoding problem with code length n , dimension k and w errors in time T and probability ϵ , then there exists an algorithm solving a random instance of the standard syndrome decoding problem with the same parameters in time T and with probability at least*

$$\frac{\binom{n/c}{w/c}^c}{\binom{n}{w}} \cdot \epsilon.$$

From [Lemma B.3](#), the complexity of c -split syndrome decoding variant is lower bounded by the complexity of solving the standard syndrome decoding problem times $\binom{n/c}{w/c}^c / \binom{n}{w}$.

We remark that the lower bound in [Lemma B.3](#) primarily works for the case that the c -split syndrome decoding variant has only one solution. When the number of solutions is taken into account, the lower bound was updated as per [\[CHT24\]](#). No existing decoding algorithm benefits from the c -split structure and thus [Lemma B.3](#) leads to relatively conservative parameters.

Remark B.4 (Regular error vector). For the optimized variants of FSS, the error vector is not only c -split but also *regular*, i.e., each block is the concatenation of t unit vectors. The regular LPN variant was used to construct hash function FSB [\[AFS03\]](#). Generally speaking, the regular variant does not significantly compromise the hardness of the problem. The regular LPN variant has renewed attention since it is used in [\[BCGI18, BCG⁺19b, BCG⁺20b, BCG⁺22, CCJ23, BCCD23\]](#) on digital signatures and secure computation. Recently, [\[ES24\]](#) pointed out that the code rate should be very small for the regular variant to have an impact on the security level and that the regular LPN variant is harder than the standard LPN in the high code-rate regime [\[ES24, Figure 1\]](#).

In our QA-SD and Ring-LPN assumption, the code rate is $1 - 1/c$ far from $1/2$ even for a constant $c \geq 3$, which makes the general attacks, for example the BKW [\[BKW00, Lyu05\]](#) and Arora-Ge [\[AG11\]](#), infeasible because both of the two algorithms make query to many LPN instances and thus are only good for a very small code rate.

Syndrome decoding instances from QA-SD and Ring-LPN can leverage the structure of quasi-cyclic and quasi-abelian codes. The Decoding One Of Many (DOOM) [Sen11] algorithm provides a $\sqrt{|G|}$ speed-up over general decoding algorithms. More importantly, the additional structure of the finite abelian group G allows an adversary to reduce an instance modulo a subgroup of G to obtain a smaller instance of the decoding problem such that the noise rate does not increase too much. As pointed out by [BCCD23, BBC⁺24], the best attack is induced from modulo subgroups of G and it is named as *folding attacks*, which will be precisely analyzed in Section B.3. At a high level, the folding attack strategy takes subgroups H of G , computes the folded code with respect to the subgroup H and then executes a general decoding algorithm on the folded instance. The folding operation maps a linear code of length $c|G|$ to a linear code of length $c|G/H|$ simultaneously keeping the code rate. The reduced error vector is still c -split because the folding operation is independently applied to each block. The number of errors in each block is upper bounded by t .

We remark that for QA-SD assumption, there are a huge number of subgroups of a given order. For $p = 2, m = 2, G = \mathbb{Z}_3^n$, there are $\begin{bmatrix} n \\ \eta \end{bmatrix}_3$ different subgroups of cardinality 3^η , where $\begin{bmatrix} n \\ \eta \end{bmatrix}_3$ is the Gaussian binomial coefficient and stands for the number of η dimensional subspaces of $\mathbb{Z}_3^n$ [And74]. For Ring-LPN assumption, it is possible to choose N being a prime factor of $p^m - 1$ to avoid the folding attacks. Even considering the subgroup in Ring-LPN, an adversary gains little advantage since the number of subgroups can be very small.

The parameters are chosen such that the general decoding algorithms against all folded instances achieve the desired security parameter, with the c -split structure and the $\sqrt{|G|/|H|}$ speed up factor from DOOM taken into account. The folding attack is only useful for large subgroups H , because a large subgroup H leads to a smaller reduced instance of the decoding problem. For the folding attack to be useful, the folded instance should have a unique solution, which requires the number of errors in the folded instance should be smaller than the Gilbert-Varshamov (GV) bound of the folded instance. Given a code of rate r , the GV bound is defined as

$$\delta_{GV} := h_q^{-1}(1 - r),$$

where $h_q(\cdot)$ is the q -ary entropy function i.e., $h_q(x) := -x \log_q(\frac{x}{q-1}) - (1-x) \log_q(1-x)$ for $x \in (0, 1 - 1/q)$. It is well-known that the GV bound provides a threshold for the number of solutions of a random instance. Given a random instance $(H, \mathbf{b})$ of code length n and distance w satisfying $w/n < \delta_{GV}$, the random instance $(H, \mathbf{b})$ has at most a unique solution with overwhelming probability over the choices of H . For $w/n > \delta_{GV}$, there will be exponentially many solutions, which was discussed in detail in [Deb23, Figure 2.2]. Specifically, once the number of errors in the folded instance is larger than the GV bound, we need to convert each of the exponential number of solutions to the folded instance to the original decoding problem to check whether it corresponds a valid solution. This procedure is equivalent to assign a small number of variables in a linear equation system induced from the original decoding problem [CT19] and it is not as efficient as

solving the decoding problem with larger code length. Note that it is sufficient to find a solution to a folded instance below the GV bound to distinguish between a QA-SD or Ring-LPN instance and a uniform instance.

In [Section B.3](#), a precise analysis of the probability distribution of the folded error vector is presented. The weight of the folded error vector is very close to the weight of the initial vector with overwhelming probability because the initial error vector distribution is extremely sparse. A formal analysis is presented to argue that the folded error has a much smaller weight $\tilde{w}$ than expected with exponentially small probability $\Pr_{\tilde{w}}$. The adversary makes use of the fact that G has a huge number of subgroups H to run the folding attack for decoding $\tilde{w}$ errors. It is worth to mention that this idea was used in [\[BBC+24\]](#) to argue the security level of the parameters in [\[BCCD23\]](#) is overestimated.

B.3 Folding Attacks

We present a formal analysis of the folding attacks. Let $H \preceq G$. The group homomorphism $\pi_H : G \rightarrow G/H$ is extended to a morphism of algebra:

$$\begin{aligned} \pi_H : \quad \text{GR}[G] &\longrightarrow \text{GR}[G/H] \\ \sum_{g \in G} a_g \cdot g &\longmapsto \sum_{\bar{g} \in G/H} \left(\sum_{h \in H} a_{h+\bar{g}} \right) \bar{g} \end{aligned}$$

A vector of length $|G|$ is mapped to a vector of length $|G/H|$. The map π_H just sums up the $|H|$ entries of each coset of G/H over G . For $\mathbf{b} := (b_1 \dots b_c) \in \text{GR}[G]^c$, the folding operation π_H is extended as

$$\pi_H(\mathbf{b}) := (\pi_H(b_1) \dots \pi_H(b_c)) \in \text{GR}[G/H]^c.$$

Given an instance $(\mathbf{a}, \langle \mathbf{a}, \mathbf{e} \rangle)$ of QA-SD or Ring-LPN, the folding operation π_H maps it to

$$\pi_H \left(e_0 + \sum_{i \in [1, c-1]} a_i \cdot e_i \right) = \pi_H(e_0) + \sum_{i \in [1, c-1]} \pi_H(a_i) \cdot \pi_H(e_i) \in \text{GR}[G/H].$$

It is indeed the syndrome of $\pi_H(e_0 \dots e_{c-1})$ over the matrix induced by $\pi_H(1, a_1 \dots a_{c-1})$.

In summary, the folding operation reduces a decoding problem with code length $c|G|$ and dimension $(c-1)|G|$ to a decoding problem with code length $c|G/H|$ and dimension $(c-1)|G|$. Thus, the code rate $1 - 1/c$ is preserved. The Hamming weight of the folded error is upper bounded by the original error weight and might reduce due to collisions. The goal is to estimate the weight of $\pi_H(e_0, \dots, e_{c-1})$, where each e_i is independent and uniformly distributed over the set of Hamming weight t .

Assume the working field is $\mathbb{F}_q$. For Galois ring, $\text{GR} := \mathbb{GR}(p^k, m)$, it holds that $\mathbb{GR}(p^k, m)/(p) \simeq \mathbb{F}_{p^m}$ and $q := p^m$. Assume $\ell := |H|$ and denote $\mathcal{C}_\ell := [\ell, \ell-1]_q$ as the parity code. Let $P_{\nu, \ell}(\mathbf{X})$ be the weight enumerator of $\mathcal{C}_\ell$ and $P_{\theta, \ell}(\mathbf{X})$ the weight enumerator of $\mathbb{F}_q^\ell \setminus \mathcal{C}_\ell$ [\[MS86, Chapter 5.2\]](#).

Lemma B.5. [[BBC⁺24](#), Proposition 18] Given $t \in [0, |G|]$ and $u \in [0, \min(t, |G/H|)]$, for e uniformly sampled over the Hamming weight t elements of $\text{GR}[G]$, it holds that

$$\Pr_e[\text{wt}(\pi_H(e)) = u] = \frac{\binom{|G/H|}{u} \cdot [\mathbf{x}^t] \left(P_{\theta, \ell}^u(\mathbf{x}) P_{\nu, \ell}^{|G/H|-u}(\mathbf{x}) \right)}{\binom{|G|}{t} (q-1)^t},$$

where $\binom{|G/H|}{u} \cdot [\mathbf{x}^t] \left(P_{\theta, \ell}^u(\mathbf{x}) P_{\nu, \ell}^{|G/H|-u}(\mathbf{x}) \right)$ is defined as the coefficient of $\mathbf{x}^t$ in the polynomial $\left(\binom{|G/H|}{u} P_{\theta, \ell}^u(\mathbf{x}) P_{\nu, \ell}^{|G/H|-u}(\mathbf{x}) \right)$.

MacWilliam's identity gives the formulae for $P_{\nu, \ell}$ and $P_{\theta, \ell}$.

Lemma B.6. [[CT19](#), Lemma 1] It holds that

$$P_{\nu, \ell}(\mathbf{x}) := \frac{1}{q} \left((1 + (q-1)\mathbf{x})^\ell + (q-1)(1-\mathbf{x})^\ell \right)$$

and

$$P_{\theta, \ell}(\mathbf{x}) := \frac{q-1}{q} \left((1 + (q-1)\mathbf{x})^\ell - (1-\mathbf{x})^\ell \right).$$

We note that [Lemma B.5](#) indicates that the probability is only related to the cardinality of H rather than H itself.

The subgroup H is chosen such that $t/|G/H|$ is below the GV bound of the folded code of rate $1 - 1/c$. For QA-SD assumption, taking $p = 2, m = 2$, G becomes $\mathbb{Z}_3^n$ and H becomes $\mathbb{Z}_3^\eta$. Then $|G/H| = 3^{n-\eta}$. Specifically, $t \leq \delta_{GV} 3^{n-\eta}$. Choosing $\eta = n - \lceil \log_3(t/\delta_{GV}) \rceil$ or $\eta = n - \lceil \log_3(t/\delta_{GV}) \rceil - 1$ yields the best complexity.

For $G = \mathbb{Z}_3^n$, there are $\begin{bmatrix} n \\ \eta \end{bmatrix}_3$ different subgroups of cardinality 3^η , where $\begin{bmatrix} n \\ \eta \end{bmatrix}_3$ is the Gaussian binomial coefficient and stands for the number of η dimensional subspaces of $\mathbb{Z}_3^n$ [[And74](#)]. An adversary could guess that the folded instance has a much smaller error weight τ with respect to H and then run the best decoding algorithm [Section B.5](#) on the folded instance. The adversary runs the decoding algorithm and aborts once the number of operations exceeds to decode τ errors. Then the adversary asserts that the current folded instance has no error vector below τ . Next the adversary takes another random subgroup H to run the decoding again. The average run time is

$$\frac{C_{\text{Dec}}(\tau) + C_{\text{fold}}}{\Pr[\text{wt}(\pi_H(\mathbf{e})) = \tau]},$$

where $C_{\text{Dec}}(\tau)$ is the complexity of the best decoding algorithm for weight $\leq \tau$ and C_{fold} is the complexity of the folding operation, scaling with code length of the original code. We choose τ to minimize the above complexity. The folding attack is very efficient for small values of c . For QA-SD and Ring-LPN assumption, we follow a SageMath script [[Bom24](#)] to choose security parameters.

B.4 New attack

Very recently, Bouillaguet et al. [BDHV25] proposed an efficient attack on the QA-SD problem over finite fields by combining techniques from sparse multivariate interpolation and convex optimization. This attack can be naturally extended to QA-SD over Galois rings by modulo p . To prevent such attacks, it suffices to choose parameters satisfying $n < 1 + (q - 1)(c - 1) \frac{m \cdot \log p}{\log(p^m - 1)}$ [BDHV25, Section 5].

B.5 Information Set Decoding (ISD)

Information Set Decoding (ISD) has been the primary technique for solving the decoding problem since Prange's work [Pra62]. Generally, the complexity of ISD is exponential in the error weight. The ISD algorithm has been widely studied [Ste88, Dum89, Wag02, MMT11, BJMM12, MO15, BM18, DEEK24]. Most of the analyses were performed for the binary field to improve the coefficient of the exponent. For large field, the recent algorithm [MO15] has large hidden constant and thus it is far from practical. Since $\mathbb{GR}(p^k, m)/(p) \simeq \mathbb{F}_{p^m}$, existing ISD decoding algorithms [BZ01, CMT24] over Galois ring essentially employ ISD over $\mathbb{F}_q$ ($q = p^m$) in an iterative way, by leveraging the p -adic representation. Therefore, we are not aware of any advantage of applying such ISD algorithms over Galois rings, and it suffices to apply ISD over $\mathbb{F}_q$ ($q = p^m$) to the Galois ring instance modulo p , where the expected noise weight is roughly reduced by $\frac{1}{q}$.

Our description of the ISD follows from the framework of [Deb23, Section 3] with respect to a parity-check matrix. Given a matrix $H \in \mathbb{F}_q^{(n-k) \times n}$, $\mathbf{b} \in \mathbb{F}_q^{n-k}$ and a weight parameter t , the goal of ISD is to find $\mathbf{e} \in \mathbb{F}_q^n$ such that $\text{wt}(\mathbf{e}) \leq t$ and $H\mathbf{e} = \mathbf{b}$. The idea is to randomly choose an information set I and bet there are only few errors in the information set.

For a set $I \subseteq [1, n]$, let $\bar{I} := [1, n] \setminus I$. Let H_I be the matrix consisting of columns indexed by the set I . An information set $I \subseteq [1, n]$ satisfies $|I| = k + \ell$ with $\ell \in \mathbb{N}$ and $H_{\bar{I}}$ of full rank, i.e., $\text{rk}(H_{\bar{I}}) = n - k - \ell$.

The ISD framework For two parameters $0 \leq \ell \leq n - k$ and $0 \leq p \leq \min(t, k + \ell)$,

1. Randomly sample a subset $I \subseteq [1, n]$ of size $k + \ell$ such that $H_{\bar{I}}$ is of full rank.
2. Compute an invertible matrix $U \in \mathbb{F}_q^{(n-k) \times (n-k)}$ such that $U \cdot H_{\bar{I}} = \begin{pmatrix} I_{n-k-\ell} \\ 0 \end{pmatrix}$ and compute $\tilde{\mathbf{b}} := U\mathbf{b} \in \mathbb{F}_q^{(n-k) \times (n-k)}$. Define two vectors $\tilde{\mathbf{b}}_1 \in \mathbb{F}_q^{(n-k-\ell)}$, $\tilde{\mathbf{b}}_2 \in \mathbb{F}_q^\ell$ as $\begin{pmatrix} \tilde{\mathbf{b}}_1 \\ \tilde{\mathbf{b}}_2 \end{pmatrix} := \tilde{\mathbf{b}} = U\mathbf{b}$ and define two matrices $H_1 \in \mathbb{F}_q^{(n-k-\ell) \times (k+\ell)}$, $H_2 \in \mathbb{F}_q^{\ell \times (k+\ell)}$ as $\begin{pmatrix} H_1 \\ H_2 \end{pmatrix} := UH_I$.
3. Compute a set $L \subset \{\mathbf{e}_2 \in \mathbb{F}_q^{k+\ell} : \text{wt}(\mathbf{e}_2) = p \text{ and } H_2\mathbf{e}_2 = \tilde{\mathbf{b}}_2\}$ from an ISD subroutine.

4. Filter an $\mathbf{e}_2 \in L$ such that $\mathbf{e}_1 := \tilde{\mathbf{b}}_1 - H_1 \mathbf{e}_2$ is of weight $t - p$. If no such $\mathbf{e}_2$, go to Step 1 to sample another information set I . If $\mathbf{e}_2$ is found, return the vector $\mathbf{e} := \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{pmatrix} \in \mathbb{F}_q^n$.

Note that after Step 2, the original instance $H\mathbf{e} = \mathbf{b}$ is transformed to

$$UH\mathbf{e} = \begin{pmatrix} I_{n-k-\ell} & H_1 \\ 0 & H_2 \end{pmatrix} \cdot \begin{pmatrix} \mathbf{e}_1 \\ \mathbf{e}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{e}_1 + H_1 \mathbf{e}_2 \\ H_2 \mathbf{e}_2 \end{pmatrix} = \tilde{\mathbf{b}} = \begin{pmatrix} \tilde{\mathbf{b}}_1 \\ \tilde{\mathbf{b}}_2 \end{pmatrix}.$$

As matrix U is non-singular, the output $\mathbf{e}$ is valid a solution of $H\mathbf{e} = \mathbf{b}$. Different variant of ISD differs in Step 3.

The above ISD framework involves a huge number of independent iterations and each iteration is a randomized algorithm. The complexity of the ISD framework is dominated by the ISD subroutine, Gaussian elimination of Step 3 and 2, and expected number of iterations, which corresponds the inverse probability of a random information set that leads to a solution. The complexity of the ISD framework is determined by the parameter (ℓ, p) and the ISD subroutine.

Let $C(n, k, q, t, \ell, p)$ be the complexity of the above ISD framework. Then it is

$$C(n, k, q, t, \ell, p) := \frac{C_{\text{sub}}(n, k, q, \ell, p) + C_{\text{Gaussian}}(n, k, q, \ell, p)}{\Pr(n, k, q, t, \ell, p)},$$

where $C_{\text{sub}}(n, k, q, \ell, p)$ is the complexity of the subroutine of Step 3, $C_{\text{Gaussian}}(n, k, q, \ell, p)$ the complexity of the Gaussian elimination of Step 2 and $\Pr(n, k, q, t, \ell, p)$ the probability of a random information set that leads to a solution.

The complexity of Gaussian elimination $C_{\text{Gaussian}}(n, k, q, \ell, p)$ is lowered bounded by $(n - k - \ell) \cdot n$.

According to [Sen23, Section 3.1], the complexities of concrete ISD variants are given as follows.

The Stern/Dumer algorithm The Stern/Dumer [Ste88, Dum89] algorithm employs the birthday paradox to find $\mathbf{e}_2$ of weight p .

Lemma B.7 (Stern/Dumer Algorithm Complexity [Ste88, Dum89]).
The complexity of an iteration of Stern's algorithm is given as

$$C_{\text{sub}}(n, k, q, \ell, p) = O(|L|(n - k - \ell)(k + \ell) + \max(|L_1|, |L|) + |L_1|),$$

where

$$|L_1| = \binom{(k + \ell)/2}{p/2} (q - 1)^{p/2} \text{ and } |L| = \binom{(k + \ell)/2}{p/2}^2 (q - 1)^p q^{-\ell}.$$

The probability of finding a solution is

$$\Pr(n, k, q, t, \ell, p) = \frac{\binom{(k + \ell)/2}{p/2}^2 \binom{n - k - \ell}{t - p}}{\binom{n}{t}}.$$

We remark that the Prange's algorithm [Pra62] can be viewed as a special case of Stern algorithm with $\ell = p = 0$ and thus no subroutine is called. The complexity of Prange's algorithm [Pra62] is explicitly given as

$$C_{\text{sub}}(n, k, q) = 0 \text{ and } \Pr(n, k, q, t) = \frac{\binom{n-k}{t}}{\binom{n}{t}}.$$

The Lee-Brickell algorithm [LB88] only set $\ell = 0$ and left p to optimize. The complexity of Lee-Brickell algorithm [LB88] is

$$C_{\text{sub}}(n, k, q) = 0 \text{ and } \Pr(n, k, q, t) = \frac{\binom{k}{p} \binom{n-k}{t-p}}{\binom{n}{t}}.$$

The MMT algorithm May et al. [MMT11] showed a representation technique to improve subroutine of the Stern algorithm.

Lemma B.8 (MMT Algorithm Complexity [MMT11]). *The complexity of an iteration of MMT algorithm is given as*

$$C_{\text{sub}}(n, k, q, \ell, p) = O(|L|(n - k - \ell)(k + \ell) + \max(|L_{00}|, |L_{02}|, |L|) + |L_{00}|),$$

where

$$|L_{00}| = \binom{(k + \ell)/2}{p/4} (q - 1)^{p/4}, |L_{02}| = \binom{(k + \ell)/2}{p/4}^2 (q - 1)^{p/2} q^{-r}$$

$$|L| = \binom{(k + \ell)/2}{p/4}^4 (q - 1)^p q^{-\ell - r}, R = \binom{p}{p/2}, r = \log_q R.$$

The probability of finding a solution is

$$\Pr(n, k, q, t, \ell, p) = \frac{\binom{k+\ell}{p} \binom{n-k-\ell}{t-p}}{\binom{n}{t}} \cdot \frac{\binom{(k+\ell)/2}{p/4}^4}{\binom{k+\ell}{p/2}^2}.$$

The BJMM algorithm Becker et al. [BJMM12] proposed the BJMM algorithm to improve the MMT algorithm. In the subroutine of MMT algorithm, the weight p is exactly divided as $(p/2, p/2)$ while the BJMM algorithm [BJMM12] chooses an additional parameter ϵ to divide p as $(p/2 + \epsilon, p/2 - \epsilon)$.

Lemma B.9 (BJMM Algorithm Complexity [BJMM12]). *The complexity of an iteration of BJMM algorithm is given as*

$$C_{\text{sub}}(n, k, q, \ell, p, \epsilon) = O(|L|(n - k - \ell)(k + \ell) + \max(|L_{00}|, |L_{02}|, |L|) + |L_{00}|),$$

where

$$|L_{00}| = \binom{(k + \ell)/2}{p_1/2} (q - 1)^{p_1/2}, |L_{02}| = \binom{(k + \ell)/2}{p_1/2}^2 (q - 1)^{p_1} q^{-r},$$

$$|L| = \left(\frac{(k+\ell)/2}{p_1/2} \right)^4 (q-1)^{2p_1} q^{-\ell-r}, p_1 = p/2 + \epsilon, r = \log_q R,$$

$$R = \sum_{\epsilon_1 + \epsilon_2 = \epsilon} \binom{p}{p/2 - \epsilon_1} \binom{p/2 + \epsilon_1}{2\epsilon_1} \binom{k + \ell - p}{\epsilon_2} (q-1)^{\epsilon + \epsilon_1}.$$

The probability of finding a solution is

$$\Pr(n, k, q, t, \ell, p) = \frac{\binom{k+\ell}{p} \binom{n-k-\ell}{t-p}}{\binom{n}{t}} \cdot \frac{\left(\frac{(k+\ell)/2}{p_1/2} \right)^4}{\binom{k+\ell}{p_1}^2}.$$

Note that for our parameter selection, the memory requirement of the MMT [MMT11] and BJMM algorithm [BJMM12] is huge and can not be neglected.

B.6 Statistical Decoding Attacks

Jabri [Jab01] introduced the statistical decoding algorithm, which was extensively studied [Ove06, DT17, CDMT22, MT23, CDMT24]. Recent research indicates that the statistical decoding algorithm improves over the classical ISD algorithm for sufficiently small code rate [CDMT22, CDMT24]. However, for code rate ≥ 0.42 , ISD is still the best known decoding algorithm [CDMT24, Figure 1].

C Deferred Proofs for Generalized Frobenius Maps and Trace Functions

C.1 Discriminant for Galois Ring Elements

Definition C.1 (Frobenius Matrix of a Basis). Let $(b_1 \dots b_m)$ be a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$. The matrix

$$M_{b_1 \dots b_m} := \sigma^j(b_i)_{i \in [1, m], j \in [0, m-1]} = \begin{pmatrix} \sigma^0(b_1) & \sigma^1(b_1) & \dots & \sigma^{m-1}(b_1) \\ \sigma^0(b_2) & \sigma^1(b_2) & \dots & \sigma^{m-1}(b_2) \\ \vdots & \vdots & \ddots & \vdots \\ \sigma^0(b_m) & \sigma^1(b_m) & \dots & \sigma^{m-1}(b_m) \end{pmatrix}$$

is defined as the Frobenius matrix of $(b_1 \dots b_m)$ with respect to σ with $\langle \sigma \rangle = \text{Gal}(\mathbb{GR}(p^k, m)/\mathbb{Z}_{p^k})$.

Definition C.2 (Discriminant Matrix of a Tuple). Let $\mathbb{GR}(p^k, m)$ be a Galois extension over $\mathbb{Z}_{p^k}$. The discriminant matrix of a tuple $(b_1 \dots b_m) \in (\mathbb{GR}(p^k, m))^m$ is defined as

$$\text{Disc}(b_1 \dots b_m) := \begin{pmatrix} \text{Tr}(b_1 b_1) & \dots & \text{Tr}(b_1 b_m) \\ \text{Tr}(b_2 b_1) & \dots & \text{Tr}(b_2 b_m) \\ \vdots & \ddots & \vdots \\ \text{Tr}(b_m b_1) & \dots & \text{Tr}(b_m b_m) \end{pmatrix}.$$

The determinant of $(b_1 \dots b_m) \in (\mathbb{GR}(p^k, m))^m$ is defined as

$$\Delta(b_1 \dots b_m) := \det(\text{Disc}(b_1 \dots b_m)) \in \mathbb{Z}_{p^k}.$$

Lemma C.3. Assume $\mathbb{GR}(p^k, m) = \mathbb{Z}_{p^k}[\zeta]$ for $\zeta \in \mathbb{GR}(p^k, m)$. Define matrix $M_\zeta := (\sigma^j(\zeta^i))_{i,j \in [0, m-1]}$. Then $\det(M_\zeta) \in \mathbb{Z}_{p^k}^*$.

Proof. From [Definition 2.11](#), the matrix M_ζ becomes

$$M_\zeta = (\zeta^{ip^j})_{i,j \in [0, m-1]} = \begin{pmatrix} 1 & 1 & \dots & 1 \\ \zeta^{p^0} & \zeta^{p^1} & \dots & \zeta^{p^{m-1}} \\ \vdots & \vdots & \ddots & \vdots \\ \zeta^{p^0(m-1)} & \zeta^{p^1(m-1)} & \dots & \zeta^{p^{m-1}(m-1)} \end{pmatrix}.$$

Hence, $\det(M_\zeta) = \prod_{i < j} (\zeta^{p^j} - \zeta^{p^i}) \in \mathbb{Z}_{p^k}^*$. $\square$

Lemma C.4. Let $(b_1 \dots b_m)$ be a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$. We have that $\det(M_{b_1 \dots b_m}) \in \mathbb{Z}_{p^k}^*$.

Proof. Let $P \in \mathbb{Z}_{p^k}^{m \times m}$ be a matrix such that $(b_1 \dots b_m)^\top = P \cdot (\zeta^0 \dots \zeta^{m-1})^\top$ and $\det(P) \in \mathbb{Z}_{p^k}^*$ as $(b_1 \dots b_m)$ is a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$. Note that the generalized Frobenius map $\sigma \in \text{Gal}(\mathbb{GR}(p^k, m)/\mathbb{Z}_{p^k})$ fixes $\mathbb{Z}_{p^k}$. Applying $\sigma^j(\cdot)$ for $j \in [0, m-1]$ on $(b_1 \dots b_m)P \cdot (\zeta^0 \dots \zeta^{m-1})^\top$ iteratively yields

$$M_{b_1 \dots b_m} = P \cdot M_\zeta.$$

Hence, $\det(M_{b_1 \dots b_m}) = \det(P) \cdot \det(M_\zeta) \in \mathbb{Z}_{p^k}^*$ because $\det(P), \det(M_\zeta) \in \mathbb{Z}_{p^k}^*$. $\square$

Theorem C.5. If $(b_1 \dots b_m)$ is a basis of $\mathbb{GR}(p^k, m)$ over $\mathbb{Z}_{p^k}$, it holds that $\det(\text{Disc}(b_1 \dots b_m)) \notin (p)\mathbb{Z}_{p^k}$ or $\det(\text{Disc}(b_1 \dots b_m)) \in \mathbb{Z}_{p^k}^*$.

Proof. Note that the discriminant matrix $\text{Disc}(b_1 \dots b_m)$ can be represented as

$$\text{Disc}(b_1 \dots b_m) = M_{b_1 \dots b_m} \cdot M_{b_1 \dots b_m}^\top.$$

Therefore, $\Delta(b_1 \dots b_m) = \det(\text{Disc}(b_1 \dots b_m)) = \det(M_{b_1 \dots b_m}) \cdot \det(M_{b_1 \dots b_m}^\top) = \det(M_{b_1 \dots b_m})^2 \in \mathbb{Z}_{p^k}^*$. $\square$

Note that the above results and proofs can be readily extended to general Galois ring extensions.